

# Historic tree asymptotics and a high order obstruction

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## Abstract

For the all-one family  $H_r^{(r)} = H_r^2$ , we prove the  $m = 2$  and  $m = 3$  instances of Burghart and Wagner’s 2026 historic-tree coefficient conjecture, where  $r = m + 1$ , and disprove every instance with  $m \geq 29$  for the specified counting sequences. For reduced 5- and 7-historic trees, the respective equivalents are  $30(n + 2)!\rho_3^{-n-3}$  and  $140(n + 3)!\rho_4^{-n-4}$ . Both have rigorously nonzero corrections oscillating in  $\log n$ , convergent local complex-power expansions, all fixed coefficient grades with exact Gamma ratios, and qualified eventual inverse brackets. Astashova’s earlier real-axis blowup laws are credited. For every ODE order  $r \geq 30$ , an elementary spectral estimate and a closed sign-variation cone exclude convergence of the actual all-one orbit to the normalized pole equilibrium. Exact rational certificates cover the finite bridge at orders 30 through 37. The spectrum itself has antecedents in B-urns and higher-order Emden–Fowler theory; the obstruction uses the specific initial data. Exact checks and replayable certificates are separated from non-certified numerical illustrations. No replacement high-order asymptotic or periodic limiting profile is asserted.

[write] **Added 5 October 2026, batch 103.** The report now has four Parts. Part I is the manuscript summarized above. Parts II–IV add three further manuscripts that prove the orders  $m = 2$ ,  $m = 3$  and  $m = 29$  again by different routes, and prove that *A333497* and *A336009* are not *P-recursive* (their generating functions are not D-finite; Theorem 30.1). The note at the end of Section 1 lists what they add.

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## Part I

# Historic tree asymptotics and a high order obstruction (source 01)

## 1 The counting problem and the results

### 1.1 The published target

Burghart and Wagner's model of reduced  $(2m+1)$ -historic trees has exponential generating function

$$H^{(m+1)} = H^2, \quad H^{(j)}(0) = 1 \quad (0 \leq j \leq m). \quad (1.1)$$

A reduced tree of size  $n$  corresponds to a B-tree history of length  $n+m$ . Their Conjecture 1 predicts

$$h_n \sim \frac{(2m+1)!}{(m!)^2} (n+m)! \rho_m^{-n-m-1}. \quad (1.2)$$

The journal statement appears in Section 4 of [5]; its predecessor is Conjecture 10 of [4]. The detailed argument in Sections 2–7 addresses  $m=2$ , for which

$$H'''(z) = H(z)^2, \quad H(0) = H'(0) = H''(0) = 1. \quad (1.3)$$

Equating exponential coefficients gives

$$h_0 = h_1 = h_2 = 1, \quad h_{n+3} = \sum_{j=0}^n \binom{n}{j} h_j h_{n-j}, \quad H(z) = \sum_{n \geq 0} h_n \frac{z^n}{n!}. \quad (1.4)$$

This recurrence identifies the sequence with OEIS A333497 [11]. The identification is by recurrence; it is not an assertion that the journal paper names this OEIS entry. The beginning is

$$1, 1, 1, 1, 2, 4, 8, 18, 48, 144, 456, 1560, 5808, 23184, 98160, 440832.$$

### 1.2 The conclusions across orders

Two low-order instances hold, but the conjecture as stated for all orders is false. ODE order  $r$  equals  $m+1$ ; the historic-tree order is  $2m+1$ .

| $m$       | $r$       | Conclusion for the all-one sequence                        |
|-----------|-----------|------------------------------------------------------------|
| 2         | 3         | A333497: equivalent, nonzero oscillation, all fixed grades |
| 3         | 4         | A336009: equivalent, nonzero oscillation, all fixed grades |
| $\geq 29$ | $\geq 30$ | Every corresponding proposed equivalent is false           |

The first two cases also admit eventual inverse brackets. Their real-axis leading powers use or recover Astashova's prior results; the coefficient conclusions require complex continuation. The last case is an obstruction for the specified initial jet, rather than an inference from generic instability. We first establish the third-order case in full detail and then isolate the additional arguments needed for the fourth-order result and the uniform high-order obstruction.

### 1.3 The third order theorem

Throughout, powers of  $t = \rho - z$  use a logarithm that is real on  $t > 0$ . Put

$$\lambda = \frac{13 + i\sqrt{71}}{2}, \quad \bar{\lambda} = \frac{13 - i\sqrt{71}}{2}, \quad \alpha = \frac{13}{2}, \quad \omega = \frac{\sqrt{71}}{2}, \quad \nu_{kl} = k\lambda + l\bar{\lambda}. \quad (1.5)$$

A *grade* means the total degree  $k + l$  in the two local complex powers. It is held fixed when  $n \rightarrow \infty$ .

**Theorem 1.1** (Counting asymptotics and convergent local expansion). *The solution of (1.3) has a finite positive blowup point  $\rho$ , satisfying*

$$180^{1/4} \leq \rho \leq 60^{1/3}. \quad (1.6)$$

*Its Taylor series at zero has radius  $\rho$ , with no other singularity on  $|z| = \rho$ . There is a uniquely normalized  $C \in \mathbb{C} \setminus \{0\}$  such that*

$$H(z) = 60t^{-3}\mathcal{F}(Ct^\lambda, \bar{C}t^{\bar{\lambda}}), \quad \mathcal{F}(X, Y) = \sum_{k, l \geq 0} a_{kl} X^k Y^l, \quad (1.7)$$

where  $\mathcal{F}$  is holomorphic in a bidisk at  $(0, 0)$ ,

$$a_{00} = a_{10} = a_{01} = 1, \quad a_{lk} = \overline{a_{kl}}.$$

*For each fixed finite angular interval on a chosen logarithmic sheet, (1.7) converges for all sufficiently small  $|t|$ , uniformly on closed subsectors.*

*For every fixed integer  $d \geq 0$ ,*

$$\frac{h_n}{30(n+2)!\rho^{-n-3}} = Q_d(n) + O_d(n^{-\alpha(d+1)}), \quad (1.8)$$

where, for all sufficiently large real  $x$ ,

$$Q_d(x) = 1 + \sum_{\substack{1 \leq k+l \leq d \\ k \neq l}} \frac{2a_{kl} C^k \bar{C}^l \rho^{\nu_{kl}}}{\Gamma(3 - \nu_{kl})} \frac{\Gamma(x + 3 - \nu_{kl})}{\Gamma(x + 3)}. \quad (1.9)$$

*In particular,  $h_n \sim 30(n+2)!\rho^{-n-3}$ .*

The factor 2 in each summand of (1.9) is present before conjugate pairing. Both  $(k, l)$  and  $(l, k)$  are included. All diagonal terms  $k = l \geq 1$  are excluded because they are analytic monomials at the singularity. The exact Gamma ratios preserve every integer inverse- $n$  correction belonging to a retained grade.

**Corollary 1.2** (A nonzero logarithmic oscillation). *With*

$$K = \frac{2C\rho^\lambda}{\Gamma(3 - \lambda)} \neq 0, \quad R_n = \frac{h_n}{30(n+2)!\rho^{-n-3}}, \quad (1.10)$$

one has

$$R_n = 1 + 2 \operatorname{Re} \left\{ K \frac{\Gamma(n+3-\lambda)}{\Gamma(n+3)} \right\} + O(n^{-13}), \quad (1.11)$$

$$= 1 + 2|K|n^{-13/2} \cos(\arg K - \omega \log n) + O(n^{-15/2}). \quad (1.12)$$

Consequently,

$$\limsup_{n \rightarrow \infty} n^{13/2}(R_n - 1) = 2|K|, \quad \liminf_{n \rightarrow \infty} n^{13/2}(R_n - 1) = -2|K|. \quad (1.13)$$

*The error  $R_n - 1$  changes sign infinitely often and is not  $O(n^{-7})$ .*

[write] **Added 5 October 2026, batch 103.** Part II (Theorem 13.1) proves Theorem 1.1 and this corollary again by an explicit convergent surface instead of Poincaré linearization, and Part III (Theorem 30.1) proves that  $H$  is not D-finite and A333497 is not P-recursive.

## 1.4 Attribution and the logical route

Astashova’s Theorem 3 [2] already gives the real-axis law  $H(x) \sim 60(\rho - x)^{-3}$  for positive blowup solutions of this third-order equation. Her theorem covers orders 3 and 4 under the stated Emden–Fowler hypotheses; it is not a theorem for all orders. The self-contained real argument below also proves finite blowup for our selected initial data. It is an alternate derivation, not a claim of priority for the real-axis law.

Real-axis blowup alone does not justify individual coefficient asymptotics. The remaining work is to obtain a convergent holomorphic local representation, prove that the first complex amplitude cannot vanish, exclude competing singularities on the convergence circle, and establish a domain in which singularity analysis applies. Holomorphic Poincaré linearization [8], Pringsheim’s theorem, and the standard transfer theorems [7] are used in their usual forms. The fourth-order extension and the high-order obstruction follow in Sections 8 and 9. The arguments are conventional analytic proofs accompanied by exact checks, not proof-assistant formalizations.

[write] **Added 2 October 2026, batch 77.** This report is a single manuscript, printed in full: manuscript 24 of batch 77 of ProveIt’s incoming-reports intake, delivered as the archive `historic-tree-report.zip` (wrapper directory `historic-tree-release/`) in commit 096ee7b87, and placed in commit 34f1acd4b in the research-report collection, among the OEIS-sequence asymptotics reports. Its main file `historic-trees.tex` is now `article.tex`; the `\input` file `extensions.tex` (Sections 8 and 9) and `references.bib` stay beside it. The manuscript names no ProveIt commit and continues no repository path, so it has no pin; its author line (above) names neither an author nor a tool, and its PDF metadata leave the author empty. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. No other report of the repository treats historic trees, A333497, A336009 or the Burghart–Wagner conjectures. The writing step prefixed every label (in both `.tex` files) with `his:`, added section labels, and added these [write] notes; no statement, proof or number of the manuscript was altered.

*Claims about external sources.* Theorem 9.1 (Burghart and Wagner’s Conjecture 1 fails for every  $m \geq 29$ ) and the remark in Section 11 that the journal’s Table 1 mislabels  $\rho_m$  as  $\rho_m^{-1}$  are this manuscript’s claims about a published paper. The intake checked the conjectured constants  $30 = 5!/(2!)^2$  and  $140 = 7!/(3!)^2$ , the exact gaps  $20/2997$  and  $392864/776223$  of Section 9.2, and reran the exact certificates; it did not re-read the journal’s table, and the theorem itself is an unrefereed proof.

*Notation.* The manuscript reuses several letters with section-local meanings. Each symbol is printed as delivered; the table fixes its meaning by section, with the tempting false readings. (Sections 1–7 treat  $r = 3$ , Section 8 treats  $r = 4$ , Section 9 treats a general order  $r$ .)

| symbol                          | meaning here (and what it is <i>not</i> )                                                                                                                                                                                                                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $m, r$                          | $m$ is Burghart–Wagner’s parameter, $r = m + 1$ the ODE order, $2m + 1 = 2r - 1$ the historic-tree order. “ $m \geq 29$ ” and “ $r \geq 30$ ” are the same range; the trees at $r = 30$ are 59-historic.                                                                                                                     |
| $\rho, \rho_m, \rho_4$          | $\rho$ is the blowup point of A333497 ( $r = 3$ ) in Sections 1–7, 10; $\rho_m$ is the conjecture’s singularity in (1.2); $\rho_4$ belongs to A336009; in Section 9, $\rho$ is the radius in the <i>alleged</i> equivalent (9.1) and in the jet (9.15). None of them is a reciprocal (see the Table 1 remark in Section 11). |
| $h_n, b_n, b_j, b(a), b_*$      | $h_n$ are exponential coefficients ( $H = \sum h_n z^n / n!$ ); $b_n$ are the <i>ordinary</i> Taylor coefficients of Section 5; $b_j$ is the initial jet (9.15); $b(a)$ the imaginary part solving (9.6); $b_* = 228/25$ .                                                                                                   |
| $a, a_{kl}, a_4$                | $a = p_*/3 = 60^{-1/3}$ (Section 3); $a_{kl}$ the universal coefficients of $\mathcal{F}$ ; $a_4 = 840^{-1/4}$ ; $a \in [0, 1]$ is a real part in Section 9.2. They are unrelated.                                                                                                                                           |
| $A, A_4, A_r$                   | $A = 4/3$ in the proof of Lemma 2.1; the Jacobian $A$ of (3.1) and $A_4$ of (8.7); $A_r = (r)_r$ of (9.2), a number.                                                                                                                                                                                                         |
| $B, \mathcal{B}, \mathcal{B}_4$ | $B = 5/3$ in the proof of Lemma 2.1; $B(u)$ the time series (3.8); $\mathcal{B}(x), \mathcal{B}_4(x)$ the main terms of the inverse problem.                                                                                                                                                                                 |
| $C, C_4, \mathcal{C}_3$         | $C, C_4$ the complex amplitudes; $\mathcal{C}_3$ the sign-variation cone (9.13); $\mathbb{C}$ the complex numbers.                                                                                                                                                                                                           |
| $D, D(\nu), D_4$                | $D(\nu)$ the indicial polynomial (4.2); $D = -1/18052070400$ the real-mode coefficient of A336009 (Theorem 8.1); $D_4(\nu)$ its indicial polynomial.                                                                                                                                                                         |
| $K, K_{\pm}, K_4$               | $K$ of (1.10) is the oscillation amplitude; $K_{\pm}$ in (3.7) are orbit constants, <i>not</i> $K$ ; $K_4$ is the order-four amplitude.                                                                                                                                                                                      |
| $P, P_s, P_r, p$                | $P(u)$ the linearized coordinate (3.3); $P_s$ the exact pole solutions (Section 2); $P(u)$ the arctangent polynomial of the finite bridge (Section 9.2); $P_r(\mu)$ the product polynomial; $p, p_j$ the desingularized coordinates.                                                                                         |
| $Q_d, Q(v), q$                  | $Q_d, Q_d^{(4)}$ the fixed-grade correction factors; $Q(v)$ the logarithm polynomial of the finite bridge; $q$ a desingularized coordinate, and also the quotient $\chi_{30}/(\mu - 1)$ .                                                                                                                                    |
| $G, G_r$                        | $G = H_4$ is the A336009 generating function (Section 8); $G_r(x) = A_r(\rho - x)^{-r}$ in Section 9 is the comparison pole, not a solution of the ODE.                                                                                                                                                                      |
| $R, R_n, \mathcal{R}$           | $R_n$ the normalized ratio (1.10); $R$ the Taylor radius (Section 5) and the implicit unknown at order four; $\mathcal{R}(\eta)$ the implicit function of Section 3.                                                                                                                                                         |
| $W, Z, Y$                       | $W(\cdot)$ is Lambert’s function in (7.6); $W = Y - 1$ at order four and $Z = Y - 1$ at order $r$ are deviations of the normalized jet $Y$ ; $Z(u)$ is also the backward trajectory in Section 8.                                                                                                                            |
| $s, S, s^-$                     | $s = \log(\rho/t)$ (Sections 8–9); $S(u)$ the homological solution (3.5); $S$ a finite time in the Metzler argument; $s^-(v)$ the number of sign changes.                                                                                                                                                                    |
| $L, L_r, L_8, \mathcal{L}$      | $L = \log X$ (Section 7); $L_r(a)$ the logarithmic modulus (9.7); $L_8$ a rational lower bound for $\log 2$ ; $\mathcal{L}$ the Euler operator.                                                                                                                                                                              |
| $\eta, \varepsilon$             | $\eta_{\pm}, \eta_j$ the local variables of the implicit equation; $\eta > 0$ the width of the Delta domain (5.4). $\varepsilon$ is a majorization margin; $\varepsilon_d(X)$ the inverse window (7.4).                                                                                                                      |
| $\lambda, \lambda_j, \alpha$    | $\lambda = (13 + i\sqrt{71})/2$ , $\lambda_{\pm}$ the pair $\lambda, \bar{\lambda}$ ; at order four $\lambda_4 = (11 + i\sqrt{159})/2$ and $\lambda_0 = 12$ . $\alpha = 13/2$ and $\alpha_4 = 11/2$ are grade weights, not exponents of a single term.                                                                       |

[write] **Added 5 October 2026, batch 103.** Since batch 103 the report has four Parts. Part I is the manuscript described above, unchanged except for dated notes. Parts II–IV print three manuscripts of the session bundle, Reports 113, 115 and 117 (sources 02, 03, 04; arrival 60f54ea06, placement 9c995cefe), written without sight of Part I. They prove again, by different routes, Theorem 1.1 with Corollary 1.2 (Part II), Theorem 8.1 (Part III) and the case  $r = 30$  of Theorem 9.1 (Part IV). They add: *A333497 and A336009 are not P-recursive and their generating functions are not D-finite* (Theorem 30.1); the bounds  $16800^{1/6} \leq \rho_4 \leq 840^{1/4}$ ; a blowup enclosure for every order (Proposition 35.1); explicit convergence radii 1183/960 and 121/6720 of the local series; and the complete spectrum at  $r = 30$  as a proved statement. Each Part opens with its provenance, a map of what it proves again, and its own notation table; its labels carry the prefixes `his:o5:`, `his:o7:`, `his:o59:`. The sentence above that no other report treats historic trees remains true outside this report. An independent check of the write, made afterwards, is recorded at the end of Section 41.

## 2 A global real solution and finite blowup

### 2.1 Desingularization

For a positive solution set

$$p = H' H^{-4/3}, \quad q = H'' H^{-5/3}, \quad \frac{d\tau}{dx} = H^{1/3}. \quad (2.1)$$

Direct differentiation gives the autonomous planar system

$$\dot{p} = q - \frac{4}{3}p^2, \quad \dot{q} = 1 - \frac{5}{3}pq, \quad (p(0), q(0)) = (1, 1), \quad (2.2)$$

where dots mean  $\tau$  derivatives. Its positive quadrant is forward invariant: on  $p = 0$  the first component equals  $q > 0$ , and on  $q = 0$  the second equals  $1 > 0$ . The unique positive equilibrium is

$$p_* = (9/20)^{1/3}, \quad q_* = \frac{4}{3}p_*^2, \quad p_*q_* = \frac{3}{5}. \quad (2.3)$$

**Lemma 2.1** (Strict Lyapunov dissipation). *On  $p \geq 0$ ,  $q > 0$ , let*

$$V(p, q) = \frac{5}{6}(p - p_*)^2 + q - q_* - q_* \log(q/q_*). \quad (2.4)$$

*Then  $V \geq 0$ , vanishing only at the equilibrium, and*

$$\dot{V} = -\frac{20}{9}(p - p_*)^2(p + p_*) - \frac{5}{3}p_* \frac{(q - q_*)^2}{q}. \quad (2.5)$$

*The orbit in (2.2) exists for every  $\tau \geq 0$  and converges to  $(p_*, q_*)$ .*

*Proof.* Write  $A = 4/3$  and  $B = 5/3$ . At the equilibrium,

$$\dot{p} = (q - q_*) - A(p^2 - p_*^2), \quad \dot{q} = -B\{q(p - p_*) + p_*(q - q_*)\}.$$

Multiply these equations by  $B(p - p_*)$  and  $1 - q_*/q$ , respectively. Their cross terms cancel, giving (2.5). The logarithmic term in  $V$  tends to infinity both as  $q \downarrow 0$  and as  $q \rightarrow \infty$ ; the quadratic term bounds  $p$  above. Thus the orbit stays in a compact sublevel in  $p \geq 0$ ,  $q > 0$ . Such a set may meet  $p = 0$ , but the polynomial vector field is regular there and points inward. This causes no loss of continuation.

On this sublevel, (2.5) bounds the integrals of  $(p - p_*)^2$  and  $(q - q_*)^2$ . Their derivatives are bounded, hence the squared deviations are uniformly continuous. A nonnegative uniformly continuous integrable function tends to zero: otherwise separated intervals of fixed positive area contradict integrability. Applying this observation twice proves convergence. Equivalently, LaSalle's invariance principle applies because the zero set of (2.5) is a single point.  $\square$

### 2.2 Reconstruction of the original time

The construction can be reversed without any blowup assumption. Define

$$H(\tau) = \exp\left(\int_0^\tau p(s) ds\right), \quad x(\tau) = \int_0^\tau H(s)^{-1/3} ds. \quad (2.6)$$

For  $x = x(\tau)$ , direct differentiation using (2.2) verifies (1.3) and its initial data. Since  $p(\tau) \rightarrow p_* > 0$ ,  $H$  grows exponentially in  $\tau$ , and

$$0 < \rho := \int_0^\infty H(s)^{-1/3} ds < \infty. \quad (2.7)$$

Thus  $x(\tau) \uparrow \rho$  and  $H(x) \rightarrow \infty$ . No earlier real obstruction occurs.

Put  $v = H^{-1/3}$ . In the original variable,

$$v'(x) = -\frac{p(x)}{3}, \quad v(x) = \frac{1}{3} \int_x^\rho p(s) ds \sim \frac{p_*}{3}(\rho - x). \quad (2.8)$$

Here  $p(x)$  means  $p(\tau(x))$ . It follows that

$$H(x) \sim \left(\frac{3}{p_*}\right)^3 (\rho - x)^{-3} = 60(\rho - x)^{-3}. \quad (2.9)$$

All three components required in Astashova's real-axis hypotheses diverge, since  $p \rightarrow p_* > 0$  and  $q \rightarrow q_* > 0$ .

### 2.3 Analytic bounds for the radius

The rectangle

$$\frac{3}{5} \leq p \leq 1, \quad \frac{12}{25} \leq q \leq 1 \quad (2.10)$$

is forward invariant and contains  $(1, 1)$ . On its four faces,

$$\dot{p}|_{p=3/5} = q - 12/25 \geq 0, \quad \dot{p}|_{p=1} \leq -1/3, \quad \dot{q}|_{q=12/25} = 1 - 4p/5 \geq 1/5, \quad \dot{q}|_{q=1} \leq 0.$$

Integrating (2.8) gives the useful tail bounds

$$x + 3H(x)^{-1/3} \leq \rho \leq x + 5H(x)^{-1/3} \quad (0 \leq x < \rho). \quad (2.11)$$

At zero these imply  $3 < \rho < 5$ , with strictness because  $p$  is not identically an endpoint.

A tighter initial bound comes from the exact pole solutions

$$P_s(x) = 60(s - x)^{-3}, \quad (P_s(0), P'_s(0), P''_s(0)) = (60s^{-3}, 180s^{-4}, 720s^{-5}).$$

For  $s = 180^{1/4}$  this initial vector dominates  $(1, 1, 1)$  componentwise; for  $s = 60^{1/3}$  it is dominated by  $(1, 1, 1)$ . The system

$$(H, H', H'')' = (H', H'', H^2)$$

is cooperative in the nonnegative octant. Its comparison principle, up to the first blowup of either solution, therefore yields (1.6). Numerically, only for orientation,

$$3.66284150148 \dots \leq \rho \leq 3.91486764116 \dots$$

These bounds are analytic and independent of every fitted value below.

[write] **Added 5 October 2026, batch 103.** The same comparison works in every order: Part IV's Proposition 35.1 encloses the blowup point of  $H_r^{(r)} = H_r^2$  for every  $r$ , and Part III proves  $16800^{1/6} \leq \rho_4 \leq 840^{1/4}$ .

### 3 A convergent expansion at the singularity

#### 3.1 Holomorphic linearization

The Jacobian of (2.2) at the equilibrium is

$$A = \begin{pmatrix} -8p_*/3 & 1 \\ -5q_*/3 & -5p_*/3 \end{pmatrix}, \quad \det(\mu I - A) = \mu^2 + \frac{13p_*}{3}\mu + \frac{20p_*^2}{3}. \quad (3.1)$$

Its eigenvalues are  $p_*(-13 \pm i\sqrt{71})/6$ . Put

$$a = \frac{p_*}{3} = 60^{-1/3}. \quad (3.2)$$

Label the eigenvalues  $-a\lambda$  and  $-a\bar{\lambda}$ . Their convex hull excludes zero. A resonance of degree  $k+l \geq 2$  would require  $k(-a\lambda) + l(-a\bar{\lambda})$  to equal one eigenvalue; equality of real parts would force  $k+l=1$ . Thus no resonance exists.

The holomorphic Poincaré linearization theorem [8] gives holomorphic coordinates  $u_+, u_-$  near the equilibrium with

$$\dot{u}_+ = -a\lambda u_+, \quad \dot{u}_- = -a\bar{\lambda} u_-, \quad p = p_* + P(u), \quad P(u) = u_+ + u_- + O(\|u\|^2). \quad (3.3)$$

Every eigenvector of  $A$  has a nonzero first component, so this normalization is possible. Choose conjugate eigenvectors; uniqueness of the normalized nonresonant conjugacy then preserves the real structure. Along our real orbit,  $u_- = \overline{u_+}$ .

#### 3.2 Removing the multiplicative correction

The variable  $v = H^{-1/3}$  satisfies

$$\dot{v} = -(a + P(u)/3)v. \quad (3.4)$$

Let  $\mathcal{L} = \lambda u_+ \partial_{u_+} + \bar{\lambda} u_- \partial_{u_-}$ . The homological equation

$$\mathcal{L}S = -P/p_*, \quad S(0) = 0 \quad (3.5)$$

has the unique holomorphic solution

$$S_{kl} = -\frac{P_{kl}}{p_*(k\lambda + l\bar{\lambda})} \quad (k+l \geq 1).$$

Indeed, the real part of each denominator apart from  $p_*$  is  $\alpha(k+l)$ , so division preserves convergence of the power series. Now

$$w = v \exp S(u) \implies \dot{w} = -aw. \quad (3.6)$$

For a sufficiently late point on the real orbit,  $w > 0$  and

$$u_+ = K_+ w^\lambda, \quad u_- = K_- w^{\bar{\lambda}}, \quad K_- = \overline{K_+}. \quad (3.7)$$

If either constant vanished, both would vanish, making the local orbit the equilibrium. Uniqueness would then make the whole planar orbit the equilibrium, contradicting  $(p(0), q(0)) = (1, 1)$ . Hence both constants are nonzero.

Let  $E(u) = \exp[-S(u)]$  and define

$$B(u) = \sum_{k,l \geq 0} \frac{E_{kl}}{1 + k\lambda + l\bar{\lambda}} u_+^k u_-^l. \quad (3.8)$$

Again the denominators preserve convergence;  $B(0) = 1$  and  $(1 + \mathcal{L})B = E$ . The remaining original time is exactly

$$t = \rho - x = \int_{\tau}^{\infty} v(s) \, ds = \frac{w}{a} B(u). \quad (3.9)$$

For example, differentiating the right-hand side gives  $-wE = -v$ , and its limit at infinity is zero. This proves the identity without a merely formal integration.

### 3.3 Analytic inversion and a nonzero linear term

Set  $\eta_+ = K_+(at)^\lambda$ ,  $\eta_- = K_-(at)^{\bar{\lambda}}$  and seek  $w = at\mathcal{R}(\eta)$ . Equation (3.9) becomes

$$\mathcal{R} B(\eta_+ \mathcal{R}^\lambda, \eta_- \mathcal{R}^{\bar{\lambda}}) = 1. \quad (3.10)$$

Powers of  $\mathcal{R}$  use the holomorphic logarithm near 1. At  $(\mathcal{R}, \eta) = (1, 0)$  the derivative with respect to  $\mathcal{R}$  equals 1. The analytic implicit-function theorem gives a unique holomorphic  $\mathcal{R}$  with  $\mathcal{R}(0) = 1$ . Therefore

$$\begin{aligned} H &= 60t^{-3} F(\eta), \\ F(\eta) &= \mathcal{R}(\eta)^{-3} \exp \left[ 3S(\eta_+ \mathcal{R}(\eta)^\lambda, \eta_- \mathcal{R}(\eta)^{\bar{\lambda}}) \right]. \end{aligned} \quad (3.11)$$

This is a holomorphic function of two independent variables near zero.

The linear coefficients, with  $e_j$  denoting a coordinate multi-index, are

$$S_{e_j} = -\frac{1}{p_* \lambda_j}, \quad B_{e_j} = \frac{1}{p_* \lambda_j (1 + \lambda_j)}, \quad F_{e_j} = -\frac{3}{p_* (1 + \lambda_j)}, \quad (3.12)$$

where  $\lambda_+ = \lambda$ ,  $\lambda_- = \bar{\lambda}$ . Thus

$$C = -\frac{3K_+ a^\lambda}{p_* (1 + \lambda)} \neq 0. \quad (3.13)$$

Absorbing these two nonzero linear coefficients into the arguments of  $F$  produces exactly the normalization  $\mathcal{F}$  in (1.7).

On every bounded interval of  $\arg t$  on a logarithmic sheet,

$$|t^{\lambda_j}| = |t|^{13/2} \exp(-\operatorname{Im}(\lambda_j) \arg t) \longrightarrow 0.$$

Hence the two-variable Taylor series converges throughout a sufficiently small corresponding sector. It agrees with the original solution along the positive real approach and therefore, by analytic uniqueness, on their common complex domain. The radius may shrink with the chosen angular interval; no radius uniform over arbitrarily many logarithmic sheets is claimed. In particular,

$$H(z) = 60t^{-3} \left( 1 + Ct^\lambda + \bar{C}t^{\bar{\lambda}} + O(|t|^{13}) \right). \quad (3.14)$$

The normalization and distinct exponents determine  $C$  uniquely on the specified branch.

[write] **Added 5 October 2026, batch 103.** Part II reaches (1.7) by a second route (Section 16), with an explicit bidisk of convergence,  $\max(|q_1|, |q_2|) < 1183/960$  in the variables  $q_1 = Ct^\lambda$ ,  $q_2 = \bar{C}t^{\bar{\lambda}}$ , and proves  $C \neq 0$  by an exact-pole contradiction.

## 4 Universal coefficients and analytic diagonal terms

Substitution of (1.7) into  $H''' = H^2$  is justified by local convergence. For a monomial  $t^{\nu-3}$ , three  $z$  derivatives multiply it by  $(3-\nu)(4-\nu)(5-\nu)$  and reduce its exponent by 3. Isolating the two products with the constant term gives, for  $k+l \geq 2$ ,

$$D(\nu_{kl})a_{kl} = 60 \sum_{\substack{0 \leq i \leq k, 0 \leq j \leq l \\ (i,j) \neq (0,0), (k,l}}} a_{ij}a_{k-i,l-j}, \quad (4.1)$$

where

$$D(\nu) = (3-\nu)(4-\nu)(5-\nu) - 120 = -(\nu+1)(\nu-\lambda)(\nu-\bar{\lambda}). \quad (4.2)$$

This recurrence is universal: initial data select  $\rho$  and  $C$ , whereas the normalized  $a_{kl}$  depend only on the equation. All denominators at degree at least two are nonzero, because  $\operatorname{Re} \nu_{kl} \geq 13$  whereas the roots of  $D$  have real parts  $-1$  and  $13/2$ . There are no logarithmic resonance terms in the local expansion.

For example,

$$\begin{aligned} a_{11} &= -\frac{1}{7}, & a_{20} &= \frac{359 + 57i\sqrt{71}}{17978}, \\ a_{30} &= \frac{-17559 + 4223i\sqrt{71}}{78725662}, & a_{21} &= \frac{2571}{1312394} - \frac{42477i\sqrt{71}}{119427854}. \end{aligned} \quad (4.3)$$

The reflected coefficients are their conjugates. The code calculates and checks exact values through degree four and numerical values through degree six.

The diagonal is exceptional for coefficient transfer. If  $k = l \geq 1$ , then

$$\nu_{kk} = 13k, \quad 60a_{kk}|C|^{2k}t^{13k-3} \quad (4.4)$$

is an analytic polynomial in  $z$  of degree  $13k-3$ . Its  $n$ th Taylor coefficient is zero once  $n > 13k-3$ . The convergent sum of the diagonal terms is an analytic *local* background at  $t = 0$ . Its local analyticity does not, by itself, give a global exponentially small error. For the fixed-grade theorem, only finitely many such polynomials need be removed, and their eventual contribution is exactly zero.

The indexing  $(k, l) \mapsto \nu_{kl}$  is injective: its real and imaginary parts determine  $k+l$  and  $k-l$ . Later, after integer inverse- $n$  corrections are introduced, different labels can have equal final exponents because  $\lambda + \bar{\lambda} = 13$ . Such coefficients must be combined; the coincidence introduces no logarithms.

## 5 The dominant circle and a transfer domain

### 5.1 Identification of the radius

Write  $H(z) = \sum_{n \geq 0} b_n z^n$  to avoid confusing ordinary coefficients with  $a_{kl}$ . Then

$$b_0 = b_1 = 1, \quad b_2 = \frac{1}{2}, \quad b_{n+3} = \frac{\sum_{j=0}^n b_j b_{n-j}}{(n+1)(n+2)(n+3)}. \quad (5.1)$$

Every  $b_n$  is strictly positive. The radius  $R$  of this Taylor series equals  $\rho$ . Indeed, real blowup implies  $R \leq \rho$ . If  $R < \rho$ , Pringsheim's theorem would force a singularity at the positive point  $R$ , whereas analytic ODE continuation along the finite real solution makes  $H$  regular there. This contradiction proves equality. The real solution and  $\rho$  were established before this argument, so there is no circular definition.

## 5.2 Strict majorization away from the positive point

Fix  $0 < r < \rho$  and  $\theta \notin 2\pi\mathbb{Z}$ . Each of  $H, H', H''$  has strictly positive coefficients at all powers, including consecutive powers. Equality in the triangle inequality is therefore impossible, and

$$|H^{(j)}(re^{i\theta})| < H^{(j)}(r) \quad (j = 0, 1, 2). \quad (5.2)$$

By continuity, choose one  $0 < \varepsilon < r$  so that all three left sides are at most  $H^{(j)}(r - \varepsilon)$ .

Let  $z_0 = re^{i\theta}$ . The Taylor recurrence at  $z_0$  has the same polynomial convolution form as (5.1). Its first three coefficients, after taking absolute values, are bounded by those of  $H(r - \varepsilon + w)$ . Induction in the recurrence gives coefficientwise domination at every degree. The real-centered comparison series has nonnegative coefficients and radius  $\rho - r + \varepsilon$ : this follows from the same real continuation, blowup, and Pringsheim argument used above. The series at  $z_0$  consequently converges in

$$|z - z_0| < \rho - r + \varepsilon. \quad (5.3)$$

Since  $|\rho e^{i\theta} - z_0| = \rho - r$ , this disk contains  $\rho e^{i\theta}$  in its interior. Every point on the dominant circle other than  $\rho$  is regular.

## 5.3 Gluing into a single dented domain

The continued germs from (5.3) agree with the original solution on their intersections with  $|z| < \rho$ . By uniqueness they agree on sufficiently small overlapping neighborhoods along the circle. Any compact arc avoiding  $\rho$  has a finite covering by such neighborhoods and hence an analytic collar. Near  $\rho$ , the convergent representation of Section 3 supplies the needed punctured sector. Shrinking the collar and the sector and using compactness gives some  $\eta > 0$  and  $0 < \vartheta < \pi/2$  such that  $H$  is single-valued and holomorphic in the standard Delta domain

$$\{z : |z| < \rho + \eta, z \neq \rho, |\arg(z - \rho)| > \vartheta\}. \quad (5.4)$$

The excluded wedge points outwards along the positive real axis. This local-to-global collar argument is sufficient for every fixed-order transfer. It makes no classification of more distant singularities.

# 6 All fixed coefficient grades and the oscillation

## 6.1 Exact transfer of each retained monomial

For a noninteger complex  $\nu$ , the binomial identity gives

$$[z^n](\rho - z)^{\nu-3} = \rho^{\nu-3-n} \frac{\Gamma(n+3-\nu)}{\Gamma(3-\nu)\Gamma(n+1)}. \quad (6.1)$$

All off-diagonal  $\nu_{kl}$  are nonreal, so this identity applies directly. The leading pole has coefficient

$$[z^n]60(\rho - z)^{-3} = 30(n+1)(n+2)\rho^{-n-3}. \quad (6.2)$$

Multiplying (6.1) by  $60a_{kl}C^k\overline{C}^l$  and dividing by (6.2) gives each summand of (1.9), including its factor 2. Multiplication by  $n!$  then gives the exact factorial normalization in (1.8).

## 6.2 A remainder valid for every fixed grade

Fix  $d \geq 0$ . Expand the convergent  $\mathcal{F}$  through degree  $d + 1$ , one grade beyond the desired truncation. Every off-diagonal term of degree  $d + 1$  has coefficient

$$O_d(\rho^{-n} n^{2-\alpha(d+1)}),$$

by (6.1) and the standard Gamma-ratio estimate. Every diagonal term is eventually zero. The remaining analytic-sector tail is

$$O_d(|t|^{\alpha(d+2)-3}).$$

Choose a real noninteger  $\beta$  strictly between  $\alpha(d + 1) - 3$  and  $\alpha(d + 2) - 3$ . The tail is also  $O_d(|t|^\beta)$ . The Delta-domain  $O$ -transfer theorem [7, Chapter VI] yields

$$O_d(\rho^{-n} n^{-\beta-1}) = o_d(\rho^{-n} n^{2-\alpha(d+1)}).$$

After division by the exact pole coefficient, this proves (1.8). Keeping one extra grade and using a noninteger tail avoids any borderline integer-exponent transfer convention.

The quantifier is important. The local two-variable series converges near the singularity, but the infinite sum obtained by transferring all grades is *not asserted to converge at fixed  $n$* . The theorem is an all-fixed-grade asymptotic result.

## 6.3 Nonvanishing and phase winding

Take  $d = 1$  in (1.8). Conjugate pairing gives (1.11). Since  $C \neq 0$ ,  $\rho > 0$ , and the Gamma function has no zeros or pole at  $3 - \lambda$ , one has  $K \neq 0$ . The fixed-parameter expansion

$$\frac{\Gamma(n + 3 - \lambda)}{\Gamma(n + 3)} = n^{-\lambda} \left( 1 + \frac{\lambda(\lambda - 5)}{2n} + O(n^{-2}) \right) \quad (6.3)$$

proves (1.12). The phase  $\arg K - \omega \log n$  winds without bound, while its increment tends to zero. Integers nearest the exponential targets for phase 0 or  $\pi$  approach those phases along subsequences. The cosine therefore has  $\limsup 1$  and  $\liminf -1$ , proving (1.13) and the sign changes.

No pure integer inverse- $n$  corrections precede this oscillation when the exact factor  $(n + 2)!$  is retained. Expanding that factorial by Stirling would introduce the usual separate Stirling corrections. In contrast, expanding every exact Gamma ratio to any fixed order supplies all requested relative algebraic orders of the form

$$\nu_{kl} + j, \quad k + l \geq 1, \quad k \neq l, \quad j \geq 0. \quad (6.4)$$

Only finitely many grades and integer corrections are needed below any fixed real-order cutoff. Coincident exponents in (6.4) are combined as noted in Section 4.

## 7 Inverse thresholds with controlled rounding

### 7.1 An eventual inverse bracket

The coefficient equivalent implies

$$\frac{h_{n+1}}{h_n} \sim \frac{n+3}{\rho} > 1,$$

so choose  $n_0$  after which  $h_n$  is strictly increasing. For sufficiently large  $X$ , define

$$N(X) = \min\{n \geq n_0 : h_n \geq X\}. \quad (7.1)$$

Put

$$\mathcal{B}(x) = 30\Gamma(x+3)\rho^{-x-3}, \quad F_d(x) = \log \mathcal{B}(x) + \log Q_d(x). \quad (7.2)$$

For fixed  $d$ ,  $Q_d(x) = 1 + O(x^{-13/2}) > 0$  eventually and

$$F'_d(x) = \psi(x+3) - \log \rho + O_d(x^{-15/2}) \sim \log(x/\rho) > 0. \quad (7.3)$$

Thus there is a unique sufficiently large root  $x_d = x_d(X)$  of  $F_d(x_d) = \log X$ .

**Theorem 7.1** (Eventual shrinking ceiling window). *For each fixed  $d \geq 0$ , there are constants  $M_d > 0$  and  $X_d$  such that, for  $X \geq X_d$ , setting*

$$\varepsilon_d(X) = M_d \frac{x_d(X)^{-\alpha(d+1)}}{\log x_d(X)} \quad (7.4)$$

*gives*

$$\lceil x_d(X) - \varepsilon_d(X) \rceil \leq N(X) \leq \lceil x_d(X) + \varepsilon_d(X) \rceil. \quad (7.5)$$

*Proof.* Taking logarithms in (1.8), with  $Q_d$  bounded away from zero, gives  $\log h_n = F_d(n) + O_d(n^{-\alpha(d+1)})$ . For integers near  $x_d$ , the mean value theorem and (7.3) show that displacement of size a sufficiently large constant times  $x_d^{-\alpha(d+1)}/\log x_d$  dominates this error. Thus integers strictly below the lower endpoint have  $h_n < X$ , and integers at or above the upper endpoint have  $h_n \geq X$ , after increasing the constant if needed. Eventual monotonicity handles all more distant integers, proving the stated ceiling window.  $\square$

The constants  $M_d$  and  $X_d$  are existence constants here. This is not a certified finite- $X$  numerical enclosure. If  $x_d$  lies exceptionally near an integer, its ceiling alone need not equal  $N(X)$ ; both sides of (7.5) must be retained. The theorem is deliberately formulated to avoid an unjustified rounding rule.

## 7.2 A Lambert W starting point

For a simple initial estimate, write  $L = \log X$  and use the positive branch of Lambert  $W$ :

$$y_0 = \frac{L}{W(L/(e\rho))}, \quad c = \log(30\sqrt{2\pi}) - 3\log \rho. \quad (7.6)$$

The shifted Stirling expansion is

$$\log \mathcal{B}(x) = x(\log(x/\rho) - 1) + \frac{5}{2} \log x + c + \frac{37}{12x} + O(x^{-2}). \quad (7.7)$$

The coefficient is  $37/12$ , not the unshifted  $1/12$ ; equivalently it is  $B_2(3)/2$ . Since  $y_0(\log(y_0/\rho) - 1) = L$ , one correction gives

$$y_1 = y_0 - \frac{(5/2) \log y_0 + c}{\log(y_0/\rho)}, \quad x_0 = y_1 + O\left(\frac{1}{y_0 \log y_0}\right). \quad (7.8)$$

Indeed  $y_1 - y_0 = O(1)$ , and the residual after this step in (7.7) is  $O(1/y_0)$ , while the derivative is asymptotic to  $\log y_0$ .

### 7.3 The first oscillatory displacement

Let  $x_0$  solve the exact main equation  $\log \mathcal{B}(x_0) = L$  and let  $x_1$  solve  $F_1(x_1) = L$ . Expansion of  $\log Q_1$  and the mean value theorem yield

$$x_1 - x_0 = -\frac{2 \operatorname{Re}\{K\Gamma(x_0 + 3 - \lambda)/\Gamma(x_0 + 3)\}}{\psi(x_0 + 3) - \log \rho} + O\left(\frac{x_0^{-13}}{\log x_0}\right). \quad (7.9)$$

The squared first correction is  $O(x_0^{-13})$  in the logarithm. Evaluation shifts and derivative corrections are smaller at this scale. Higher grades are handled by the exact  $F_d$  equations, or by fixed-order Newton expansions. The inverse accuracy improves with every fixed grade, while integer recovery continues to obey (7.5).

[write] **Added 2 October 2026, batch 77.** The inversion steps of this section are instances of general results of the repository's transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`), and no novelty is claimed for them. The Lambert start (7.6) is the case  $a = b = 1$  of `p0:thm:lambert-core` applied to  $u = \log(x/(e\rho))$ : the equation  $x(\log(x/\rho) - 1) = L$  becomes  $u + \log u = \log(L/(e\rho))$ , whose positive root is  $u = W_0(L/(e\rho))$ , branch  $W_0$ . The ceiling window of Theorem 7.1 has the form of the separation condition, part (2) of `p0:thm:staircase`, with  $x_* = x_d(X)$  and  $\eta = \varepsilon_d(X)$ : when the two ceilings in (7.5) coincide,  $N(X)$  is their common value. Part (1) of that theorem (exact rounding) is not used, because  $F_d$  is a model of  $\log h_n$  rather than an interpolation of it, which is why the manuscript keeps both ceilings. The first oscillatory displacement (7.9) is the first-order term  $-\varepsilon E/F'$  of `p0:thm:perturbed-inversion` with  $F = \log \mathcal{B}$  and  $\varepsilon E = \log Q_1$ . The generic staircase arithmetic is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` (file `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`); those lemmas concern an arbitrary monotone function, not  $h_n$ , and give this report no formal status. The same remarks apply to the order-four inverse of Section 8.6.

## 8 The fourth order equation and reduced seven historic trees

We now write  $G = H_4$  for the all-one solution of  $G^{(4)} = G^2$ , with exponential coefficients  $g_n$ . Its recurrence is (1.4) with shift four and four initial ones, identifying OEIS A336009 [12]. Write  $\rho_4$  for its positive real blowup point and put

$$\lambda_4 = \frac{11 + i\sqrt{159}}{2}, \quad \alpha_4 = \frac{11}{2}, \quad a_4 = 840^{-1/4}. \quad (8.1)$$

**Theorem 8.1** (The second proved instance). *There are  $\rho_4 > 0$  and  $C_4 \neq 0$  such that*

$$g_n = 140(n+3)!\rho_4^{-n-4}(1 + O(n^{-11/2})). \quad (8.2)$$

*The point  $\rho_4$  is the unique singularity on the convergence circle. With  $t = \rho_4 - z$ , a convergent local expansion is*

$$G(z) = 840t^{-4}\mathcal{F}_4(Dt^{12}, C_4t^{\lambda_4}, \overline{C}_4t^{\overline{\lambda}_4}), \quad D = -\frac{1}{18052070400}, \quad (8.3)$$

*where  $\mathcal{F}_4$  is holomorphic near zero and has constant and all three linear coefficients equal to one. The fixed-grade coefficient and inverse statements in Sections 8.5 and 8.6 hold. In particular, the first oscillatory relative correction has nonzero amplitude and order  $n^{-11/2}$ .*

[write] **Added 5 October 2026, batch 103.** Part III proves this theorem again (Theorem 22.1), adds  $16800^{1/6} \leq \rho_4 \leq 840^{1/4}$ , an explicit tridisk of convergence and a scalar-majorant proof that  $C_4 \neq 0$ , and proves that  $G$  is not D-finite and A336009 is not P-recursive (Theorem 30.1).

## 8.1 Finite blowup and the derivative upgrade

The following elementary observation works in every order, without claiming a universal power-law profile.

**Lemma 8.2** (Finite positive blowup in every order). *For  $r \geq 2$ , the all-one solution of  $H_r^{(r)} = H_r^2$  has a finite positive right endpoint  $\rho_r$  and  $H_r(x) \rightarrow +\infty$  as  $x \uparrow \rho_r$ .*

*Proof.* All state derivatives are positive and increasing. Taylor's formula with integral remainder gives

$$H_r(x+u) \geq H_r(x) + H_r(x)^2 u^r / r!$$

whenever the interval exists. If the solution were global, recursively take  $u_j = (r!/H_r(x_j))^{1/r}$  and  $x_{j+1} = x_j + u_j$ . Then  $H_r(x_j) \geq 2^j$  but  $x_j \leq (r!)^{1/r} \sum_{k \geq 0} 2^{-k/r} < \infty$ , contradicting continuity at the finite limit of the  $x_j$ . At a finite endpoint, bounded  $H_r$  would bound  $H_r^{(r)}$  and, by repeated integration, every lower derivative; the ODE would extend. Thus the increasing  $H_r$  diverges there.  $\square$

[write] **Added 5 October 2026, batch 103.** Part IV's Proposition 35.1 sharpens this lemma to the enclosure  $\min_j r_j \leq \rho_r \leq \max_j r_j$ ,  $r_j = [A_r(r)_j]^{1/(r+j)}$ .

For  $r = 4$ , Astashova's Theorem 3 [2] now gives

$$G(x) \sim 840(\rho_4 - x)^{-4}. \quad (8.4)$$

To upgrade derivatives, use the following monotone-density argument. If  $f(x) \sim c(\rho - x)^{-\beta}$  with  $c, \beta > 0$  and  $f'$  nondecreasing, sandwich  $f'(x)$  between the backward and forward secants at  $x \pm \varepsilon(\rho - x)$ . First let  $x \uparrow \rho$  for fixed  $0 < \varepsilon < 1$ , then let  $\varepsilon \downarrow 0$ . This gives  $f'(x) \sim c\beta(\rho - x)^{-\beta-1}$ . Repeating for  $G, G', G''$  is legitimate because  $G'', G''', G^{(4)}$  are positive. Therefore

$$G^{(j)}(x) \sim 840(4)_j(\rho_4 - x)^{-4-j}, \quad 0 \leq j \leq 3, \quad (8.5)$$

where  $(r)_j = r(r+1) \cdots (r+j-1)$  and  $(r)_0 = 1$ .

## 8.2 Three dimensional linearization and time inversion

Set  $p_j = G^{(j)}G^{-1-j/4}$  for  $j = 1, 2, 3$  and  $d\tau/dx = G^{1/4}$ . Then

$$\dot{p}_1 = p_2 - \frac{5}{4}p_1^2, \quad \dot{p}_2 = p_3 - \frac{3}{2}p_1p_2, \quad \dot{p}_3 = 1 - \frac{7}{4}p_1p_3. \quad (8.6)$$

Equation (8.5) implies convergence to  $(4a_4, 20a_4^2, 120a_4^3)$ , and  $\tau \rightarrow \infty$ . The Jacobian and its polynomial are

$$A_4 = \begin{pmatrix} -10a_4 & 1 & 0 \\ -30a_4^2 & -6a_4 & 1 \\ -210a_4^3 & 0 & -7a_4 \end{pmatrix}, \quad (8.7)$$

$$\det(sI - A_4) = (s + 12a_4)(s^2 + 11a_4s + 70a_4^2).$$

The three positive-real-part exponents are  $\lambda_0 = 12$ ,  $\lambda_+ = \lambda_4$ , and  $\lambda_- = \overline{\lambda_4}$ . Their negatives times  $a_4$  are the eigenvalues. They are nonresonant: the real part of any sum of at least two exponents exceeds  $11/2$ ; at degree two the only candidates below 12 have real part 11, and at degree three or more the minimum is  $33/2 > 12$ .

Holomorphic Poincaré linearization gives  $\dot{u}_j = -a_4\lambda_j u_j$  and  $p_1 = 4a_4 + P(u)$ , normalized so the three linear coefficients of  $P$  equal one. Every eigenvector has nonzero first component, as the first two rows of  $A_4$  show. With  $v = G^{-1/4}$ , let

$$\mathcal{L} = \sum_j \lambda_j u_j \partial_{u_j}, \quad \mathcal{L}S = -P/(4a_4), \quad w = ve^{S(u)}.$$

Then  $\dot{w} = -a_4 w$  exactly. Division by  $\mathbf{m} \cdot \lambda$ , whose real part is at least  $(11/2)|\mathbf{m}|$ , preserves convergence. If  $E = e^{-S}$  and  $B_{\mathbf{m}} = E_{\mathbf{m}}/(1 + \mathbf{m} \cdot \lambda)$ , then

$$t = (w/a_4)B(u), \quad u_j = K_j w^{\lambda_j}.$$

The implicit equation

$$RB(\eta_0 R^{12}, \eta_+ R^{\lambda_4}, \eta_- R^{\bar{\lambda}_4}) = 1, \quad \eta_j = K_j (a_4 t)^{\lambda_j},$$

has derivative one in  $R$  at  $(1, 0)$ . Exactly as in Section 3, it yields a holomorphic three-variable function and a convergent expansion in every fixed finite-angle sector. Its linear coefficients before normalization are

$$F_{e_j} = -\frac{1}{a_4(1 + \lambda_j)} \neq 0. \quad (8.8)$$

After absorbing them into the three amplitudes, one obtains (8.3). At this point the amplitudes are real/conjugate as appropriate; their precise real value and complex nonvanishing are established next.

### 8.3 The conserved energy fixes the real mode

Direct differentiation shows that

$$\mathcal{E} = G'''G' - \frac{1}{2}(G'')^2 - \frac{1}{3}G^3 \equiv \frac{1}{6}. \quad (8.9)$$

In the convergent expansion, let  $D$  be the relative coefficient of  $t^{12}$ . The three linear contributions of  $840Dt^8$  to the constant term of (8.9) are, respectively,  $-2304 \cdot 840^2 D$ ,  $-1120 \cdot 840^2 D$ , and  $-840 \cdot 840^2 D$ . Their sum is  $-4264 \cdot 840^2 D$ . No complex-mode combination contributes to this constant term: a real exponent in the relative semigroup has the form  $12m + 11k$ , and  $12m + 11k = 12$  has only the solution  $(m, k) = (1, 0)$  in nonnegative integers. Hence

$$-4264 \cdot 840^2 D = \frac{1}{6},$$

which gives the exact nonzero value in (8.3).

### 8.4 A backward positive system excludes a zero complex amplitude

Put  $s = \log(\rho_4/t)$  and

$$Y_j = \frac{t^{4+j}G^{(j)}}{840(4)_j}, \quad 0 \leq j \leq 3.$$

Then  $Y'_j = (4+j)(Y_{j+1} - Y_j)$  for  $j < 3$  and  $Y'_3 = 7(Y_0^2 - Y_3)$ . Subtracting the equilibrium,  $W = Y - \mathbf{1}$  satisfies  $W' = M(s)W$ , where

$$M(s) = \begin{pmatrix} -4 & 4 & 0 & 0 \\ 0 & -5 & 5 & 0 \\ 0 & 0 & -6 & 6 \\ 7(Y_0(s) + 1) & 0 & 0 & -7 \end{pmatrix}.$$

All  $Y_j$  are positive. For  $J_0 = \text{diag}(1, -1, 1, -1)$ , the matrix  $-J_0 M(s) J_0$  is Metzler, meaning that all its off-diagonal entries are nonnegative. If  $W$  were strictly alternating at some finite time  $S$ , then  $J_0 W(S)$  or its negative would be strictly positive. The backward trajectory  $Z(u) = J_0 W(S - u)$  solves  $Z' = -J_0 M(S - u) J_0 Z$ . A continuous Metzler system preserves a strictly positive initial vector over finite time, so strict alternation would already hold at  $s = 0$ .

However,

$$Y_j(0) = \frac{\rho_4^{4+j}}{840(4)_j}, \quad \frac{Y_{j+1}(0)}{Y_j(0)} = \frac{\rho_4}{4+j}.$$

The ratios decrease, so  $\log Y_j(0)$  is strictly concave. The set of indices for which  $Y_j(0) > 1$  is an interval. After any zeros are deleted,  $W(0)$  therefore has at most two sign changes and is not strictly alternating.

If the complex amplitudes vanished, convergence and the nonzero real mode would give

$$G = 840t^{-4}(1 + Dt^{12} + O(t^{24})), \quad W = Dt^{12} \left(1, -2, \frac{14}{5}, -\frac{14}{5}\right) + O(t^{24}).$$

This is strictly alternating for small positive  $t$ , a contradiction. Thus  $C_4 \neq 0$ , using (8.8). This proof concerns the actual initial data and needs no numerical amplitude estimate.

## 8.5 All fixed coefficient grades

The strict-majorization proof of Section 5 applies with four initial derivatives. The ordinary coefficients of  $G$  and its first three derivatives are strictly positive. A common strict margin bounds their values at  $re^{i\theta}$  by those at  $r - \varepsilon$ ; the fourth-order Taylor convolution then supplies a disk of radius  $\rho_4 - r + \varepsilon$ . Thus  $\rho_4$  is the unique point on its dominant circle, and the convergent sector supplies a Delta domain.

Expand (8.3) as

$$G = 840t^{-4} \sum_{m,k,l \geq 0} c_{mkl} D^m C_4^k \bar{C}_4^l t^{\nu_{mkl}}, \quad \nu_{mkl} = 12m + k\lambda_4 + l\bar{\lambda}_4. \quad (8.10)$$

The constant and linear coefficients are one. Universally, at total degree at least two,

$$D_4(\nu_{\mathbf{m}})c_{\mathbf{m}} = 840 \sum_{\mathbf{0} < \mathbf{i} < \mathbf{m}}^* c_{\mathbf{i}} c_{\mathbf{m}-\mathbf{i}}, \quad D_4(\nu) = (\nu + 1)(\nu - 12)(\nu - \lambda_4)(\nu - \bar{\lambda}_4). \quad (8.11)$$

Here the starred sum means all componentwise  $0 \leq \mathbf{i} \leq \mathbf{m}$  except the two endpoints; it does not require strict inequality in each component. Equivalently,  $D_4(\nu) = (4 - \nu)(5 - \nu)(6 - \nu)(7 - \nu) - 1680$ . The nonresonance check above proves that every denominator in (8.11) is nonzero. The recurrence may be justified in the three independent mode parameters before specializing to this orbit, so collisions among numerical exponents at high degree cause no ambiguity.

Every term with  $k = l$  has relative exponent  $12m + 11k$ . Except for the constant, this is an integer at least 11 and gives an analytic polynomial in  $G$ . These terms must be excluded from the coefficient expansion. For fixed  $d \geq 0$ , define

$$Q_d^{(4)}(x) = 1 + \sum_{\substack{1 \leq m+k+l \leq d \\ k \neq l}} \frac{6c_{mkl} D^m C_4^k \bar{C}_4^l \rho_4^{\nu_{mkl}}}{\Gamma(4 - \nu_{mkl})} \frac{\Gamma(x + 4 - \nu_{mkl})}{\Gamma(x + 4)}. \quad (8.12)$$

The factor is  $\Gamma(4) = 6$  before pairing. The same extra-grade/noninteger-tail argument proves

$$\frac{g_n}{140(n+3)!\rho_4^{-n-4}} = Q_d^{(4)}(n) + O_d(n^{-\alpha_4(d+1)}). \quad (8.13)$$

Total degree gives the convenient minimum weight  $\alpha_4 = 11/2$ ; the real mode has the larger weight 12. Thus this is a valid fixed-degree truncation even when a retained real-mode term has higher weight than an omitted focus term. For a sharper ordering one may instead retain all exponents up to a fixed real-part cutoff. Equal exponents, including those caused by  $\lambda_4 + \bar{\lambda}_4 = 11$ , are combined after specialization. No logarithms result.

With  $K_4 = 6C_4\rho_4^{\lambda_4}/\Gamma(4 - \lambda_4) \neq 0$ , the first grade is

$$\frac{g_n}{140(n+3)!\rho_4^{-n-4}} = 1 + 2\operatorname{Re}\left\{K_4\frac{\Gamma(n+4-\lambda_4)}{\Gamma(n+4)}\right\} + O(n^{-11}). \quad (8.14)$$

Its leading cosine has amplitude  $2|K_4|$ , angular frequency  $\sqrt{159}/2$  in  $\log n$ , and relative order  $n^{-11/2}$ . The corresponding scaled limsup and liminf are  $\pm 2|K_4|$ . In particular the relative error changes sign infinitely often and is not  $O(n^{-6})$ .

## 8.6 Inverses at every fixed grade

Define  $\mathcal{B}_4(x) = 140\Gamma(x+4)\rho_4^{-x-4}$  and  $F_d^{(4)} = \log \mathcal{B}_4 + \log Q_d^{(4)}$ . Then

$$(F_d^{(4)})'(x) = \psi(x+4) - \log \rho_4 + O_d(x^{-13/2}) > 0$$

eventually. If  $x_d^{(4)}$  solves  $F_d^{(4)}(x) = \log X$ , the eventual inverse of  $g_n$  lies between

$$\left[ x_d^{(4)} - M_d \frac{(x_d^{(4)})^{-\alpha_4(d+1)}}{\log x_d^{(4)}} \right] \quad \text{and} \quad \left[ x_d^{(4)} + M_d \frac{(x_d^{(4)})^{-\alpha_4(d+1)}}{\log x_d^{(4)}} \right].$$

This follows exactly as in Theorem 7.1, with eventual existence constants. No finite numerical certificate or unconditional ceiling rule is asserted. The first oscillatory displacement is (7.9) with  $K, \lambda, \rho, 3$  replaced by  $K_4, \lambda_4, \rho_4, 4$ , and error  $O(x^{-11}/\log x)$ .

The Lambert- $W$  start also carries over. In place of (7.7),

$$\log \mathcal{B}_4(x) = x(\log(x/\rho_4) - 1) + \frac{7}{2} \log x + \log(140\sqrt{2\pi}) - 4 \log \rho_4 + \frac{73}{12x} + O(x^{-2}).$$

Thus replace  $\rho$  by  $\rho_4$  and  $5/2$  by  $7/2$  in (7.6)–(7.8), with the displayed new constant. The one-step error remains  $O(1/(y_0 \log y_0))$ .

## 9 The proposed equivalent fails at every sufficiently high order

The positive low-order conclusions cannot be extended to the whole family.

**Theorem 9.1** (Failure for every parameter at least twenty nine). *For every integer  $r \geq 30$ , let  $H_r^{(r)} = H_r^2$  with  $H_r^{(j)}(0) = 1$  for  $0 \leq j < r$ , and let  $h_n^{(r)}$  be its exponential coefficients. There is no  $\rho > 0$  for which*

$$h_n^{(r)} \sim \frac{(2r-1)!}{((r-1)!)^2} (n+r-1)! \rho^{-n-r}. \quad (9.1)$$

*Consequently Burghart and Wagner's Conjecture 1 is false for every  $m = r-1 \geq 29$ . In particular, at  $r = 30$  it fails for the specified sequence of reduced 59-historic trees.*

The distinction between the ODE order  $r$ , conjecture parameter  $m = r - 1$ , and historic-tree order  $2m + 1 = 2r - 1$  is essential. The theorem identifies no replacement asymptotic, proves no periodic limiting profile, and makes no assertion that  $m = 29$  is the first failing parameter. The even-order historic-tree Conjecture 2 is not addressed.

[write] **Added 5 October 2026, batch 103.** Part IV (bundle Report 117, “B-tree order 59”) proves the first case  $r = 30$  again, that is  $m = 29$  and historic-tree order 59, by a cyclic-sign cone, a complete spectral count and the ordinary stable manifold; it claims nothing for  $r > 30$  and nothing about a first transition. Dictionary: Section 33.1.

## 9.1 What a coefficient equivalent would force

For general  $r \geq 2$ , put

$$A_r = (r)_r = \frac{(2r-1)!}{(r-1)!}, \quad G_r(x) = A_r(\rho - x)^{-r}. \quad (9.2)$$

The conjectured comparison coefficients are exactly  $G_r^{(n)}(0) = A_r(n+r-1)!\rho^{-n-r}/(r-1)!$ . Suppose for contradiction that the equivalent holds for one fixed  $r \geq 30$ . The root test makes the Taylor radius of  $H_r$  equal to this  $\rho$ . For each fixed  $j \geq 0$ , coefficient equivalence gives

$$H_r^{(j)}(x) \sim G_r^{(j)}(x) = A_r(r)_j(\rho - x)^{-r-j} \quad (x \uparrow \rho).$$

Indeed, for any  $\varepsilon > 0$ , all but finitely many coefficients of the derivative series are within relative  $\varepsilon$  of those of  $G_r^{(j)}$ . The finite head stays bounded at  $\rho$ , while  $G_r^{(j)}$  diverges. Divide the tail bounds by  $G_r^{(j)}$  and let  $\varepsilon \downarrow 0$ . This is a direct Abelian argument, not a Tauberian converse or a complex transfer theorem.

With  $t = \rho - x$  and  $s = \log(\rho/t)$ , define

$$Y_j(s) = \frac{t^{r+j} H_r^{(j)}(x)}{A_r(r)_j}, \quad 0 \leq j < r. \quad (9.3)$$

All coordinates are positive, and the alleged equivalent forces  $Y(s) \rightarrow \mathbf{1}$ . Exact differentiation yields

$$\begin{aligned} Y_j' &= (r+j)(Y_{j+1} - Y_j), & 0 \leq j < r-1, \\ Y_{r-1}' &= (2r-1)(Y_0^2 - Y_{r-1}). \end{aligned} \quad (9.4)$$

The Jacobian  $J_r$  at  $\mathbf{1}$  has diagonal  $-(r+j)$ , superdiagonal  $r+j$ , and bottom-left entry  $2(2r-1)$ . Its characteristic polynomial is

$$\chi_r(\mu) = \prod_{j=r}^{2r-1} (\mu + j) - 2 \prod_{j=r}^{2r-1} j, \quad \chi_r(1) = 0. \quad (9.5)$$

The mode  $\mu = 1$  corresponds to changing the blowup location.

## 9.2 An unstable conjugate pair for every order at least thirty

Divide (9.5) by  $A_r$ . A root obeys

$$\prod_{j=r}^{2r-1} (1 + \mu/j) = 2.$$

For  $a \in [0, 1]$ , define  $b(a) > 0$  by the phase equation

$$\Phi_r(a, b(a)) := \sum_{j=r}^{2r-1} \arctan \frac{b(a)}{j+a} = 2\pi. \quad (9.6)$$

The left side increases continuously from 0 to  $r\pi/2$  with  $b$ , so  $b(a)$  exists uniquely and depends continuously on  $a$ . Define the logarithmic modulus

$$L_r(a) = \frac{1}{2} \sum_{j=r}^{2r-1} \log \frac{(j+a)^2 + b(a)^2}{j^2}. \quad (9.7)$$

At  $a = 1$ , since  $b(1) > 0$ , telescoping gives

$$L_r(1) > \sum_{j=r}^{2r-1} \log \frac{j+1}{j} = \log 2.$$

We show  $L_r(0) < \log 2$ . Continuity then gives a root  $\mu = a + ib(a)$  with  $0 < a < 1$  and its conjugate. Together with  $\mu = 1$ , this supplies at least three unstable eigenvalues.

**The finite bridge from thirty to thirty seven.** Set

$$b_* = \frac{228}{25}, \quad P(u) = u - \frac{u^3}{3} + \frac{u^5}{5} - \frac{u^7}{7}, \quad Q(v) = v - \frac{v^2}{2} + \frac{v^3}{3},$$

and  $L_8 = 2 \sum_{k=0}^7 ((2k+1)3^{2k+1})^{-1} < \log 2$ . For nonnegative arguments,  $P(u) \leq \arctan u$  and  $\log(1+v) \leq Q(v)$ : the derivatives of the differences are  $u^8/(1+u^2)$  and  $v^3/(1+v)$ , respectively, and both differences vanish at zero. For each of the eight integers  $30 \leq r \leq 37$ , exact rational arithmetic gives

$$\sum_{j=r}^{2r-1} P(b_*/j) - \frac{44}{7} > \frac{1}{200}, \quad (9.8)$$

$$L_8 - \frac{1}{2} \sum_{j=r}^{2r-1} Q((b_*/j)^2) > \frac{1}{1000}. \quad (9.9)$$

The archive contains the exact fractions and two independent replay implementations. These are finite rational inequalities, without rounded transcendental evaluations. Since  $2\pi < 44/7$ , (9.8) implies  $b(0) < b_*$ . Monotonicity in  $b$  and (9.9) then imply  $L_r(0) < \log 2$ .

**The infinite range from thirty eight onward.** For  $r \geq 38$ , use  $\arctan u \geq u - u^3/3$ , the integral bounds

$$\sum_{j=r}^{2r-1} \frac{1}{j} \geq \log 2, \quad \sum_{j=r}^{2r-1} \frac{1}{j^3} \leq \frac{1}{2(r-1)^2},$$

and  $\log 2 > 56/81$ , from the first two positive terms of its atanh series. The phase at  $b = 10$  is at least

$$10 \log 2 - \frac{1000}{6(r-1)^2} > 10 \frac{56}{81} - \frac{1000}{6 \cdot 37^2} = \frac{753140}{110889} > \frac{44}{7} > 2\pi. \quad (9.10)$$

Thus  $b(0) < 10$ . Since  $\log(1 + u) < u$  for  $u > 0$ ,

$$L_r(0) < 50 \sum_{j=r}^{2r-1} j^{-2} \leq 50 \left( \frac{1}{r-1} - \frac{1}{2r-1} \right) \leq \frac{76}{111} < \frac{56}{81} < \log 2. \quad (9.11)$$

The rational function in the middle decreases for  $r \geq 2$ . The strict gaps are exactly  $56/81 - 76/111 = 20/2997$  and  $753140/110889 - 44/7 = 392864/776223$ . This completes the unstable-pair proof for every  $r \geq 30$ .

### 9.3 Simple spectra without exponential aliasing

Let  $P_r(\mu) = \prod_{j=r}^{2r-1} (\mu + j)$ . Its derivative has  $r - 1$  distinct real zeros interlacing the roots. At any such critical point  $\xi$ , every factor has modulus at most  $r - 1$ , so

$$|P_r(\xi)| \leq (r - 1)^r < r^r \leq A_r < 2A_r.$$

Therefore  $P_r - 2A_r$  has no repeated root: all eigenvalues of  $J_r$  are simple, uniformly in  $r$ .

Gershgorin disks bound every imaginary part by  $2(2r - 1) < 4r$ . Choose

$$\delta = \frac{1}{100r}, \quad T_r = \exp(\delta J_r). \quad (9.12)$$

The imaginary spectral span after scaling is less than  $0.08 < 2\pi$ , so distinct eigenvalues cannot alias under exponentiation. The matrix  $T_r$  is nonsingular with simple spectrum and at least three eigenvalues of modulus greater than one.

### 9.4 A closed cone containing the actual orbit

For a real vector  $v$ , let  $s^-(v)$  count ordinary left-to-right sign changes after deleting zero coordinates, with  $s^-(0) = 0$ . Define the closed, generally nonconvex cone

$$\mathcal{C}_3 = \{v \in \mathbb{R}^r : s^-(v) \leq 2\}. \quad (9.13)$$

To see closedness, a limit vector with three sign changes would have four ordered nonzero coordinates witnessing them; their signs persist in every sufficiently close vector.

Writing  $Z = Y - \mathbf{1}$ , subtraction in (9.4) gives the exact time-varying linear equation  $Z' = M(s)Z$ . The matrix  $M$  has the same diagonal and superdiagonal as  $J_r$ , but bottom-left entry

$$(2r - 1)(Y_0(s) + 1) > 0. \quad (9.14)$$

Its third additive compound  $M^{[3]}$  is Metzler. In a basis indexed by sorted three-element subsets, each off-diagonal move replaces a selected coordinate by its cyclic predecessor when that site is empty. An ordinary adjacent replacement has positive exterior sign; a wraparound crosses two selected coordinates and has sign  $(-1)^2 = +1$ .

The graph is strongly connected. Identify a vertex with three particles on a directed cycle; an edge moves one particle into a vacant neighboring site. The outgoing and incoming degrees count the cyclic 10 and 01 interfaces and are equal. The underlying undirected graph is connected because ordinary adjacent exchanges connect all three-element subsets. A finite weakly connected balanced directed graph is strongly connected: a source component in its condensation graph has no incoming edges, hence by degree balance no outgoing edges, contradicting weak connectedness unless it is the whole graph.

The continuous-time strong  $k$ -positivity theorem [13, Theorems 4–5] now implies that every positive-time transition matrix has all third-order minors positive and preserves  $\mathcal{C}_3 \setminus \{0\}$ . Its hypotheses hold on every compact time interval: coefficients are continuous and the fixed compound graph is irreducible with positive edges. In particular this applies to the constant matrix  $J_r$ , so  $\exp(uJ_r)$  is strictly sign-regular of order three for  $u > 0$ . Equivalently,

$$(\exp(uJ_r))^{(3)} = \exp(uJ_r^{[3]}) > 0$$

entrywise, where the superscript (3) denotes the multiplicative compound.

At  $s = 0$  the actual initial data give

$$b_j := Y_j(0) = \frac{\rho^{r+j}}{A_r(r)_j}, \quad \frac{b_{j+1}}{b_j} = \frac{\rho}{r+j}. \quad (9.15)$$

The decreasing ratios make  $\log b_j$  strictly concave. Its superlevel set  $\{j : b_j > 1\}$  is an interval, so  $b_j - 1$  has at most the block pattern negative/positive/negative, with zeros deleted. Therefore  $Z(0) \in \mathcal{C}_3$ . It is nonzero:  $b_0 = b_1 = b_2 = 1$  would require both  $\rho = r$  and  $\rho = r + 1$ . Uniqueness prevents  $Z$  from reaching zero at finite time. Consequently

$$Z(s) \in \mathcal{C}_3 \setminus \{0\} \quad (s \geq 0). \quad (9.16)$$

[write] **Added 5 October 2026, batch 103.** An elementary proof of (9.16), in every order and without [13], follows from Part IV: the cone  $\mathcal{C}_3$  equals Part IV’s cyclic cone (Remark 33.1), and the system  $Z' = M(s)Z$  has the cyclic form to which Part IV’s Euler-step Lemma 36.2 applies (Remark 33.2). Part I still uses [13] for  $T_r$  below; for the cyclic count see Remark 33.3.

## 9.5 A spectral gap and a generalized stable manifold

The spectral separation theorem for nonsingular strictly sign-regular matrices [1, Theorem 2(ii)–(iii)] applies to  $T_r$  with  $k = 3$ . Let  $E_{\text{fast}}$  be its top-three real spectral subspace and  $E_{\text{slow}}$  the invariant complement. It gives a strict modulus gap and

$$E_{\text{slow}} \cap \mathcal{C}_3 = \{0\}. \quad (9.17)$$

This conclusion concerns every real linear combination in the entire complementary subspace. The uniform simplicity and nonaliasing check justifies the eigenvector formulation. Write  $\sigma_-$  for the largest slow spectral modulus and  $\sigma_+$  for the smallest fast modulus. Then

$$\sigma_- < \sigma_+, \quad \sigma_+ > 1. \quad (9.18)$$

Some slow eigenvalues may also be expanding or lie on the unit circle. No absence of imaginary-axis roots in all orders is assumed.

Working in the centered coordinates  $Z = Y - \mathbf{1}$ , the time- $\delta$  map has derivative  $T_r$  at zero. Lanford’s generalized stable-manifold theorem [10, Theorem 7.3.1(1),(3)] gives a local  $C^1$  invariant manifold tangent to  $E_{\text{slow}}$  that contains every sufficiently small forward orbit staying near zero. The hypotheses are exactly the strict gap (9.18),  $\sigma_+ > 1$ , a  $C^1$  map, and a smooth cutoff available in finite dimension. In particular, the tail of any convergent orbit lies on this manifold.

A sufficiently small punctured part of that manifold is disjoint from  $\mathcal{C}_3$ . Otherwise, normalizing a sequence of nonzero intersection points approaching zero would give a unit subsequential limit in both the tangent space  $E_{\text{slow}}$  and the closed cone, contradicting (9.17). The nonzero orbit  $Z$

stays in  $\mathcal{C}_3$  by (9.16). Thus  $Z$  cannot converge to 0, equivalently  $Y$  cannot converge to  $\mathbf{1}$ . This contradicts (9.1) and proves Theorem 9.1 for every  $r \geq 30$ .

The same argument excludes the pure real blowup law  $H_r(x) \sim A_r(\rho_r - x)^{-r}$  at the actual finite endpoint: the monotone-density argument used in (8.5) would force convergence of the entire normalized jet. The proof does not identify its alternative behavior.

## 9.6 Independent finite spectral checks

The essential high-order proof uses the elementary phase estimates above. As an independent check, the archive also includes exact Routh calculations. At  $r = 30$ , the degree-29 quotient  $q = \chi_{30}/(\mu - 1)$  has first-column signs

$$\underbrace{+, +, \dots, +}_{28 \text{ entries}}, \quad -, \quad +,$$

with no zero first-column entry or all-zero row. Thus  $q$  has exactly two open-right-half-plane roots. At  $r = 29$  there are none. Polynomial greatest-common-divisor checks exclude repeated and imaginary-axis roots in these two finite cases; independent exact complex root counts agree. The order-30 compound graph replay visits all  $\binom{30}{3} = 4060$  vertices in each direction and checks all 11340 transitions. These checks support the specific spectral crossing; they do not prove that a counting equivalent holds at every lower order or that  $m = 29$  is the first false case.

[write] **Added 5 October 2026, batch 103.** At  $r = 30$  the complete spectral count is now a proved statement: Part IV’s Proposition 37.1 and Corollary 38.1 (a modulus curve with monotone phase and seven exact inequalities).

## 9.7 What is prior and what the obstruction adds

The polynomial (9.5) is already the characteristic polynomial in Chauvin, Gardy, Pouyanne, and Ton-That’s *B-urns* [6, Section 4.1]. Their well-known urn transition at parameter 60 concerns a real-part threshold  $1/2$ . The present blowup issue concerns crossing real part 0, with the ODE-order parameter  $r = 30$ . These are different thresholds for the same polynomial family; no affine identification with the parameter-60 polynomial is asserted.

Kozlov [9, Lemma 5.3] already proves extra unstable eigenvalues at sufficiently large quadratic order. Astashova’s later work studies atypicality of power-law Cauchy data [3]. Spectral instability or a measure-zero statement alone cannot exclude one specified initial vector. The additional ingredient here is the initial jet’s low-sign-variation constraint, its forward invariance, and the separation of the whole stable subspace from that cone. This supplies the orbit-specific contradiction for each of the counting sequences with  $r \geq 30$ .

# 10 Reproducibility and numerical evidence

## 10.1 Exact checks and independent calculations

The accompanying archive contains the article, its LaTeX and bibliography, original research code and data, adapted replay paths, and build instructions. It omits third-party papers and raw working material. The checks have different evidential roles:

1. **Exact symbolic checks.** Rational and algebraic arithmetic verifies the Lyapunov identity modulo  $p_*^3 = 9/20$ , the indicial factorization, conjugation of  $a_{kl}$ ,  $a_{11} = -1/7$ , and the degree-four coefficient recurrence. A second implementation checks the Jacobian characteristic polynomial and  $37/12$  in (7.7).

2. **Exact integer checks.** One implementation computes (1.4) through  $n = 202$  and agrees with a rational ordinary-EGF recurrence through  $n = 72$ . An independent implementation uses multiplicatively updated binomial coefficients through  $n = 600$ , agreeing with a separate rational recurrence through  $n = 80$ . The first 29 terms agree with the inspected OEIS record.
3. **Fourth-order exact checks.** A separate script verifies the normalized Jacobian spectrum, both indicial factorizations, the conserved energy and its value  $1/6$ , the exact real-mode coefficient  $D$ , the alternating real-mode vector, and two independent exact count recurrences through  $n = 100$ .
4. **High-order exact certificates.** Two independent rational implementations check all eight finite phase/modulus pairs for  $30 \leq r \leq 37$ . Exact rational checks verify the infinite-range margins in (9.10)–(9.11). Supplemental Routh tables, polynomial greatest-common-divisor checks, exact complex root counts, and the order-30 compound graph are replayed separately.
5. **Exploratory numerical checks.** High-precision coefficient calculations fit  $\rho$ ,  $\operatorname{Re} C$ , and  $\operatorname{Im} C$  at three training indices. Disjoint testing indices and changed training sets check consistency. A separate DOP853 integration of the desingularized system estimates  $\rho$ . Neither procedure gives interval certificates.

The exact file of  $h_0, \dots, h_{600}$ , one index and integer per line, has SHA-256

aa02b70dab0ee9522c1a5dea137dd6cd2158b0401671e85da9db160d774bf635.

The replay checks this file and the archived exact JSON outputs. Exact arithmetic validates the stated identities and recurrence outputs; it does not independently formalize every analytic argument in the article.

## 10.2 Fitted constants and held-out indices

The numerical illustration in this subsection concerns the third-order sequence A333497 only. The arithmetic precisions are 120 and 150 decimal digits. Both runs retain grade four in the fit; their respective training sets are  $\{400, 550, 700\}$  and  $\{450, 650, 850\}$ . The displayed fitted constants are deliberately rounded:

$$\begin{aligned}\rho &\approx 3.77462757572068, \\ C &\approx -7.35007549061487 \times 10^{-6} - 5.54631650011815 \times 10^{-6}i, \\ K &\approx -15603.3656213 - 1986.37940722i.\end{aligned}\tag{10.1}$$

They are exploratory approximations, not certified digits. Agreement between high-precision fits is eventually limited by omitted grades and finite- $n$  effects, not solely by arithmetic precision. Printing all working digits would not turn them into proven constants. The proof of  $C \neq 0$  is (3.13), not a numerical fit.

The independent floating-point ODE calculation gives  $\rho \approx 3.774627575720685$ , consistent with (10.1). Table 1 reports errors using the 150-digit fit and independently computed exact integers; all listed indices are outside that fit’s training set.

Finite- $n$  error is not necessarily monotone in truncation degree, particularly at small indices. Near a zero of an oscillatory term, the error can also look unusually small. At very high grades, uncertainty in the fitted constants sets a floor. These effects are consistent with a fixed-grade theorem and should not be presented as failures or numerical proofs of its asymptotic remainder.

Table 1: Exploratory relative discrepancies at held-out indices.  $R_n$  is defined in (1.10);  $Q_1$  retains the exact first Gamma ratio. The normalization uses fitted constants.

| $n$ | $R_n - 1$                  | $R_n - Q_1(n)$             |
|-----|----------------------------|----------------------------|
| 80  | $-9.49027 \times 10^{-9}$  | $-6.24477 \times 10^{-13}$ |
| 120 | $-4.60494 \times 10^{-10}$ | $2.02329 \times 10^{-15}$  |
| 200 | $3.38059 \times 10^{-11}$  | $2.71211 \times 10^{-18}$  |
| 300 | $-6.92182 \times 10^{-13}$ | $-1.81857 \times 10^{-20}$ |
| 400 | $-3.79476 \times 10^{-13}$ | $2.41521 \times 10^{-22}$  |
| 500 | $-5.62532 \times 10^{-14}$ | $1.53402 \times 10^{-23}$  |
| 600 | $2.52383 \times 10^{-15}$  | $-1.82740 \times 10^{-24}$ |

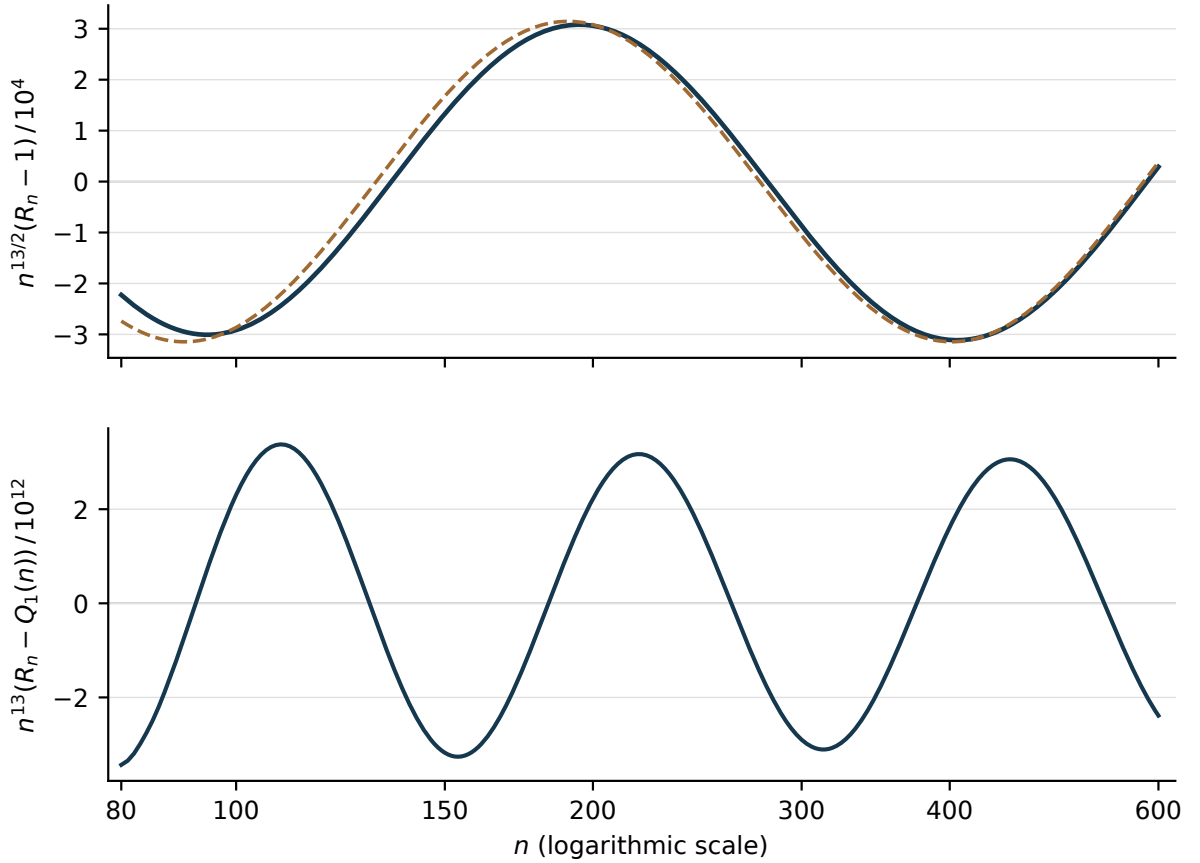


Figure 1: Exploratory illustration using exact counts for every integer  $80 \leq n \leq 600$  and the fitted constants in (10.1). The upper panel shows the logarithmically oscillating normalized discrepancy (solid) and its leading cosine (dashed). The lower panel scales the residual after the exact first Gamma-ratio correction by  $n^{13}$ . The figure illustrates the proven orders but does not certify the constants or a uniform numerical bound.

The inverse replay evaluates exact thresholds  $X = h_n$  and geometric midpoints  $X = \sqrt{h_n h_{n+1}}$  for  $n = 75, 100, 150, 200$ . Midpoint ceilings agree with the exact inverse in all four tests. At thresholds, tiny signed errors in the continuous root demonstrate why an unconditional ceiling claim would be unsafe. The code labels these computations as exploratory.

### 10.3 How to replay and rebuild

From the extracted archive root, the two principal commands are

```
bash replay.sh
bash build.sh
```

The first uses Python 3 with `mpmath`, `sympy`, and `scipy`; it runs in an isolated output directory, preserving the shipped data, and produces a comparison summary and environment information. The replay reruns both precision fits, the original exact and inverse checks, the fourth-order supplement, and the uniform high-order certificates. The report figure can be regenerated from the supplied exact counts and fitted data with `make_figure.py`; this additionally needs `matplotlib`.

The second uses pdfLaTeX and BibTeX. If the TeX installation has its package tree but lacks a prebuilt format or standard font map, the script creates the missing format and map inside the package-local `.build` directory. It does not change the system TeX installation. The build fixes the source date for stable PDF metadata. A manifest records the released file hashes; check it before replay with `sha256sum -c MANIFEST.sha256`. Reproducibility of exact data is byte-for-byte. Floating-point comparisons are reported separately, and cross-platform last digits are not claimed to be invariant.

[write] **Added 2 October 2026, batch 77.** This section describes the delivered archive. In the repository the files keep their bytes but not their places: `replay.sh`, `build.sh` and `make_figure.py` are in `code/`, the scripts of `code/extensions/` are `code/extensions-*.py`, the data subdirectories are flattened to `data/producer-*`, `data/independent-*` and `data/extensions-*` (the file of  $h_0, \dots, h_{600}$  above is `data/independent-exact_h_0_600.txt`, with the stated SHA-256), and the requirements files are in `data/`. The manifest `MANIFEST.sha256` and the delivered PDF were verified at placement and not shipped; both survive in the arrival commit. The scripts expect the delivered layout, so the report's `README.md` rebuilds it on a scratch copy before replaying. On Windows the producers write CRLF line endings, so the driver's byte comparison reports a mismatch although the content agrees; the intake's rerun (2 October 2026) matched all twelve exact artifacts after CRLF was stripped. The article's `bash build.sh` compiles `historic-trees.tex`; the report is built from `article.tex` instead (see the README).

## 11 Scope and further questions

### 11.1 What has been established

The  $m = 2$  and  $m = 3$  instances of the published coefficient conjecture hold, with nonzero oscillatory corrections, convergent local expansions, all fixed coefficient grades, and qualified eventual inverses. Every instance with  $m \geq 29$  is false for its specified all-one sequence. These results leave the intervening orders unresolved here, make no claim that 29 is the first failing parameter, and do not settle the separate even-order historic-tree Conjecture 2.

[write] **Added 5 October 2026, batch 103.** Parts II–IV prove these conclusions again for  $m = 2$ ,  $m = 3$  and  $m = 29$ . Part III adds that A333497 and A336009 are not P-recursive (Theorem 30.1). The orders  $4 \leq m \leq 28$  and Conjecture 2 remain unresolved after batch 103.

The journal's Table 1 is headed by  $\rho_m^{-1}$  but lists 3.7746 at  $m = 2$ . Under its definition of  $\rho_m$  as the positive singularity, the independent recurrence and ODE computations agree with that number

as the radius rather than its reciprocal. This appears to be a reciprocal-label error. The intrinsic definition (2.7) and analytic bounds (1.6) remove any dependence on that convention.

A bounded source check on 2 October 2026 considered the primary journal paper, its predecessor, accessible OEIS records, the cited Emden–Fowler literature, B-urns, and relevant public repository searches. It located no prior resolution of the specific counting statements presented here. That is a bounded negative finding, not an exhaustive priority certificate. The real-axis results, underlying spectral family, sign-regularity results, linearization, and transfer methods are explicitly credited.

## 11.2 Questions that follow naturally

1. **The actual high-order asymptotic.** What replaces the proposed factorial-pole equivalent for  $m \geq 29$ ? The nonconvergence result alone does not distinguish periodic profiles, more complicated bounded behavior, or other possibilities. Any claim about a periodic limit needs an additional global dynamical argument.

[write] **Added 5 October 2026, batch 103.** Still open. A pointer: the normalized system (9.4) is a monotone cyclic feedback system, so the Poincaré–Bendixson theorem of Mallet-Paret and Smith for such systems may restrict the possible limit sets once boundedness of the orbit is established (Remark 33.3).

2. **The intervening orders.** Which of the cases  $4 \leq m \leq 28$  satisfy the conjectured equivalent? The exact spectral check at ODE order 29 removes one particular local instability, but local spectral stability does not prove that the selected orbit enters the equilibrium’s basin. Is  $m = 29$  actually the first failure?

[write] **Added 5 October 2026, batch 103.** Still open; Part IV treats  $m = 29$  only and claims no first transition.

3. **Certified constants and thresholds.** Can validated integration, effective linearization bounds, and quantitative Delta domains give interval enclosures for the low-order radii and amplitudes, and explicit finite- $X$  inverse brackets? This would make the asymptotic inverse statements directly usable for certified integer recovery.

[write] **Added 5 October 2026, batch 103.** Still open (Parts II and III ask the same); the only certified enclosures are (1.6) and Part III’s (22.4).

4. **Global singularities and effective convergence.** Quantitative majorants could make the local bidisk or tridisk radius explicit. Locating more distant complex singularities could also identify contributions beyond every fixed algebraic order; local analyticity of diagonal terms is insufficient for such a claim.

[write] **Added 5 October 2026, batch 103.** Partly answered: the polydisk radii are explicit in the variables of the local series, 1183/960 at  $r = 3$  (Part II, Section 16) and 121/6720 at  $r = 4$  (Part III, Section 25); radii in  $t$  need a bound on  $|C|$ , which is not available. Part III’s Theorem 30.1 shows that continuation around  $\rho$  and  $\rho_4$  gives infinitely many independent branches; more distant singularities remain unlocated.

5. **Broader initial data and combinatorial meaning.** The cone argument suggests studying other positive jets with log-concave normalized coordinates. Separately, is there a direct interpretation of the low-order logarithmic oscillations in the evolution of B-tree histories, beyond the nonlinear ODE and its focus eigenvalues?

[write] **Added 5 October 2026, batch 103.** Still open; Parts II and III ask the same about other initial data, including the exceptional exact-pole family and the real-mode family.

## Part II

# Order five by an explicit psi series (source 02, bundle Report 113)

## 12 Reading Part II: provenance, credit and notation

*This section and every [write] note of Part II were added when bundle Report 113 joined the report (batch 103, 5 October 2026). Sections 13–20 print the manuscript in full.*

### 12.1 Provenance

Source 02 is Report 113 of the session bundle of the repository’s incoming research reports: the archive `A333497_B_Tree_Histories_All_Orders_and_Inversion_Source.zip` (477,780 bytes, wrapper directory `report113/`), which arrived in commit `60f54ea06` and was placed in this report by commit `9c995cefe` (batch 103) as the files with prefix `02-order5-`. Its main file `report113.tex` (788 lines; 13-page PDF) is titled *Order five B tree histories: Convergent psi series and coefficient asymptotics*, is dated 2 October 2026, and carries the status line “A self contained proof with a reproducibility companion”. It names no author or tool (the PDF Author field is empty), contains no “private review” wording, names no ProveIt commit and continues no repository path, so it has no pin. The manuscript, its PDF, its delivery `README.txt`, its `MANIFEST.json` and its 501-term table were not staged (the table is the first 501 lines of `data/independent-exact_h_0_600.txt`, byte for byte); all survive in the arrival commit. The README of this report lists the shipped files and the rerun routes.

*The archive name.* “All\_Orders\_and\_Inversion” does not mean all B-tree orders. The manuscript treats one B-tree order, five ( $m = 2$ , A333497); “all orders” means every fixed *coefficient* order of that one sequence (Theorem 13.1(iii)).

*What was changed in printing.* Every label carries the prefix `his:o5:` and every reference was updated. Citation keys were mapped to this report’s bibliography: `bw`→[5], `oeis`→[11], `fs`→[7]; the source’s reference `fo` (P. Flajolet and A. Odlyzko, *Singularity analysis of generating functions*, SIAM J. Discrete Math. **3** (1990), 216–240), which is not in this report’s bibliography, is printed as “[FO]”. The macros `\Rea`, `\Ima`, `\dd` are rendered by this report’s definitions (`\dd` adds a thin space). Section labels were added for the cross-references below, and the abstract is printed in Section 12.4. No statement, proof or number of the manuscript was altered; text added in writing is marked [write].

### 12.2 Credit: what Part II proves again and what it adds

Source 02 was written without sight of Part I, whose manuscript came from another delivery (batch 77, arrival `096ee7b87`, 2 October 2026; placed in `34f1acd4b`). Both manuscripts are dated 2 October 2026. In this repository Part I holds the first proof of the order-five case; source 02 is an independent second proof by different devices, and its abstract’s “We prove the order-five case” is to be read in that sense (dated note in Section 12.4).

| Source 02                                                                           | Relation to Part I                                                                                                                |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Theorem 13.1(i)–(iii) and (13.7)                                                    | second proof of Theorem 1.1, the bounds (1.6) and Corollary 1.2; the same $a_{k\ell}$ ( $a_{20}, a_{11} = -1/7$ agree with (4.3)) |
| Lemma 14.1                                                                          | the Lyapunov function of Lemma 2.1 after rescaling ( $V_{\text{here}} = V/p_*^2$ )                                                |
| Section 16: Catalan majorant, explicit invariant surface, transverse coordinate $p$ | second route to Section 3, replacing holomorphic Poincaré linearization [8] by an explicit convergent surface                     |
| The radius 1183/960 of the bidisk in $(q_1, q_2)$                                   | <b>new</b> ; partly answers Question 4 of Section 11 (dated note there)                                                           |
| $c \neq 0$ from the exact pole (Section 16)                                         | second route to (3.13)                                                                                                            |
| Analytic-sector cancellation and (17.2)                                             | the case $d = 2$ of (1.8); diagonal exclusion as in (4.4)                                                                         |
| Bernoulli-polynomial algorithm (17.3)–(17.4), $d_2(b)$                              | <b>new</b> explicit form; $d_1$ is (6.3)                                                                                          |
| Section 18: inverses, Lambert corrections, all-order recursion, rounding window     | instances of Theorem 7.1 and of the transseries volume (see the note at the end of Section 7); no novelty claimed                 |
| Table 1 remark (Section 14)                                                         | the same observation as Section 11                                                                                                |

The independent agreements are worth recording: source 02's fitted  $\rho$  (Section 19) agrees with all 38 printed digits of Part I's 120- and 150-digit fits (`data/producer-numeric_{120,150}dps.json`), and its fitted  $C$  equals Part I's fitted  $C\rho^\lambda$  to 28 digits (intake computation). These are agreements of two uncertified fits, not certified digits.

## 12.3 Notation of Part II

Part II keeps source 02's letters. The table gives each letter's meaning here, its counterpart in Part I, and the reading to avoid.

| Symbol                                              | In Part II (source 02)                                                                                  | Part I; false reading to avoid                                                                                       |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| $\alpha, \beta_{k\ell}$                             | $(13 + i\sqrt{71})/2$ ; $k\alpha + \ell\bar{\alpha}$                                                    | Part I's $\lambda$ and $\nu_{kl}$ ; <i>not</i> Part I's real weight $\alpha = 13/2$                                  |
| $C, c$                                              | $C$ multiplies $(1 - z/\rho)^\alpha$ ; $c$ multiplies $t^\alpha$                                        | $c$ is Part I's $C$ ; $C_{\text{here}} = C_{\text{Part I}}\rho^\lambda$                                              |
| $U, a_{k\ell}$                                      | $U = \mathcal{F} - 1$ , so $a_{00} = 0$                                                                 | Part I's $\mathcal{F}$ has $a_{00} = 1$ ; otherwise the same $a_{k\ell}$                                             |
| $G_n(b)$                                            | $\Gamma(n+3-b)/(\Gamma(3-b)\Gamma(n+3))$                                                                | <i>not</i> Part I's $G = H_4$ or $G_r$                                                                               |
| $Q(b), D(b)$                                        | $(3-b)(4-b)(5-b)$ ; $Q(b) - 120$                                                                        | $D$ is Part I's $D(\nu)$ ; <i>not</i> $Q_d$ or $Q(v)$                                                                |
| $u, v, s$ ; $X, Y, \tau$                            | $u = y'y^{-4/3}$ , $v = y''y^{-5/3}$ , $ds/dx = y^{1/3}$ ;<br>$X = u/u_*$ , $Y = v/v_*$ , $\tau = u_*s$ | Part I's $p, q, \tau$ are $u, v, s$ here; $v$ is <i>not</i> Part I's $H^{-1/3}$ ; $Y$ is <i>not</i> a normalized jet |
| $V$                                                 | Lyapunov function of Lemma 14.1                                                                         | Part I's $V$ divided by $p_*^2$                                                                                      |
| $t, \sigma, F, w$                                   | $t = \rho - z$ , $\sigma = -\log t$ , $F = t^3H/60$ ,<br>$w = F - 1$                                    | Part I's $s = \log(\rho/t) = \sigma + \log \rho$ ; $w$ is <i>not</i> Part I's $w$ of (3.6)                           |
| $E, \Theta, p, e_u$                                 | Euler operator; invariant surface; transverse coordinate; unstable eigenvector                          | $E$ is Part I's $\mathcal{L}$ , <i>not</i> $E = e^{-S}$ ; $p$ is <i>not</i> $H'H^{-4/3}$                             |
| $M, A_N, A(r); A$                                   | Catalan constant 120/1183, majorant; modulus in Section 17                                              | <i>not</i> $A, A_r$ of Part I                                                                                        |
| $r$                                                 | a radius (majorant variable; $0 < r < \rho$ )                                                           | <i>not</i> the ODE order $r$ of Part I                                                                               |
| $J$                                                 | truncation degree                                                                                       | Part I's grade $d$                                                                                                   |
| $\ell_r, d_j, B_j$                                  | gamma-ratio algorithm; Bernoulli polynomials                                                            | $d_j$ is <i>not</i> a grade                                                                                          |
| $M(x), D_x, q, q_J, S_J, M_J$                       | smooth inverse model and its corrections                                                                | $M(x)$ is Part I's $\mathcal{B}(x)$ ; $S_J$ is Part I's $Q_J$                                                        |
| $L, w, N_0, d, a; \delta_j, \mathcal{F}(\delta, u)$ | Lambert data and formal residual                                                                        | <i>not</i> $w = F - 1$ , the grade $d$ , Part I's $a$ or $\mathcal{F}$                                               |
| $T(X), K_0, q_\pm$                                  | threshold and its envelopes                                                                             | $T(X)$ equals Part I's $N(X)$ for large $X$ ; $K_0$ is <i>not</i> Part I's $K$                                       |

## 12.4 The source's abstract

The exponential generating function of OEIS A333497 satisfies  $H''' = H^2$  with  $H(0) = H'(0) = H''(0) = 1$ . We prove the order-five case,  $m = 2$ , of the asymptotic conjecture of Burghart and Wagner:  $h_n \sim 30(n+2)!\rho^{-n-3}$ . The proof first establishes finite positive-real blow-up, then excludes all competing singularities on the convergence circle. A constructive analytic stable surface yields a convergent local complex-power expansion. Its first nonzero correction is log-periodic, with exponent  $\alpha = (13 + i\sqrt{71})/2$ . Finite sector truncations give coefficient expansions to every fixed order, including exact gamma-ratio constants and analytic-sector cancellations. We also give smooth inversion, controlled Lambert  $W$  approximations, and integer threshold bounds. The proof is independent of numerical fitting. Decimal estimates of the singularity and amplitude are exploratory, and no claim of a convergent large-index asymptotic expansion or of a solution for every  $m$  is made.

[write] **Added 5 October 2026, batch 103.** The order-five case was already proved in this repository by Part I (Theorem 1.1, Corollary 1.2); the theorem below is an independent second proof. What source 02 adds is listed in Section 12.2. Its non-claims (no  $m \geq 3$ , no convergent large- $n$  expansion, no summation of infinitely many sectors, no certified decimals, no quantified  $|C|$ , no priority) are kept with the text below.

## 13 The exact model and the result

### 13.1 Sequence and counting convention

Let  $h_0 = h_1 = h_2 = 1$  and define

$$h_{n+3} = \sum_{k=0}^n \binom{n}{k} h_k h_{n-k} \quad (n \geq 0). \quad (13.1)$$

Then

$$(h_n)_{n \geq 0} = 1, 1, 1, 1, 2, 4, 8, 18, 48, 144, 456, 1560, 5808, \dots,$$

which is the defining recurrence and offset-zero sequence A333497 in the OEIS [11]. Its exponential generating function is

$$H(z) = \sum_{n \geq 0} h_n \frac{z^n}{n!}, \quad H'''(z) = H(z)^2, \quad H(0) = H'(0) = H''(0) = 1. \quad (13.2)$$

Local analytic existence for a polynomial differential system supplies a positive Taylor radius, and coefficient comparison proves (13.1). The OEIS description is the equivalent statement that squaring the EGF shifts its sequence three places left.

Burghart and Wagner [5, Section 4, equation (4)] associate  $H^{(m+1)} = H^2$  with reduced  $(2m+1)$ -historic trees. The reduction removes the first  $m$  vertices. A reduced tree with  $n$  internal vertices therefore corresponds to a B-tree history of length  $n+m$ . Here  $m=2$ :  $h_n$  counts reduced 5-historic trees with  $n$  internal vertices, or histories of length  $n+2$ . It is not the number of insertion permutations; distinct permutations can produce the same history.

The paper's previously settled  $m=1$  case concerns A007558 and the second-order equation  $H'' = H^2$ , whose solution is expressed by an elliptic function. It is a different sequence. The third-order problem treated here is separate from that solved case.

[write] **Added 5 October 2026, batch 103.** The statement about the case  $m=1$  is source 02's reading of [5]; the intake did not check it (Section 20.1, item 6).

## 13.2 Main theorem

Set

$$\alpha = \frac{13 + i\sqrt{71}}{2}, \quad \beta_{k\ell} = k\alpha + \ell\bar{\alpha}, \quad G_n(b) = \frac{\Gamma(n+3-b)}{\Gamma(3-b)\Gamma(n+3)}. \quad (13.3)$$

For nonnegative integral  $b-3$ , the last expression is interpreted through its entire continuation in  $b$ , equivalently as a binomial coefficient; for all sufficiently large  $n$  it is zero.

**Theorem 13.1.** *There are a positive real number  $\rho$  and a nonzero complex number  $C$  with the following properties.*

- (i)  $\rho$  is the positive-real blow-up time, the Taylor radius of  $H$ , and its unique singularity on  $|z| = \rho$ . Moreover,

$$180^{1/4} \leq \rho \leq 60^{1/3}. \quad (13.4)$$

- (ii) In a slit neighborhood of  $\rho$ , on the branch real for  $z < \rho$ ,

$$H(z) = 60(\rho - z)^{-3} \left[ 1 + U \left( C(1 - z/\rho)^\alpha, \bar{C}(1 - z/\rho)^{\bar{\alpha}} \right) \right], \quad (13.5)$$

where  $U(q_1, q_2) = \sum_{k+\ell \geq 1} a_{k\ell} q_1^k q_2^\ell$  is a convergent bivariate power series with  $a_{10} = a_{01} = 1$ .

- (iii) For every fixed integer  $J \geq 0$ ,

$$\frac{h_n}{30\Gamma(n+3)\rho^{-n-3}} = 1 + 2 \sum_{1 \leq k+\ell \leq J} a_{k\ell} C^k \bar{C}^\ell G_n(\beta_{k\ell}) + O\left(n^{-13(J+1)/2}\right). \quad (13.6)$$

The constant in the remainder can depend on  $J$ . In particular,

$$h_n = 30(n+2)!\rho^{-n-3} \left[ 1 + 4 \operatorname{Re}\{CG_n(\alpha)\} + O(n^{-13}) \right]. \quad (13.7)$$

The leading equivalent in (13.7) is exactly the  $m = 2$  instance of Conjecture 1 of [5, p. 10], since  $(2m+1)!/(m!)^2 = 5!/(2!)^2 = 30$ . The number 60 is the local pole coefficient; the coefficient asymptotic contains 30 because  $[z^n](1-z)^{-3} = (n+1)(n+2)/2$ .

[write] **Added 5 October 2026, batch 103.** Theorem 13.1 is a second proof of Part I's Theorem 1.1 with Corollary 1.2. Part (i) is (1.6) with the identification of the radius and of the only singularity on the circle; (ii) is (1.7), with  $U = \mathcal{F} - 1$  and  $C = C_{\text{Part I}}\rho^\lambda$ ; (iii) is (1.8) with  $J = d$  (the sum in (13.6) includes the diagonal  $k = \ell$ , whose terms vanish for large  $n$  by the convention after (13.3)); and (13.7) is (1.11), since  $4 \operatorname{Re}\{CG_n(\alpha)\} = 2 \operatorname{Re}\{K\Gamma(n+3-\lambda)/\Gamma(n+3)\}$  with Part I's  $K$ .

The proof occupies Sections 14–17. Section 18 treats inversion, and Section 19 describes the accompanying checks. Nothing in the argument depends on a numerical approximation to  $\rho$  or  $C$ .

## 14 Positive real blow up

Let  $y(x) = H(x)$  be the real solution while it exists. Initially  $y, y', y''$  are positive; because  $y''' = y^2$ , all remain positive and increasing. Introduce

$$u = \frac{y'}{y^{4/3}}, \quad v = \frac{y''}{y^{5/3}}, \quad \frac{ds}{dx} = y^{1/3}.$$

Direct differentiation gives

$$\frac{du}{ds} = v - \frac{4}{3}u^2, \quad \frac{dv}{ds} = 1 - \frac{5}{3}uv. \quad (14.1)$$

The positive equilibrium is

$$u_* = (9/20)^{1/3}, \quad v_* = \frac{4}{3}u_*^2.$$

With  $X = u/u_*$ ,  $Y = v/v_*$  and  $\tau = u_*s$ ,

$$\dot{X} = \frac{4}{3}(Y - X^2), \quad \dot{Y} = \frac{5}{3}(1 - XY). \quad (14.2)$$

Here a dot means differentiation in  $\tau$ .

**Lemma 14.1.** *Every positive orbit of (14.2) is contained in a compact subset of the positive quadrant and tends to  $(1, 1)$  as  $\tau \rightarrow \infty$ .*

*Proof.* Use the explicit Lyapunov function

$$V(X, Y) = \frac{5}{6}(X - 1)^2 + \frac{4}{3}(Y - 1 - \log Y).$$

Its derivative is the exact identity

$$\dot{V} = -\frac{20}{9} \left[ \frac{(Y - 1)^2}{Y} + (X - 1)^2(X + 1) \right]. \quad (14.3)$$

A sublevel set bounds  $X$  above and bounds  $Y$  above and away from zero. Write  $Y \geq a > 0$ . At  $X = \sqrt{a}$  the first component of the vector field is nonnegative, so  $X \geq \min\{X(0), \sqrt{a}\} > 0$ . This proves compact containment. Ordinary continuation gives existence for every  $\tau \geq 0$ .

Integrating (14.3) shows that the two nonnegative summands on its right are integrable. Their derivatives are bounded on the compact trapping set, hence they are uniformly continuous. A uniformly continuous, nonnegative, integrable function tends to zero: otherwise intervals of a fixed positive width around a sequence of positive peaks would contradict integrability. Consequently  $X \rightarrow 1$  and  $Y \rightarrow 1$ .  $\square$

[write] **Added 5 October 2026, batch 103.** With  $p = p_*X$ ,  $q = q_*Y$  and  $q_* = \frac{4}{3}p_*^2$ , Part I's Lyapunov function (2.4) equals  $p_*^2V(X, Y)$ : Lemma 14.1 is Lemma 2.1 in rescaled coordinates. The intake re-verified the identity (14.3) symbolically.

The time change reconstructs  $y$  and  $x$  through

$$\frac{dy}{ds} = uy, \quad \frac{dx}{ds} = y^{-1/3}.$$

The positive upper and lower bounds on  $u$  make  $y$  grow exponentially in  $s$ . Thus  $x(s)$  tends to a finite number  $\rho$ , while  $y(s) \rightarrow \infty$ . The reconstructed solution solves the original initial-value problem, so uniqueness identifies  $\rho$  as its real maximal existence time. Furthermore,

$$\frac{d}{dx}(y^{-1/3}) = -\frac{1}{3}u \rightarrow -\frac{1}{3}u_*.$$

Integrating to  $\rho$  yields

$$y(x) \sim 60(\rho - x)^{-3}, \quad y'(x) \sim 180(\rho - x)^{-4}, \quad y''(x) \sim 720(\rho - x)^{-5}. \quad (14.4)$$

Indeed  $27/u_*^3 = 60$ ; the derivative equivalents then follow from the definitions of  $u, v$  and their limits.

## 14.1 Rigorous bounds and the source table convention

For  $R > 0$ ,  $p_R(x) = 60(R - x)^{-3}$  solves the same differential equation. Its initial data are

$$(p_R(0), p'_R(0), p''_R(0)) = (60/R^3, 180/R^4, 720/R^5).$$

The first-order system  $(y, y', y'')' = (y', y'', y^2)$  is cooperative in the positive octant. Componentwise initial inequalities therefore persist up to either blow-up time. For  $R = 180^{1/4}$  all three initial data of  $p_R$  are at least 1, so  $y \leq p_R$  and  $\rho \geq R$ . For  $R = 60^{1/3}$  all three are at most 1, so  $y \geq p_R$  and  $\rho \leq R$ . This proves (13.4). These inequalities are exact; their rough readable form is  $3.6628 < \rho < 3.9149$ .

Table 1 of [5, p. 10] labels its entries  $\rho_m^{-1}$  and lists 3.7746 for  $m = 2$ . The bounds just proved imply

$$60^{-1/3} \leq \rho^{-1} \leq 180^{-1/4} < 1,$$

so that entry cannot be  $\rho^{-1}$  in the paper's own singularity convention. The numerical recurrence calculation below places  $\rho$  near 3.7746275757, matching the displayed entry as  $\rho$ , not its reciprocal. This establishes a reciprocal-label inconsistency for the  $m = 2$  entry; it does not certify all the other entries of that table.

[write] **Added 5 October 2026, batch 103.** Part I makes the same observation (Section 11). The heading  $\rho_m^{-1}$  is the manuscripts' reading of the journal's table, not re-read at the intake.

## 15 The dominant singularity is unique

All  $h_n$  in (13.1) are strictly positive. Pringsheim's theorem says that a power series of nonnegative coefficients and finite radius has a singularity at its positive radius. The solution is analytic on the real interval before  $\rho$ , so its Taylor radius cannot be smaller than  $\rho$ . Blow-up prevents a larger radius. Thus the radius is exactly  $\rho$ .

We next prove continuation at every other point of the convergence circle. Fix  $0 < r < \rho$  and  $\theta \notin 2\pi\mathbb{Z}$ . For  $j = 0, 1, 2$ , every coefficient of  $H^{(j)}$  is positive. In particular its constant and linear terms have different arguments at  $re^{i\theta}$ . Strict triangle inequality gives

$$|H^{(j)}(re^{i\theta})| < H^{(j)}(r).$$

By continuity and the strict gaps, there is an  $\varepsilon \in (0, r)$  such that

$$|H^{(j)}(re^{i\theta})| \leq H^{(j)}(r - \varepsilon), \quad 0 \leq j \leq 2. \quad (15.1)$$

Let  $d_j = H^{(j)}(z_0)$  for  $z_0 = re^{i\theta}$ . The differential equation implies the local derivative recurrence

$$d_{n+3} = \sum_{k=0}^n \binom{n}{k} d_k d_{n-k}.$$

The corresponding real derivatives at  $r - \varepsilon$  are positive and obey the same recurrence. Induction from (15.1) bounds every  $|d_n|$  by that real derivative. Therefore the Taylor series at  $z_0$  converges at least in

$$|z - z_0| < \rho - r + \varepsilon. \quad (15.2)$$

The majorant is  $H(r - \varepsilon + w)$ ; its Taylor series has radius at least  $\rho - r + \varepsilon$ , because this disk is contained in  $|z| < \rho$  after translation. It follows that (15.2) continues  $H$  through  $\rho e^{i\theta}$ .

On compact arcs separated from  $\theta = 0$ , continuity makes the strict gaps uniform. A finite collection of these disks supplies continuation through an annulus near those arcs. The continuations

agree on overlaps: the disks have centers in the original disk, and any nonempty intersection contains a point on the segment between the centers, hence a point in the original disk. The identity theorem then glues the local solutions. Section 16 supplies the remaining punctured sector at  $\rho$ , with angular half-width greater than  $\pi/2$ . Shrink collar neighborhoods that meet this sector so their overlaps meet the original disk, and glue by the identity theorem. Compactness of the remaining boundary arc then gives a common outer radius increment. Together these give a  $\Delta$ -domain: for some  $\eta > 0$  and  $0 < \vartheta < \pi/2$ ,  $H$  is analytic in

$$\{z : |z| < \rho + \eta, z \neq \rho, |\arg(z - \rho)| > \vartheta\}. \quad (15.3)$$

A smaller domain may always be taken. This explicit continuation argument is what permits coefficient transfer; the real blow-up equivalent alone would not suffice.

## 16 A convergent local psi series

### 16.1 Rescaling and the characteristic exponents

Put  $t = \rho - z$ ,  $\sigma = -\log t$  and  $F = t^3 H/60$ . On the positive real approach, (14.4) gives  $F \rightarrow 1$ . Its first two derivatives are

$$F_\sigma = \frac{t^4 y' - 3t^3 y}{60}, \quad F_{\sigma\sigma} = \frac{t^5 y'' - 7t^4 y' + 9t^3 y}{60},$$

and both tend to zero. For  $F = 1 + w$  the exact equation is

$$w''' + 12w'' + 47w' - 60w = 60w^2, \quad (16.1)$$

where primes now mean  $\sigma$  derivatives. Its characteristic roots are  $1, -\alpha, -\bar{\alpha}$ . The positive root is the displacement of the blow-up location; the two nonreal roots are stable.

It is useful to put

$$Q(b) = (3-b)(4-b)(5-b), \quad D(b) = Q(b) - 120 = -(b+1)(b-\alpha)(b-\bar{\alpha}). \quad (16.2)$$

We construct the stable surface rather than treating a formal local expansion as if convergence were automatic.

### 16.2 Coefficient recursion and Catalan majorization

Let  $a_{00} = 0$ ,  $a_{10} = a_{01} = 1$ . For  $N = k + \ell \geq 2$ , define

$$a_{k\ell} = \frac{60}{D(\beta_{k\ell})} \sum_{p=0}^k \sum_{q=0}^\ell a_{pq} a_{k-p, \ell-q}. \quad (16.3)$$

Terms at the two endpoints vanish. Every term on the right thus has smaller total degree, making the recursion well-defined. There is no resonance: the real parts of  $\beta_{k\ell} + 1$ ,  $\beta_{k\ell} - \alpha$  and  $\beta_{k\ell} - \bar{\alpha}$  give

$$|D(\beta_{k\ell})| \geq \left(\frac{13N}{2} + 1\right) \left(\frac{13(N-1)}{2}\right)^2 \geq \frac{1183}{2}. \quad (16.4)$$

Write  $A_N = \sum_{k+\ell=N} |a_{k\ell}|$  and  $M = 120/1183$ . Then  $A_1 = 2$  and

$$A_N \leq M \sum_{i=1}^{N-1} A_i A_{N-i}.$$

The scalar majorant  $A(r) = 2r + MA(r)^2$  has solution

$$A(r) = \frac{1 - \sqrt{1 - 8Mr}}{2M} = \sum_{N \geq 1} 2^N M^{N-1} \text{Cat}_{N-1} r^N.$$

It is analytic for  $|r| < 1/(8M) = 1183/960$ . Consequently  $U(q_1, q_2) = \sum a_{k\ell} q_1^k q_2^\ell$  converges absolutely whenever  $\max(|q_1|, |q_2|) < 1183/960$ . Derivatives converge normally on smaller bidisks.

**[write] Added 5 October 2026, batch 103.** The explicit radius  $1183/960 = 1/(8M)$  is new to the repository and partly answers Part I's Question 4 (Section 11). It is a radius in the variables  $(q_1, q_2) = (ct^\alpha, \bar{c}t^{\bar{\alpha}})$ . Since  $|ct^\alpha| = |c||t|^{13/2}e^{-(\sqrt{71}/2)\arg t}$ , the series (16.8) converges whenever  $|\arg t| \leq \phi$  and  $|c||t|^{13/2}e^{(\sqrt{71}/2)\phi} < 1183/960$ ; a numerical radius in  $t$  therefore needs an upper bound for  $|c|$ , which neither source supplies.

Define  $E = \alpha q_1 \partial_{q_1} + \bar{\alpha} q_2 \partial_{q_2}$ . The coefficient recursion verifies, as an analytic identity,

$$[Q(E) - 120]U = 60U^2.$$

It follows that

$$\Theta(q_1, q_2) = (U, -EU, E^2U) \quad (16.5)$$

is an invariant analytic surface for the first-order version of (16.1), with intrinsic dynamics  $q_1' = -\alpha q_1$ ,  $q_2' = -\bar{\alpha} q_2$ . Its tangent vectors at zero are the independent stable eigenvectors  $(1, -\alpha, \alpha^2)$  and  $(1, -\bar{\alpha}, \bar{\alpha}^2)$ .

For reference, the first nonlinear coefficients are

$$a_{20} = \frac{359 + 57i\sqrt{71}}{17978}, \quad a_{11} = -\frac{1}{7}, \quad a_{02} = \overline{a_{20}}. \quad (16.6)$$

In general  $a_{\ell k} = \overline{a_{k\ell}}$  by the conjugate-symmetric recursion. All these constants lie in  $\mathbb{Q}(i\sqrt{71})$ .

### 16.3 The actual real orbit lies on this surface

Adjoin the unstable eigenvector  $e_u = (1, 1, 1)$ . The map

$$(q_1, q_2, p) \mapsto \Theta(q_1, q_2) + pe_u$$

has invertible derivative at zero, and hence is an analytic coordinate chart near zero. Invariance of  $p = 0$  forces the transverse equation to have the form

$$p' = pa(q_1, q_2, p), \quad a(0, 0, 0) = 1. \quad (16.7)$$

The factorization follows by analyticity from the vanishing of the transverse component on  $p = 0$ .

Choose the chart respecting conjugation. For a real state,  $q_2 = \bar{q}_1$  and  $p$  is real, and the value of  $a$  is real. The actual state  $(w, w', w'')$  tends to zero by (14.4), so it stays in this chart for every sufficiently large  $\sigma$ . On a sufficiently late tail  $a \geq 1/2$ . If  $p$  were nonzero there, (16.7) would make  $|p|$  grow at least exponentially at rate  $1/2$ , contradicting  $p \rightarrow 0$ . Therefore  $p = 0$  on the whole late tail. The exact stable dynamics gives a constant  $c \in \mathbb{C}$  such that

$$H(z) = 60t^{-3}\{1 + U(ct^\alpha, \bar{c}t^{\bar{\alpha}})\}. \quad (16.8)$$

The normalization in Theorem 13.1 is  $C = c\rho^\alpha$ , using the real logarithm of the positive  $\rho$ .

For each fixed  $\phi < \pi$ , both complex powers tend uniformly to zero as  $t \rightarrow 0$  in  $|\arg t| \leq \phi$ :  $|t^\alpha| = |t|^{13/2}e^{-(\sqrt{71}/2)\arg t}$ , and the conjugate power has the opposite angular factor. Thus (16.8)

converges normally in a sufficiently small slit sector and is holomorphic there on the fixed logarithm branch. It agrees with  $H$  along a real interval and hence by analytic continuation. This proves the local continuation claimed in (15.3). No formal-series convergence assumption is left over.

[write] **Added 5 October 2026, batch 103.** The first three subsections of Section 16 are a second route to Part I's Section 3: an explicit convergent invariant surface and a transverse coordinate replace holomorphic Poincaré linearization [8], the time inversion and the implicit-function step. The normalization  $C = c\rho^\alpha$  relates the two amplitudes:  $c$  is Part I's  $C$ .

## 16.4 The oscillatory amplitude cannot vanish

If  $c = 0$ , the stable coordinates vanish, and (16.8) is the exact pole  $H(z) = 60(\rho - z)^{-3}$ . The identity theorem extends this equality to the whole Taylor disk. The initial data  $H(0) = H'(0) = 1$  would then require  $\rho^3 = 60$  and  $\rho^4 = 180$ . Dividing gives  $\rho = 3$ , contradicting  $\rho^3 = 60$ . Hence  $c \neq 0$  and  $C \neq 0$ . This argument proves nonvanishing without a numerical fit or a numerically bounded amplitude.

[write] **Added 5 October 2026, batch 103.** A second route to (3.13): Part I deduces  $C \neq 0$  from nonzero orbit constants  $K_\pm$  and the linear coefficients (3.12); here the vanishing of  $c$  would make  $H$  an exact pole, contradicting the initial data.

## 17 Transfer and every fixed coefficient order

### 17.1 Exact transfer of a sector

For any complex  $b$ , generalized binomial coefficients give

$$[z^n](1 - z/\rho)^{b-3} = \rho^{-n} \frac{(3-b)_n}{n!} = \rho^{-n} \frac{\Gamma(n+3-b)}{\Gamma(3-b)\Gamma(n+1)}, \quad (17.1)$$

with the rising-factorial interpretation at exceptional integers. This is an exact coefficient identity, not an asymptotic approximation. Multiplication by  $60\rho^{-3}n!$  and division by  $30\Gamma(n+3)\rho^{-n-3}$  gives exactly  $2G_n(b)$ .

Truncate the convergent series  $U$  at total degree  $J$ . In each closed subsector of the  $\Delta$ -domain, its tail is  $O(|1 - z/\rho|^{13(J+1)/2})$ . After multiplication by the pole factor, the error in  $H$  is

$$O(|1 - z/\rho|^{13(J+1)/2-3}).$$

The big- $O$  transfer theorem for  $\Delta$ -analytic functions [FO, Theorem 1, pp. 220–221] (see also [7, Chapter VI, Theorem VI.3]) gives coefficient error  $O(\rho^{-n}n^{2-13(J+1)/2})$  in the ordinary coefficients of  $H$ . This big- $O$  theorem permits every real remainder exponent, including nonnegative integers; the integer exception in an individual-power asymptotic equivalent does not restrict this error estimate. Multiplying by  $n!$  and normalizing gives (13.6). This completes the proof of Theorem 13.1.

### 17.2 Analytic sectors cancel exactly

For  $k = \ell \geq 1$ ,  $\beta_{kk} = 13k$  is a positive integer and the corresponding local contribution is a multiple of  $(1 - z/\rho)^{13k-3}$ , a polynomial. Its coefficients vanish exactly when  $n > 13k - 3$ . Accordingly, in any fixed sector truncation all diagonal terms disappear for sufficiently large  $n$ . One must not evaluate a quotient of singular gamma factors separately; (17.1) or continuation in  $b$  specifies the answer unambiguously. Off-diagonal exponents have nonzero imaginary part, so their gamma ratios have no such qualification.

In particular, the degree-two expansion is

$$\frac{h_n}{30(n+2)!\rho^{-n-3}} = 1 + 4 \operatorname{Re}\{CG_n(\alpha) + a_{20}C^2G_n(2\alpha)\} + O(n^{-39/2}). \quad (17.2)$$

The missing  $a_{11}|C|^2$  term is a genuine analytic-sector cancellation, not a zero coefficient in the local solution. Nonlinear sectors do not produce logarithmic factors: the recursion (16.3) has no resonant denominator.

[write] **Added 5 October 2026, batch 103.** (17.2) is the case  $d = 2$  of (1.8); the cancellation of the diagonal is (4.4).

### 17.3 Expanding the gamma ratios

Let  $B_j(x)$  denote the Bernoulli polynomials with  $te^{xt}/(e^t - 1) = \sum_{j \geq 0} B_j(x)t^j/j!$ . Set

$$\ell_r(b) = \frac{(-1)^{r+1}}{r(r+1)} \{B_{r+1}(3-b) - B_{r+1}(3)\}, \quad r \geq 1, \quad (17.3)$$

and define  $d_0(b) = 1$  and

$$jd_j(b) = \sum_{r=1}^j r \ell_r(b) d_{j-r}(b). \quad (17.4)$$

For any fixed  $b$  that is not an exceptional integer, Stirling's expansion gives, for every fixed integer  $K \geq 1$ ,

$$G_n(b) = \frac{n^{-b}}{\Gamma(3-b)} \left\{ \sum_{j=0}^{K-1} \frac{d_j(b)}{n^j} + O(n^{-K}) \right\}. \quad (17.5)$$

The first two nonconstant polynomials are

$$d_1(b) = \frac{b(b-5)}{2}, \quad d_2(b) = \frac{b(b+1)(3b^2 - 29b + 74)}{24}.$$

Equations (16.3), (17.3) and (17.4) are an explicit algorithm for every fixed coefficient order.

[write] **Added 5 October 2026, batch 103.** The Bernoulli-polynomial recursion (17.3)–(17.4) is an explicit form not printed in Part I;  $d_1$  is the coefficient in (6.3). The intake checked  $d_2$  numerically: at  $b = 2.3 + 1.1i$  the residual of (17.5) with  $K = 3$ , multiplied by  $n^3$ , is 21.06, 21.10, 21.10 at  $n = 10^3, 10^4, 10^5$ .

More precisely, to obtain a relative error  $O(n^{-R})$  for any prescribed real  $R > 0$ , choose  $J$  with  $13(J+1)/2 \geq R$ . Keep the off-diagonal sectors of degree at most  $J$  with real exponent below  $R$ . In each such sector keep the integer corrections with  $\operatorname{Re} \beta_{k\ell} + j < R$ , expanding far enough that its omitted tail has order at most  $n^{-R}$ . Only finitely many sector and integer orders are involved. The procedure determines all constants exactly in terms of  $\rho$ ,  $C$ , gamma values, and the rational algebraic coefficients  $a_{k\ell}$ .

### 17.4 What log periodicity means here

Write  $C/\Gamma(3-\alpha) = Ae^{i\varphi}$ , where  $A > 0$  by nonvanishing of  $C$  and absence of zeros of the gamma function. Then

$$4 \operatorname{Re} \left\{ \frac{C}{\Gamma(3-\alpha)} n^{-\alpha} \right\} = 4An^{-13/2} \cos \left( \varphi - \frac{\sqrt{71}}{2} \log n \right).$$

Its log-period is  $4\pi/\sqrt{71}$ . Thus the envelope has order  $n^{-13/2}$ , although the oscillation has zeros and no pointwise lower bound of that order is asserted. Using just this leading oscillation in (13.7) changes its error to  $O(n^{-15/2})$ . The exact gamma ratio in (13.7) retains all integer corrections in that sector and is needed for the sharper  $O(n^{-13})$  statement.

The local bivariate expansion is convergent. The coefficient expansions just described are Poincaré expansions at arbitrary fixed finite order. Neither convergence of the resulting large- $n$  series nor summation of infinitely many transferred sectors at fixed  $n$  is claimed.

## 18 Inverting the growth law

[write] **Added 5 October 2026, batch 103.** The results of this section are instances of Part I's Theorem 7.1 and of the transseries volume named in the note at the end of Section 7; no novelty is claimed for them. (18.3) states for the smooth models  $M_J$  what Theorem 7.1 states for  $F_d$ ; the Lambert start (18.4) is `p0:thm:lambert-core` ( $a = b = 1$ , branch  $W_0$ ); the all-order coefficients  $\delta_j$  are a formal reversion of the kind treated by `p0:thm:lambert-centered`; and the window (18.9) has the two-ceiling form of (7.5) and of part (2) of `p0:thm:staircase`.

### 18.1 The smooth leading inverse

Define

$$M(x) = 30\Gamma(x+3)\rho^{-x-3}, \quad D_x = \frac{d}{dx} \log M(x) = \psi(x+3) - \log \rho,$$

where  $\psi = \Gamma'/\Gamma$ . Since  $D_x \sim \log x$ ,  $M$  is strictly increasing eventually. Let  $q(X)$  be the inverse on this increasing tail. The exact constant  $\rho$  is used in all inversion assertions. An exploratory numerical estimate in its place adds an error that these formulas do not bound.

The mean value theorem applied to (13.7) gives

$$q(h_n) = n + O\left(\frac{n^{-13/2}}{\log n}\right). \quad (18.1)$$

Keeping the first correction explicitly sharpens this to

$$q(h_n) = n + \frac{4 \operatorname{Re}\{CG_n(\alpha)\}}{D_n} + O\left(\frac{n^{-13}}{\log n}\right). \quad (18.2)$$

To justify it, take logarithms of (13.7); the squared correction is  $O(n^{-13})$ . Taylor expansion of  $\log M$  about  $n$ , with  $D_n \asymp \log n$  and  $(\log M)''(n) = O(1/n)$ , leaves a smaller remainder. For numerical inversion the smooth leading model can be solved directly with log-gamma and a monotone scalar method.

### 18.2 Corrected smooth inverses at every fixed order

For a fixed  $J \geq 1$ , let

$$S_J(x) = 1 + 2 \sum_{\substack{1 \leq k+\ell \leq J \\ k \neq \ell}} a_{k\ell} C^k \bar{C}^\ell \frac{\Gamma(x+3-\beta_{k\ell})}{\Gamma(3-\beta_{k\ell})\Gamma(x+3)}.$$

For large real  $x$ , conjugation makes  $S_J(x)$  real,  $S_J(x) = 1 + O(x^{-13/2}) > 0$ , and  $S'_J(x) = O(x^{-15/2})$ . Thus  $M_J(x) = M(x)S_J(x)$  is eventually increasing. Let  $q_J$  denote its large inverse. Equation (13.6) gives

$$q_J(h_n) = n + O\left(\frac{n^{-13(J+1)/2}}{\log n}\right). \quad (18.3)$$

These are inverses of explicitly defined smooth models, not a purported unique interpolation of the discrete sequence. They provide inversion to any prescribed fixed algebraic order after choosing  $J$  sufficiently large.

### 18.3 Explicit Lambert W approximations

Put

$$L = \log(X\rho/30), \quad w = W_0\left(\frac{L}{e\rho}\right), \quad N_0 = \frac{L}{w}, \quad d = 1 + w, \quad a = \frac{1}{2}\log(2\pi N_0). \quad (18.4)$$

All assertions concern  $X \rightarrow \infty$ , so  $L, w, N_0$  are positive. The principal real Lambert branch solves  $we^w = L/(e\rho)$ ; hence  $N_0 \log(N_0/(e\rho)) = L$  and  $d = \log(N_0/\rho)$ . Writing  $N = q(X) + 2$ , the exact inverse equation is

$$\log \Gamma(N + 1) - N \log \rho = L.$$

Stirling's formula gives

$$\log \Gamma(N + 1) - N \log \rho = N \log \frac{N}{e\rho} + \frac{1}{2} \log(2\pi N) + \frac{1}{12N} + O(N^{-3}). \quad (18.5)$$

Expansion around  $N_0$  yields

$$q(X) = N_0 - 2 - \frac{a}{d} + O\left(\frac{1}{N_0 d}\right), \quad (18.6)$$

$$q(X) = N_0 - 2 - \frac{a}{d} - \frac{a^2/(2d^2) - a/(2d) + 1/12}{N_0 d} + O\left(\frac{1}{N_0^2 d}\right). \quad (18.7)$$

Here  $a/d$  stays bounded. The uncorrected  $N_0 - 2$  alone has an order-one shift; it is not an error- $o(1)$  inverse. At  $X = h_n$ , the error in (18.1) can be added to the displayed Lambert errors and is smaller than either of them.

For completeness, the Lambert expansion also has an all-fixed-order construction. Put  $u = N_0^{-1}$  and seek  $\delta = N - N_0 = \sum_{j \geq 0} \delta_j u^j$  formally. Take  $\delta_0 = -a/d$ . Using the Bernoulli numbers  $B_{2k} = B_{2k}(0)$ , define the formal residual

$$\begin{aligned} \mathcal{F}(\delta, u) &= d\delta + a + \sum_{r \geq 2} \frac{(-1)^r \delta^r u^{r-1}}{r(r-1)} + \frac{1}{2} \sum_{r \geq 1} \frac{(-1)^{r+1} \delta^r u^r}{r} \\ &\quad + \sum_{k \geq 1} \frac{B_{2k}}{2k(2k-1)} u^{2k-1} (1 + u\delta)^{-(2k-1)}. \end{aligned}$$

For  $j \geq 1$ , let  $\delta_j$  be  $-1/d$  times the coefficient of  $u^j$  in  $\mathcal{F}(\sum_{r < j} \delta_r u^r, u)$ . Only finitely many terms are needed for each coefficient. This gives  $\delta_1 = -[\delta_0^2/2 + \delta_0/2 + 1/12]/d$ , agreeing with (18.7). At every fixed  $K \geq 0$ , Stirling's finite-order remainder and  $d \asymp \log N_0$  justify

$$q(X) = N_0 - 2 + \sum_{j=0}^K \frac{\delta_j}{N_0^j} + O\left(\frac{1}{N_0^{K+1}d}\right). \quad (18.8)$$

For example, the residual after the chosen cancellation is  $O(N_0^{-K-1})$ , and division by the derivative  $\asymp d$  gives the stated error. Boundedness of  $\delta_0$  and the recursive coefficients at each fixed order makes this estimate uniform as  $X \rightarrow \infty$ . This is again an asymptotic expansion, not a convergence assertion.

## 18.4 Integer thresholds and the rounding boundary

Let  $T(X) = \min\{n \geq 0 : h_n \geq X\}$ . The sequence is nondecreasing and strictly increasing eventually. This follows directly from (13.1): after the initial terms, increasing the binomial upper index and using the preceding monotonic inequalities compares  $h_{n+4}$  with  $h_{n+3}$ , with an additional positive endpoint. The asymptotic law also gives  $h_{n+1}/h_n \sim (n+3)/\rho$ .

A formula with an error  $o(1)$  need not determine an integer threshold by rounding when  $X$  lies very close to a jump. To state the exact limitation, choose  $K_0$  and  $n_0$  such that

$$|\log(h_n/M(n))| \leq K_0 n^{-13/2} \quad (n \geq n_0).$$

Their existence follows from (13.7); no numerical values for them are certified here. For sufficiently large  $X$ , define  $q_+(X)$  and  $q_-(X)$  by the increasing-tail equations

$$\log M(q_+) + K_0 q_+^{-13/2} = \log X, \quad \log M(q_-) - K_0 q_-^{-13/2} = \log X.$$

Then

$$\lceil q_+(X) \rceil \leq T(X) \leq \lceil q_-(X) \rceil, \quad q_-(X) - q_+(X) = O\left(\frac{q(X)^{-13/2}}{\log q(X)}\right). \quad (18.9)$$

Indeed every integer  $n < q_+$  has its upper envelope below  $X$ , while every integer  $n \geq q_-$  has its lower envelope at least  $X$ . Thus  $T(X) = q(X) + O(1)$ . The two ceilings agree except in a shrinking neighborhood of an integer boundary. Near a boundary, use exact recurrence values or certified envelopes; neither a fitted  $\rho$  nor an arbitrarily rounded Lambert truncation is a rigorous decision rule. The same envelope argument applies to each corrected model  $M_J$  with exponent  $13(J+1)/2$  and its own eventual constants.

## 19 Computation and reproducibility

### 19.1 Exact checks and their scope

The companion package contains this TeX source, the PDF, an exact checker using Python and SymPy, the first 501 recurrence terms, finite exact sector data, a deterministic build script, and a strict SHA-256 manifest. The checker uses rational arithmetic in  $\mathbb{Q}(i\sqrt{71})$  and explicit exceptions rather than removable Python assertions. It verifies:

- the recurrence and all 501 supplied, regenerated terms  $h_0, \dots, h_{500}$ , with the first 29 values independently retrieved from the OEIS displayed entry; no external 501-term b-file comparison is claimed;
- the Lyapunov identity after clearing its positive denominator, the characteristic factorization and stable eigenvector determinant;
- the exact degree-two coefficients, the finite invariant-surface recursion and conjugation symmetry, and Catalan majorant identities;

- the first gamma-ratio polynomial coefficients and the first Lambert correction by exact symbolic polynomial identities;
- the artifact inventory, file sizes and hashes under a closed manifest, with duplicate names, path traversal, symlinks, unknown keys and extra files rejected.

Finite computation does not prove the infinite recurrence construction, analytic continuation, or asymptotic transfer. Those are established in the text. The checks are independent tests of the algebra, implementation and delivered data, not a substitute for the analysis.

The validation workflow runs the checker both normally and with Python optimization enabled, exercises manifest and data mutations, rebuilds the PDF twice, and replays the archive after clean extraction. The archive does not contain downloaded papers or temporary rendered pages. Instructions and dependencies are in `README.txt`.

[write] **Added 5 October 2026, batch 103.** In the repository the companion's files keep their bytes but not their names: the checker and its scripts are `code/02-order5-checks-*.py` and `code/02-order5-{build,pack}.py`, its data `data/02-order5-checks-*.json`, and its `checks/README.txt` is `02-order5-checks-README.txt`. The TeX source, the PDF, `README.txt`, `MANIFEST.json` and `recurrence_terms_0_500.txt` were not shipped (the last is the first 501 lines of `data/independent-exact_h_0_600.txt`); the report's `README` gives the rerun routes.

## 19.2 Exploratory numerical estimates

The optional numerical script regenerates the terms from (13.1), evaluates a finite number of the sector corrections and fits  $\log \rho$ ,  $\operatorname{Re} C$ ,  $\operatorname{Im} C$  using three sequence values at a time. It uses 100-digit arithmetic, but arithmetic precision does not turn a finite-order fit into a certified enclosure. Varying the fitting points and truncation degree is a consistency check only. A high-index degree-four fit suggests

$$\begin{aligned}\rho &\approx 3.7746275757206818041751733737441633123, \\ C &\approx -0.051699371021608688299058887 + 0.002095821372019341459228944 i.\end{aligned}$$

Only the exact bounds (13.4) are certified by this report. The numerical values are not used in any proof or rounding guarantee. The included numerical output explicitly marks its results as exploratory and uncertified.

[write] **Added 5 October 2026, batch 103.** These fits agree with Part I's independent fits:  $\rho$  in all 38 printed digits, and  $C$  with Part I's fitted  $C\rho^\lambda$  to 28 digits. Two uncertified fits agreeing do not certify either; the values are exploratory.

## 20 Scope and further questions

The proof settles the specified third-order initial-value problem and therefore the  $m = 2$  instance of the cited conjecture. It does not prove Conjecture 1 for  $m \geq 3$ , the even-order B-tree conjecture, or a global priority claim over all existing literature. The source paper identifies the conjecture; the OEIS identifies the recurrence; neither alone establishes novelty of every argument in this report.

Several concrete continuations remain.

1. **Certified constants.** Combine interval integration of the real rescaled system with an explicit analytic tail bound to enclose  $\rho$  and  $C$ , then derive numerical error constants and usable threshold cutoffs. The present nonvanishing proof does not quantify  $|C|$ .

2. **Higher orders.** For  $H^{(m+1)} = H^2$ , determine when the real positive orbit approaches the pole equilibrium and whether all remaining modes have the stability and nonresonance properties needed here. A higher-dimensional system need not inherit the planar Lyapunov mechanism of (14.2).
3. **Other initial data.** The positive real dynamics and the analytic stable surface apply more broadly. Characterize the dependence of the movable singularity and stable amplitudes on positive initial conditions, including the exceptional exact-pole family.
4. **Global continuation.** Describe singularities beyond the first circle and the multivalued continuation generated by the complex powers. The local slit-sector construction does not classify the whole Riemann surface.
5. **Effective asymptotics.** Optimize finite truncations for numerical work and obtain certified uniform remainders. Convergence of the local psi series alone does not justify an infinite coefficient expansion at fixed index.

## 20.1 Further questions and research

*[write] Added 5 October 2026, batch 103, under the repository's standing rule that no unproved claim is dropped. Source 02 contains no claim found to be wrong.*

1. **Certified constants** (source 02, question 1, the same as Part I's Question 3). Interval enclosures of  $\rho$ ,  $C$  and a positive lower bound for  $|C|$ , then numerical remainder constants and threshold cutoffs. Open. Only (13.4) is certified; the fitted decimals of Section 19 are exploratory and are not to be quoted as values.
2. **Higher orders** (source 02, question 2, Part I's Question 2). For  $m \geq 29$  Part I's Theorem 9.1 answers the question negatively; the orders  $4 \leq m \leq 28$  remain open after batch 103.
3. **Other initial data** (source 02, question 3, Part I's Question 5), including the exceptional exact-pole family. Open.
4. **Global continuation** (source 02, question 4, Part I's Question 4). Re-scoped: Part III's Theorem 30.1 shows that continuation around  $\rho$  produces infinitely many linearly independent branches, so  $\rho$  is a branch point of infinite order and  $H$  is not D-finite. The location of further singularities and the sheet structure remain open.
5. **Effective asymptotics** (source 02, question 5). Optimal truncation and certified uniform remainders. Open. The bidisk radius 1183/960 of Section 16 is explicit in  $(q_1, q_2)$ ; converting it into a radius in  $t$  needs a bound on  $|c|$ , which is not available.
6. **External claim kept as source 02's.** Section 13 says that Burghart and Wagner settled the case  $m = 1$  (A007558,  $H'' = H^2$ ) by an elliptic function. The intake did not check this against the journal.
7. **The journal's Table 1.** Section 14 proves, from (13.4), that the  $m = 2$  entry 3.7746 cannot be  $\rho^{-1}$ ; that it is printed under the heading  $\rho_m^{-1}$  is source 02's reading of the journal (as in Part I, Section 11), not re-read at the intake.

## Part III

# Order seven, and non-D-finiteness of A333497 and A336009 (source 03, bundle Report 115)

## 21 Reading Part III: provenance, credit and notation

*This section and every [write] note of Part III were added when bundle Report 115 joined the report (batch 103, 5 October 2026). Sections 22–32 print the manuscript in full.*

### 21.1 The new theorem

Part III contains the one result of batch 103 that is new to the repository in substance: *the exponential generating functions of A333497 and A336009 are not D-finite, and both sequences are not P-recursive* (Theorem 30.1, printed in full in Section 30 with its complete proof). Neither Part I nor any other report of the repository says anything about D-finiteness of these functions. The proof uses only a convergent local expansion at the dominant singularity, the universal coefficient recurrence and a nonzero complex amplitude. For A333497 ( $m = 2$ ) these inputs are exactly Part I’s Theorem 1.1 (the expansion (1.7) with  $a_{10} = a_{01} = 1$ , the recurrence (4.1), and  $C \neq 0$  by (3.13)), proved again in Part II (Theorem 13.1); the source cites its companion, Report 113, which is Part II. For A336009 ( $m = 3$ ) they are proved in this Part, and also by Part I’s Theorem 8.1 (see the note after the theorem).

### 21.2 Provenance

Source 03 is Report 115 of the session bundle: the archive `A336009_B_Tree_Histories_Transseries_and_Non_D_Finiteness_Source.zip` (512,656 bytes, wrapper directory `report115/`), arrival commit `60f54ea06`, placed by `9c995cefe` (batch 103) as the files with prefix `03-order7-`. Its main file `report115.tex` (1100 lines; 17-page PDF) is titled *Order seven B tree histories: Convergent psi series and coefficient asymptotics*, dated 2 October 2026, with the status line “A proof with an exact and reproducible computational companion”. It names no author or tool (empty PDF Author), has no “private review” wording, names no ProveIt commit and continues no repository path: no pin. Not staged, and retrievable from the arrival commit: the manuscript and PDF, `README.txt`, `CHECKSUMS.sha256`, `checks/MANIFEST.json`, and the generic `checks/requirements.txt` (SymPy 1.14.0, mpmath 1.3.0).

*The archive name.* “Transseries” does not describe the content: the expansions are convergent psi series (power series in complex powers of  $t$ ), with no logarithmic or exponential transmonomials, and the manuscript does not use the word. The material therefore stays here rather than in the repository’s transseries volume.

*What was changed in printing.* Labels carry the prefix `his:o7::`; citation keys were mapped: `bw`→[5], `oeis`→[12], `ast`→[2], `fs`→[7], `fo`→“[FO]” (Flajolet–Odlyzko 1990, as in Part II), and `report113`→“Part II” (Report 113 is Part II of this report). Macros and section labels as in Part II. No statement, proof or number was altered.

### 21.3 Credit: what Part III proves again and what it adds

Source 03 was written without sight of Part I (same delivery series as source 02). Its abstract’s “We prove the order-seven case” is a second proof in this repository: Part I’s Theorem 8.1 came first

(dated note in Section 21.5).

| Source 03                                                                                                          | Relation to Part I                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 22.1, (22.7)                                                                                               | second proof of Theorem 8.1, (8.13), (8.14); $c_0$ is Part I's $D$ , and $-3008678400 = -4264 \cdot 840^2$ is the coefficient of Part I's energy computation |
| Bounds (22.4), $16800^{1/6} \leq \rho \leq 840^{1/4}$                                                              | <b>new</b> (Part I gives no bounds for $\rho_4$ ); also the case $d = 4$ of Proposition 35.1                                                                 |
| Derivative limits (23.4) by integrating the ODE                                                                    | second route to the monotone-density upgrade (8.5); the same input [2]                                                                                       |
| Section 24                                                                                                         | the argument of Section 8.5                                                                                                                                  |
| Section 25: explicit convergent surface, nonresonance bound $ D(\beta_\nu)  \geq 363/2$ , radius $121/6720$ in $q$ | second route to the linearization of Section 8; the explicit radius is <b>new</b> (partly answers Question 4, Section 11)                                    |
| Section 26, energy                                                                                                 | the computation of (8.9), by the weight argument $EJ = 12J$                                                                                                  |
| Section 27: $c \neq 0$ by a scalar majorant reaching $z = 0$ (28.4), $O(n^{-33/2})$                                | second route to Part I's backward-Metzler argument                                                                                                           |
| Section 29                                                                                                         | a real-part-cutoff instance of (8.13)                                                                                                                        |
| Theorem 30.1                                                                                                       | instances of Section 8.6 and of the transseries volume; no novelty claimed                                                                                   |
| Table 1 datum 5.1792 at $m = 3$ (Section 31)                                                                       | <b>new</b> (Section 21.1)                                                                                                                                    |
|                                                                                                                    | new datum; it rests on the source's reading of the journal's table, which the intake did not re-read                                                         |

The intake re-verified every link of the chain in Section 27:  $|c_0|840^3 = 0.03283 \dots < 1/30$ ,  $B(1/30) = 0.033341 \dots < 1/29$ ,  $B'(1/30) = 1.0005 \dots < 2$ , and  $4 + \frac{12}{30} \cdot \frac{2}{1-1/29} = \frac{169}{35}$  exactly.

## 21.4 Notation of Part III

| Symbol                             | In Part III (source 03)                                                                                                                    | Part I; false reading to avoid                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $H, h_n, \rho$                     | the A336009 function, its coefficients, its blowup point (Sections 22–31)                                                                  | Part I's $G, g_n, \rho_4$                                                                                                                                                     |
| $H_2, H_3; \rho_2, \rho_3$         | in Section 30, indexed by $m$ : $H_2$ is A333497, $H_3$ is A336009                                                                         | <i>not</i> Part I's $H_r$ , indexed by the ODE order $r = m + 1$ : $H_2$ here is Part I's $H = H_3$                                                                           |
| $\alpha, \beta_{k\ell m}$          | $(11 + i\sqrt{159})/2$ ; $12k + \alpha\ell + \bar{\alpha}m$                                                                                | Part I's $\lambda_4$ and $\nu_{mkl} = 12m + k\lambda_4 + l\bar{\lambda}_4$ : the real-mode index is first here, last there; $\alpha$ is not Part I's weight $\alpha_4 = 11/2$ |
| $c_0, A$                           | $c_0 = -1/18052070400$ ; $A = c_0\rho^{12}$                                                                                                | $c_0$ is Part I's $D$ ; $A$ is not $A_4$ or $A_r$                                                                                                                             |
| $C, c$                             | $C = c\rho^\alpha$ ; $c$ multiplies $t^\alpha$                                                                                             | $c$ is Part I's $C_4$                                                                                                                                                         |
| $U, a_{k\ell m}$                   | $U = \mathcal{F}_4 - 1$                                                                                                                    | Part I's coefficients $c_{mkl}$                                                                                                                                               |
| $Q(b), D(b)$                       | $(4 - b) \cdots (7 - b)$ ; $Q(b) - 1680$                                                                                                   | $D$ is Part I's $D_4$ ; $Q$ is not $Q_d^{(4)}$                                                                                                                                |
| $G_n(b)$                           | $\Gamma(n + 4 - b)/(\Gamma(4 - b)\Gamma(n + 4))$                                                                                           | <i>not</i> Part I's $G$                                                                                                                                                       |
| $I, J(q); J$                       | energy (Part I's $\mathcal{E}$ ) and its surface form; also a truncation degree                                                            | the two $J$ 's are unrelated                                                                                                                                                  |
| $B(r), M; B_J, M_J; B_{2r}$        | majorants with $M = 560/121$ (Section 25) and $M = 7/955$ (Section 27); model factor and model; Bernoulli numbers                          | <i>not</i> Part I's $B(u), \mathcal{B}_4$ or $M(s)$                                                                                                                           |
| $r; R_\pm$                         | a radius; comparison radii                                                                                                                 | <i>not</i> the ODE order $r$                                                                                                                                                  |
| $E, \Theta, p, a(q, p)$            | Euler operator, invariant surface, transverse coordinate and its rate                                                                      | as in Part II                                                                                                                                                                 |
| $\sigma, w$                        | $\sigma = -\log t$ , $w = F - 1$ ; in Section 30<br>$w = \log t$                                                                           | Part I's $s = \sigma + \log \rho$                                                                                                                                             |
| $K, K_\phi, s_J, s, P_j$           | envelope and sector constants; $11(J + 1)/2$ ; Lambert shift and coefficients                                                              | $K$ is <i>not</i> Part I's amplitude $K_4$                                                                                                                                    |
| $\kappa, p, V, v_k, T, P(X), d, L$ | Section 30: 60 or 840; pole order 3 or 4; pure-axis series; exponential sum; winding polynomial; order of a hypothetical ODE; its operator | $p$ is <i>not</i> $H'H^{-4/3}$ , $V$ <i>not</i> a Lyapunov function, $d$ <i>not</i> a grade                                                                                   |

## 21.5 The source's abstract

The exponential generating function of OEIS A336009 satisfies  $H^{(4)} = H^2$  with  $H^{(j)}(0) = 1$  for  $0 \leq j \leq 3$ . We prove the order-seven case,  $m = 3$ , of the asymptotic conjecture of Burghart and Wagner:  $h_n \sim 140(n + 3)!\rho^{-n-4}$ . Explicit comparison gives  $16800^{1/6} \leq \rho \leq 840^{1/4}$ . Astashova's published fourth-order theorem supplies the positive-real blow-up profile; the remaining complex-analytic argument is given here. A convergent three-variable singular expansion has relative exponents 12 and  $(11 \pm i\sqrt{159})/2$ . An energy integral fixes the real amplitude exactly, and a separate scalar-majorant argument proves that the complex amplitude is nonzero. We exclude competing singularities on the convergence circle and derive coefficient expansions to every fixed order, with their exact gamma factors and analytic-sector cancellations. Smooth inverse models, Lambert  $W$  corrections and integer threshold brackets follow. An infinite-dimensional monodromy argument also proves that the order-five and order-seven generating functions are not D-finite and their sequences are not P-recursive. Decimal fits are explicitly exploratory, and neither the general higher-order conjecture nor convergence of the large-index asymptotic expansions is asserted.

[write] **Added 5 October 2026, batch 103.** The order-seven case was already proved in this repository by Part I (Theorem 8.1); Theorem 22.1 is an independent second proof. New here are the radius bounds, the explicit tridisk radius, the scalar route to  $c \neq 0$  and, above all, Theorem 30.1. The order-five input to that theorem is Part I's Theorem 1.1 (and Part II). The non-claims (no general  $m$ ; Astashova's theorem a cited input; no convergent large- $n$  series; no rounding guarantee; no certified decimals;  $|C|$  not quantified; the 501-term fixture generated, not an OEIS table) are kept below.

## 22 The exact model and main result

### 22.1 Recurrence and counting convention

Let

$$h_0 = h_1 = h_2 = h_3 = 1, \quad h_{n+4} = \sum_{k=0}^n \binom{n}{k} h_k h_{n-k} \quad (n \geq 0), \quad (22.1)$$

and put

$$H(z) = \sum_{n \geq 0} h_n \frac{z^n}{n!}, \quad H^{(4)} = H^2, \quad H^{(j)}(0) = 1 \quad (0 \leq j \leq 3). \quad (22.2)$$

Local complex analytic existence for this polynomial differential system gives a positive Taylor radius. Coefficient comparison makes (22.1) and (22.2) equivalent. The first values are

$$1, 1, 1, 1, 1, 2, 4, 8, 16, 34, 82, 226, 678, 2108, 6892, \dots$$

This is OEIS A336009 [12]. The exact companion regenerates  $h_0, \dots, h_{500}$  and checks the 31 terms of the retrieved OEIS prefix. The longer recurrence computation is not described as a published 501-term OEIS table.

In Burghart and Wagner [5, Section 4, equation (4)], the reduced  $(2m+1)$ -historic-tree generating function satisfies  $H^{(m+1)} = H^2$ . Removing the initial  $m$  vertices identifies the reduced object with  $n$  internal vertices with a B-tree shape history of length  $n+m$ . Thus  $h_n$  here counts reduced 7-historic trees with  $n$  internal vertices, equivalently order-seven B-tree histories of length  $n+3$ . A history is a sequence of successive shapes; it is neither an insertion permutation nor merely a final shape. The interpretation uses the paper's bijection and initial-vertex removal, rather than just the OEIS description.

### 22.2 Main theorem

Write

$$\alpha = \frac{11 + i\sqrt{159}}{2}, \quad \beta_{k\ell m} = 12k + \alpha\ell + \bar{\alpha}m, \quad G_n(b) = \frac{\Gamma(n+4-b)}{\Gamma(4-b)\Gamma(n+4)}. \quad (22.3)$$

For a fixed real integer  $b \geq 4$ , the associated local power is a polynomial. Its contribution is interpreted as zero for all sufficiently large  $n$ , rather than evaluating a gamma function at its poles.

**Theorem 22.1.** *There are a positive real number  $\rho$  and a nonzero complex number  $C$  with the following properties.*

- (i) *The number  $\rho$  is the positive-real blow-up time, the Taylor radius of  $H$ , and the only singularity on  $|z| = \rho$ . Moreover,*

$$16800^{1/6} \leq \rho \leq 840^{1/4}. \quad (22.4)$$

- (ii) *Set  $c_0 = -1/18052070400$  and  $A = c_0\rho^{12}$ . There is a convergent three-variable power series*

$$U(q_0, q_1, q_2) = \sum_{k+\ell+m \geq 1} a_{k\ell m} q_0^k q_1^\ell q_2^m, \quad a_{100} = a_{010} = a_{001} = 1,$$

*for which, on the branch real at  $z < \rho$ ,*

$$H(z) = 840(\rho - z)^{-4} \left[ 1 + U \left( A(1 - z/\rho)^{12}, C(1 - z/\rho)^\alpha, \bar{C}(1 - z/\rho)^{\bar{\alpha}} \right) \right]. \quad (22.5)$$

*This identity holds in a slit neighborhood of  $\rho$ .*

(iii) For every fixed integer  $J \geq 0$ , with  $\nu = (k, \ell, m)$ ,

$$\frac{h_n}{140\Gamma(n+4)\rho^{-n-4}} = 1 + 6 \sum_{1 \leq |\nu| \leq J} a_\nu A^k C^\ell \overline{C}^m G_n(\beta_\nu) + O(n^{-11(J+1)/2}). \quad (22.6)$$

The constant in the remainder may depend on  $J$ . In particular,

$$h_n = 140(n+3)!\rho^{-n-4} \left[ 1 + 12 \operatorname{Re}\{CG_n(\alpha)\} + O(n^{-11}) \right]. \quad (22.7)$$

The leading equivalent is the  $m = 3$  case of Conjecture 1 in [5, p. 10], since  $7!/(3!)^2 = 140$ . The pole coefficient 840 and the coefficient multiplier 140 differ by  $\Gamma(4) = 6$ . The leading correction has a genuinely nonzero log-periodic envelope, but its real oscillation can approach zero. In particular, a uniform two-sided bound on the correction is not claimed.

The local expansion converges. The coefficient and inverse expansions below are asserted to every fixed finite order, not as convergent large- $n$  series. No fitted decimal value enters the proof. [write] **Added 5 October 2026, batch 103.** Theorem 22.1 is a second proof of Part I's Theorem 8.1: (i) adds the bounds (22.4); (ii) is (8.3) with  $U = \mathcal{F}_4 - 1$ ,  $A = D\rho^{12}$  and  $C = C_4\rho^{\lambda_4}$  (Part I's  $D$  and  $C_4$ ); (iii) is (8.13) with  $J = d$ ; and (22.7) is (8.14), since  $12 \operatorname{Re}\{CG_n(\alpha)\} = 2 \operatorname{Re}\{K_4\Gamma(n+4-\lambda_4)/\Gamma(n+4)\}$ .

## 23 Positive real blow up and its profile

### 23.1 Exact comparison bounds

For the real solution  $y(x) = H(x)$ , the first-order system is

$$(y, y', y'', y''')' = (y', y'', y''', y^2).$$

It is cooperative on the nonnegative orthant. All four coordinates start at one and remain positive and increasing on their maximal positive interval. Exact comparison solutions are

$$p_R(x) = 840(R-x)^{-4},$$

with initial coordinates

$$\left( \frac{840}{R^4}, \frac{3360}{R^5}, \frac{16800}{R^6}, \frac{100800}{R^7} \right). \quad (23.1)$$

Let  $R_- = 16800^{1/6}$  and  $R_+ = 840^{1/4}$ . At  $R_-$  these coordinates are  $(R_-^2/20, R_-/5, 1, 6/R_-)$ , all at least one because  $5 < R_- < 6$ . At  $R_+$  they are  $(1, 4/R_+, 20/R_+^2, 120/R_+^3)$ , all at most one. Cooperative comparison gives

$$p_{R_+}(x) \leq y(x) \leq p_{R_-}(x)$$

on the respective common intervals, including the corresponding bounds on the first three derivatives. If  $y$  ended before  $R_-$ , the upper comparison would keep all state coordinates bounded and permit continuation of the polynomial system. If it survived beyond  $R_+$ , the lower comparison would contradict finite values as  $x \uparrow R_+$ . Its finite maximal endpoint  $\rho$  therefore satisfies (22.4).

[write] **Added 5 October 2026, batch 103.** The bounds  $(22.4)$ ,  $5.0607 \dots \leq \rho_4 \leq 5.3835 \dots$ , are new to the repository: Part I gives none at order four. They are the case  $d = 4$  of Part IV's Proposition 35.1.

If  $y$  stayed bounded as  $x \uparrow \rho$ , then  $y^{(4)} = y^2$  would be bounded. Repeated integration would bound the other state coordinates and again allow continuation. Thus

$$y(x) \longrightarrow +\infty \quad (x \uparrow \rho). \quad (23.2)$$

## 23.2 The published real theorem and its hypotheses

The substantive real-profile input is Astashova's Theorem 3 [2, p. 5]. In the notation needed here, it concerns

$$y^{(r)} = p(x, y, y', \dots, y^{(r-1)})|y|^k \operatorname{sgn} y, \quad r \in \{3, 4\}, \quad k > 1.$$

The function  $p$  must be positive and continuous, Lipschitz in its state variables, and have a positive limit  $p_0$  in the regime  $x \uparrow x^*$  and all its state arguments tending to  $+\infty$ . For a positive solution with vertical asymptote  $x = x^*$ , the theorem then gives a power profile with exponent  $r/(k-1)$  and leading constant

$$\left( \frac{\gamma(\gamma+1) \cdots (\gamma+r-1)}{p_0} \right)^{1/(k-1)}, \quad \gamma = \frac{r}{k-1}.$$

Here  $r = 4$ ,  $k = 2$ , and  $p \equiv 1$ : positivity and continuity hold everywhere, its state Lipschitz constant is zero, and its limiting value is identically  $p_0 = 1$ . The solution is positive and has the finite vertical asymptote (23.2). Every hypothesis is therefore satisfied. The extra quantitative conditions labelled (A) and (B) elsewhere in the cited paper are not additional hypotheses of its Theorem 3; constant  $p$  would satisfy them anyway. The state-limit condition is a condition on the coefficient function, so there is no need to presume derivative asymptotics in order to apply the theorem here.

Consequently,

$$y(x) \sim 840(\rho - x)^{-4}. \quad (23.3)$$

We do not differentiate this equivalent. Instead, the ODE gives  $y^{(4)} \sim 705600(\rho - x)^{-8}$ . If  $f(x) \sim K(\rho - x)^{-p}$  with  $p > 1$ , elementary upper and lower integral comparisons give

$$\int_{x_0}^x f(u) \, du \sim \frac{K}{p-1} (\rho - x)^{-p+1}.$$

The divergent power integral makes the finite integration constant negligible. Three applications yield

$$y''' \sim 100800(\rho - x)^{-7}, \quad y'' \sim 16800(\rho - x)^{-6}, \quad y' \sim 3360(\rho - x)^{-5}. \quad (23.4)$$

Thus every normalized state coordinate needed below has a proved limit. Astashova's result supplies this real input only; it does not supply our complex continuation or coefficient transfer.

[write] **Added 5 October 2026, batch 103.** A second route to Part I's monotone-density upgrade (8.5): here the top derivative is known from the equation and is integrated down, as in Part IV's (35.4). Both routes use the same real input [2].

## 24 The convergence circle and other boundary points

All  $h_n$  are strictly positive by (22.1). By Pringsheim's theorem, a finite Taylor radius is a positive-real singularity. Real analytic existence before  $\rho$  and blow-up at  $\rho$  imply that the Taylor radius is exactly  $\rho$ .

Fix  $0 < r < \rho$  and  $\theta \not\equiv 0 \pmod{2\pi}$ . Each of  $H, H', H'', H'''$  has positive constant and linear Taylor coefficients and nonnegative remaining coefficients. The constant and linear terms alone force strict triangle inequality at  $re^{i\theta}$ , so

$$|H^{(j)}(re^{i\theta})| < H^{(j)}(r) \quad (0 \leq j \leq 3).$$

Choose a common  $\varepsilon \in (0, r)$  small enough that

$$|H^{(j)}(re^{i\theta})| \leq H^{(j)}(r - \varepsilon) \quad (0 \leq j \leq 3). \quad (24.1)$$

For ordinary Taylor coefficients  $a_n$  at any regular center, (22.2) gives

$$a_{n+4} = \frac{\sum_{k=0}^n a_k a_{n-k}}{(n+1)(n+2)(n+3)(n+4)}. \quad (24.2)$$

This recurrence preserves absolute coefficientwise majorization. The Taylor expansion at  $z_0 = re^{i\theta}$  is therefore dominated by that of  $H(r - \varepsilon + w)$ . The latter has radius exactly  $\rho - r + \varepsilon$ : the original disk gives analyticity at least that far, and the positive-axis endpoint is singular. The resulting disk centered at  $z_0$  contains  $\rho e^{i\theta}$  in its interior. This proves analytic continuation across every other point of  $|z| = \rho$ , rather than just boundedness on a ray.

These continuations glue. Intersecting comparison disks have a connected intersection, and a point of their intersection can be chosen on the segment joining their centers, which lies in the original disk. Their two analytic functions agree there, hence throughout the intersection. A compactness argument supplies an outer collar away from a small neighborhood of  $\rho$ . A slit sector at  $\rho$ , constructed next, will complete the domain needed for coefficient extraction.

## 25 A convergent three variable stable surface

### 25.1 Normalization and characteristic exponents

Put

$$t = \rho - z, \quad \sigma = -\log t, \quad F = \frac{t^4 H}{840}, \quad w = F - 1.$$

A dot denotes differentiation in  $\sigma$  in this section. Since  $d/dz = t^{-1} d/d\sigma$ , the ODE becomes

$$(\partial_\sigma + 4)(\partial_\sigma + 5)(\partial_\sigma + 6)(\partial_\sigma + 7)F = 840F^2,$$

or

$$\ddot{w} + 22\dot{w} + 179w + 638\dot{w} - 840w = 840w^2. \quad (25.1)$$

Along the real solution, (23.3) gives  $w \rightarrow 0$ . To verify convergence of the entire first-order state, the quantities  $840\dot{F}$ ,  $840\ddot{F}$ , and  $840\ddot{\ddot{F}}$  equal respectively

$$\begin{aligned} t^5 y' - 4t^4 y, \\ t^6 y'' - 9t^5 y' + 16t^4 y, \\ t^7 y''' - 15t^6 y'' + 61t^5 y' - 64t^4 y. \end{aligned}$$

Substitution of (23.3)–(23.4) makes all three limits zero. Hence  $(w, \dot{w}, \ddot{w}, \ddot{\ddot{w}}) \rightarrow 0$ .

Define

$$Q(b) = (4-b)(5-b)(6-b)(7-b), \quad D(b) = Q(b) - 1680 = (b+1)(b-12)(b^2 - 11b + 70). \quad (25.2)$$

The four state eigenvalues are  $1, -12, -\alpha, -\bar{\alpha}$ . The positive eigenvalue is the unstable direction associated with moving the pole; the other three determine the expansion below.

### 25.2 Coefficient construction and convergence

For  $\nu = (k, \ell, m) \in \mathbb{N}_0^3$  let  $|\nu| = k + \ell + m$  and  $\beta_\nu = 12k + \alpha\ell + \bar{\alpha}m$ . Set  $a_0 = 0$  and  $a_{100} = a_{010} = a_{001} = 1$ . For  $|\nu| \geq 2$  define

$$a_\nu = \frac{840}{D(\beta_\nu)} \sum_{\mu+\eta=\nu} a_\mu a_\eta. \quad (25.3)$$

The endpoints with a zero multi-index contribute nothing. We verify both nonresonance and convergence, rather than treating this as only a formal recursion.

For  $N = |\nu| \geq 2$  we have  $\operatorname{Re} \beta_\nu \geq 11N/2$ , and thus

$$|\beta_\nu + 1| \geq \frac{11N}{2} + 1, \quad |\beta_\nu - \alpha|, |\beta_\nu - \bar{\alpha}| \geq \frac{11(N-1)}{2}.$$

The equation  $\operatorname{Re} \beta_\nu = 12$  would imply  $24k + 11(\ell + m) = 24$ , whose only nonnegative integral solution is  $(k, \ell, m) = (1, 0, 0)$  of degree one. Since these real parts are half-integers,  $|\beta_\nu - 12| \geq 1/2$  for  $N \geq 2$ . Consequently

$$|D(\beta_\nu)| \geq \left( \frac{11N}{2} + 1 \right) \frac{1}{2} \left( \frac{11(N-1)}{2} \right)^2 \geq \frac{363}{2}. \quad (25.4)$$

Every nonlinear denominator is nonzero and uniformly bounded away from zero.

If  $A_N = \sum_{|\nu|=N} |a_\nu|$ , then  $A_1 = 3$  and

$$A_N \leq M \sum_{i=1}^{N-1} A_i A_{N-i}, \quad M = \frac{560}{121}.$$

The scalar majorant  $B(r) = 3r + MB(r)^2$  is analytic for  $|r| < 1/(12M)$ . Therefore

$$U(q) = \sum_{|\nu| \geq 1} a_\nu q^\nu \quad (25.5)$$

converges absolutely in a genuine tridisk; all its differentiated series converge on smaller tridisks. The deliberately conservative bound (25.4) suffices for this purpose.

**[write] Added 5 October 2026, batch 103.** Explicitly,  $B$  is analytic for  $|r| < 1/(12M) = 121/6720$ , so  $U$  converges absolutely on the tridisk  $\max_j |q_j| < 121/6720$ . This radius, like the nonresonance bound (25.4), is new to the repository (Part I proves nonresonance only) and partly answers Part I's Question 4. In  $t$  it needs a bound on  $|c|$ ; the real-mode variable  $q_0 = c_0 t^{12}$  is explicit.

### 25.3 Why the actual solution belongs to the surface

Let

$$E = 12q_0 \partial_{q_0} + \alpha q_1 \partial_{q_1} + \bar{\alpha} q_2 \partial_{q_2}.$$

The convergent recursion verifies  $[Q(E) - 1680]U = 840U^2$ . It follows that

$$\Theta(q) = (U, -EU, E^2U, -E^3U)$$

is an analytic invariant surface of (25.1), with exact restricted dynamics

$$\dot{q}_0 = -12q_0, \quad \dot{q}_1 = -\alpha q_1, \quad \dot{q}_2 = -\bar{\alpha} q_2. \quad (25.6)$$

Its tangent vectors are the three distinct stable eigenvectors  $(1, -r, r^2, -r^3)$ , for  $r = 12, \alpha, \bar{\alpha}$ . Adding the unstable eigenvector  $e_u = (1, 1, 1, 1)$  makes  $(q, p) \mapsto \Theta(q) + p e_u$  an analytic local coordinate chart. Invariance of  $p = 0$  forces the transverse equation to have the form

$$\dot{p} = p a(q, p), \quad a(0, 0) = 1,$$

with  $a$  analytic. Conjugation symmetry makes  $p$  and  $a$  real on the real fixed locus  $q_0 \in \mathbb{R}$ ,  $q_2 = \bar{q}_1$ .

The normalized actual orbit tends to zero. On a sufficiently late real tail,  $a \geq 1/2$ . If  $p$  were nonzero anywhere on that tail, its absolute value would then grow at least exponentially, contrary to convergence to zero. Thus  $p = 0$  on the tail. Solving (25.6) supplies real  $c_0$  and complex  $c$  such that

$$H(z) = 840t^{-4} \left[ 1 + U(c_0t^{12}, ct^\alpha, \bar{c}t^{\bar{\alpha}}) \right]. \quad (25.7)$$

This is membership in an explicitly convergent surface, with an explicitly linear restricted flow. An unstated analytic linearization or formal stable-manifold assertion is not needed.

[write] **Added 5 October 2026, batch 103.** A second route to the three-dimensional linearization and time inversion of Section 8.

## 26 Sectorial continuation and the energy amplitude

### 26.1 The domain for singularity analysis

For every fixed  $\phi < \pi$  and a fixed branch of  $\log t$ , the inputs in (25.7) tend uniformly to zero as  $t \rightarrow 0$  in  $|\arg t| \leq \phi$ . Indeed,

$$|t^\alpha|, |t^{\bar{\alpha}}| \leq K_\phi |t|^{11/2}, \quad |t^{12}| = |t|^{12}.$$

Thus the convergent expression is holomorphic in that punctured sector and agrees with the original function on the overlap with its disk. Choose  $\pi/2 < \phi < \pi$ . Together with the collar in Section 24, it gives a standard  $\Delta$ -domain at  $\rho$ : for suitable  $R > \rho$  and  $0 < \theta < \pi/2$  the domain can be chosen to contain

$$\{z : |z| < R, \ z \neq \rho, \ |\arg(z - \rho)| > \theta\}.$$

Boundary neighborhoods close to  $\rho$  may be shrunk to lie inside the sector; a finite covering deals with the remaining compact arc. All overlaps agree by the identity theorem through the original disk. This supplies the local and global geometry needed for coefficient transfer without a claim about all continuation sheets.

### 26.2 An exact first integral

The ODE has the first integral

$$I = H'''H' - \frac{(H'')^2}{2} - \frac{H^3}{3} = \frac{1}{6}, \quad (26.1)$$

because  $I' = H'(H^{(4)} - H^2) = 0$  and the initial values give  $1 - 1/2 - 1/3 = 1/6$ . On the analytic invariant surface, all terms have the same  $t$ -weight, so  $I = t^{-12}J(q)$  with  $J$  analytic. Conservation along  $\dot{q} = -Eq$  and  $\dot{t} = -t$  gives  $EJ = 12J$ . A Taylor monomial in  $J$  can survive only when  $\beta_\nu = 12$ . The only such multi-index is  $(1, 0, 0)$ . Hence  $J$  is exactly a constant multiple of  $q_0$ .

The coefficient is computed by substituting  $H = 840t^{-4}(1 + c_0t^{12})$  to first order in  $c_0$ :

$$J(q) = -3008678400q_0, \quad c_0 = -\frac{1}{18052070400}. \quad (26.2)$$

For clarity, the energy of this truncated test profile is

$$-3008678400c_0 + 197568000c_0^2t^{12} - 197568000c_0^3t^{24}.$$

The test profile is not asserted to solve the ODE. Its constant term computes the linear coefficient of  $J$ , and  $EJ = 12J$  eliminates all higher contributions on the full surface. In particular, the nonlinear diagonal sectors introduce no additional energy constant.

Set

$$A = c_0 \rho^{12} = -\frac{\rho^{12}}{18052070400}, \quad C = c\rho^\alpha. \quad (26.3)$$

Since  $\rho > 0$ , this normalization is unambiguous. The recursion gives  $a_{kml} = \overline{a_{k\ell m}}$  and, for example,

$$a_{200} = \frac{7}{955}, \quad a_{020} = \frac{-201 + 143i\sqrt{159}}{78376}, \quad a_{011} = -2. \quad (26.4)$$

The energy argument alone does not prove  $C \neq 0$ : even with  $c = 0$ , a nontrivial real-mode family remains. It must be excluded separately.

[write] **Added 5 October 2026, batch 103.** The value of  $c_0$  is Part I's  $D$ , obtained there from the same energy (8.9) by the semigroup argument  $12m + 11k = 12$ ; here the weight identity  $EJ = 12J$  does the same work.

## 27 The complex amplitude is nonzero

Suppose, for contradiction, that  $c = 0$ , and write

$$U_0(q) = U(q, 0, 0) = \sum_{k \geq 1} a_{k00} q^k.$$

For  $k \geq 2$ , all factors of  $D(12k)$  are positive and increasing, so

$$D(12k) \geq D(24) = 114600, \quad \frac{840}{D(12k)} \leq \frac{7}{955}.$$

The coefficients of  $U_0$  are positive and are majorized by the coefficients of

$$B(r) = r + MB(r)^2 = \frac{2r}{1 + \sqrt{1 - 4Mr}}, \quad M = \frac{7}{955}. \quad (27.1)$$

Thus this restricted series converges for  $|q| < 955/28$ . This new one-variable bound is much larger than the earlier generic tridisk bound; it is essential for reaching  $z = 0$  below.

The exact comparison and energy bounds give, for  $0 \leq t \leq \rho$ ,

$$|c_0|t^{12} \leq \frac{840^3}{18052070400} < \frac{1}{30}. \quad (27.2)$$

At  $r = 1/30$ , the rational inequality  $1 - 4Mr > (14/15)^2$  implies

$$B(1/30) < \frac{1}{29}, \quad B'(1/30) = \frac{1}{\sqrt{1 - 4M/30}} < 2. \quad (27.3)$$

Coefficient positivity extends these absolute bounds to negative real arguments of absolute value below  $1/30$  as well.

The expression  $840t^{-4}[1 + U_0(c_0t^{12})]$  is meromorphic in a  $t$ -disk of radius strictly greater than  $\rho$ . It agrees with  $H$  near the real blow-up point by (25.7). Its overlap with  $|z| < \rho$  is connected: it is an intersection of two disks, and the only possible pole  $z = \rho$  lies on the boundary of the original disk. The identity theorem therefore extends equality all the way to  $z = 0$ .

At  $z = 0$  let  $q = c_0\rho^{12}$ . Differentiation of this convergent expression, followed by (27.2)–(27.3), gives

$$\begin{aligned}\frac{H'(0)}{H(0)} &= \frac{4 - 12qU'_0(q)/(1 + U_0(q))}{\rho} \\ &< \frac{4 + 12(1/30)2/(1 - 1/29)}{\rho} = \frac{169}{35\rho} < \frac{169}{175} < 1.\end{aligned}\tag{27.4}$$

Here  $\rho \geq 16800^{1/6} > 5$ , and  $1 + U_0(q) > 28/29$ . This contradicts  $H'(0)/H(0) = 1$ . Hence  $c \neq 0$  and  $C \neq 0$ . The proof uses only exact inequalities and a convergent majorant; fitted digits play no role. [write] **Added 5 October 2026, batch 103.** A second route to Part I's  $C_4 \neq 0$ , which uses a backward Metzler system and the concavity of the initial jet. Every link of the chain above was re-verified at the intake (Section 21.3).

## 28 Coefficient expansions to every fixed order

### 28.1 Exact transfer and polynomial sectors

Substitute  $t = \rho(1 - z/\rho)$  into (25.7). The pole term has the exact EGF coefficient

$$n![z^n] 840(\rho - z)^{-4} = 840\rho^{-n-4}n!\binom{n+3}{3} = 140(n+3)!\rho^{-n-4}.\tag{28.1}$$

For a nonreal exponent  $\beta = \beta_\nu$ , generalized binomial extraction yields the relative contribution

$$6a_\nu A^k C^\ell \overline{C}^m \frac{\Gamma(n+4-\beta)}{\Gamma(4-\beta)\Gamma(n+4)}.\tag{28.2}$$

This explains the factor six in (22.6).

If  $\ell = m$  and  $|\nu| \geq 1$ , then  $\beta_\nu = 12k + 11\ell$  is an integer at least 11. The power  $(1 - z/\rho)^{\beta_\nu-4}$  is a polynomial, whose coefficients are exactly zero for  $n > \beta_\nu - 4$ . If  $\ell \neq m$ , the exponent is nonreal and the ordinary gamma quotient applies. This identifies every analytic integer sector rather than treating its formal gamma poles as numerical terms. For a fixed truncation only finitely many such polynomials occur, so they all vanish for sufficiently large  $n$ .

### 28.2 Remainder transfer

Absolute convergence of  $U$  implies that truncation at degree  $J$  has a uniform local remainder  $O(|t|^{11(J+1)/2})$  inside the fixed sector. After multiplication by  $t^{-4}$ , the absolute remainder in  $H$  is  $O(|t|^{s-4})$ , where  $s = 11(J+1)/2$ . The big- $O$  transfer theorem of Flajolet and Odlyzko [FO, Theorem 1, pp. 220–221], applied in the  $\Delta$ -domain already proved, gives the ordinary coefficient bound

$$O(\rho^{-n}n^{3-s}).$$

The theorem permits real exponents, including nonnegative integers; there is no gap when  $s - 4$  is integral. Multiplication by  $n!$  and division by (28.1) give the relative  $O(n^{-s})$  bound in (22.6). This completes the proof of Theorem 22.1.

### 28.3 First oscillations and arbitrary algebraic precision

At degree one the real term is eventually zero, and the conjugate pair gives (22.7). For its gamma ratio,

$$G_n(\alpha) = \frac{n^{-\alpha}}{\Gamma(4-\alpha)} \left[ 1 + \frac{\alpha(\alpha-7)}{2n} + O(n^{-2}) \right]. \quad (28.3)$$

The frequency in  $\log n$  is  $\sqrt{159}/2$  and the relative envelope has order  $n^{-11/2}$ , because  $C \neq 0$  and  $\Gamma(4-\alpha)$  is finite and nonzero. The exact gamma ratio retains all integer inverse-power corrections from this first sector.

At the next useful precision, the pure second harmonics have real exponent 11; the diagonal sector is polynomial, and the mixed real/complex modes have real exponent  $35/2$ . Consequently

$$\frac{h_n}{140(n+3)!\rho^{-n-4}} = 1 + 12 \operatorname{Re}\{CG_n(\alpha) + a_{020}C^2G_n(2\alpha)\} + O(n^{-33/2}). \quad (28.4)$$

Degree order and real-exponent order are different. Formula (22.6) is safe because it may retain terms smaller than its stated error. Alternatively, for any fixed  $B > 0$ , retain the finitely many nonpolynomial terms with  $\operatorname{Re}\beta_\nu < B$ . First take a sufficiently long degree truncation, then discard the finitely many remaining terms with real exponent at least  $B$ . The relative error is  $O(n^{-B})$ .

[write] **Added 5 October 2026, batch 103.** (28.4) is an instance of (8.13) with a real-part cutoff, as Part I suggests after that display. The intake checked the exponent bookkeeping: after  $\operatorname{Re}\beta = 11/2$  and 11, the next nonpolynomial real part is  $33/2$ .

Finite-order Stirling expansion of each retained gamma ratio now provides every prescribed fixed algebraic order. All exponents and coefficients are determined by (25.3),  $A$  and  $C$ . No logarithmic multipliers are forced by a nonlinear resonance, because every denominator in (25.3) is nonzero. The nonreal powers themselves generate the oscillations. Convergence of this large-index expansion has not been proved or used.

## 29 Smooth inverse models and Lambert corrections

[write] **Added 5 October 2026, batch 103.** As in Part II, this section consists of instances of Part I's Section 8.6 and of the transseries volume (note at the end of Section 7); no novelty is claimed.

### 29.1 Inverse approximation at the sequence values

For fixed  $J \geq 0$ , let  $B_J(x)$  be the finite expression on the right of (22.6), without its remainder and with all polynomial sectors omitted. For large real  $x$ , use the ordinary gamma quotients for the remaining terms. Conjugation gives

$$B_J(x) = 1 + O(x^{-11/2}), \quad B'_J(x) = O(x^{-13/2}).$$

The differentiated bound follows from fixed-parameter gamma-ratio asymptotics or the corresponding digamma difference. In particular,  $B_0 = 1$ . Define

$$M_J(x) = 140\Gamma(x+4)\rho^{-x-4}B_J(x). \quad (29.1)$$

It is eventually positive, and

$$\frac{d}{dx} \log M_J(x) = \psi(x+4) - \log \rho + \frac{B'_J(x)}{B_J(x)} \sim \log x > 0.$$

Let  $q_J$  be its inverse on this increasing tail. The proved relative error becomes a logarithmic error of the same order. The inverse mean-value theorem then gives

$$q_J(h_n) = n + O\left(\frac{n^{-11(J+1)/2}}{\log n}\right). \quad (29.2)$$

These are approximations at the values of the discrete sequence; they do not assert a canonical continuous inverse of that sequence.

## 29.2 Two explicit Lambert approximations

For the leading model  $M_0$ , set

$$L = \log \frac{X\rho}{140}, \quad w = W_0\left(\frac{L}{e\rho}\right), \quad N_0 = \frac{L}{w}, \quad d = 1 + w, \quad a = \frac{1}{2} \log(2\pi N_0). \quad (29.3)$$

These quantities are real and positive where appropriate for sufficiently large  $X$ . The change of variable  $N = x + 3$  turns  $\log M_0(x) = \log X$  into

$$\log \Gamma(N + 1) - N \log \rho = L. \quad (29.4)$$

The leading term  $N \log(N/(e\rho))$  takes the value  $L$  at  $N = N_0$  and has derivative  $d$  there. Taylor expansion of Stirling's formula gives

$$q_0(X) = N_0 - 3 - \frac{a}{d} + O\left(\frac{1}{N_0 d}\right), \quad (29.5)$$

and the next correction is

$$q_0(X) = N_0 - 3 - \frac{a}{d} - \frac{a^2/(2d^2) - a/(2d) + 1/12}{N_0 d} + O\left(\frac{1}{N_0^2 d}\right). \quad (29.6)$$

The signs, the shift by  $-3$ , and the factor  $\rho/140$  in  $L$  all follow from the exact normalization in (29.4).

## 29.3 An all fixed order recursion

Write  $\varepsilon = 1/N_0$  and  $N = N_0 + s$ . Expanding (29.4) around  $N_0$  gives the formal equation

$$\begin{aligned} 0 = ds + a + \sum_{k \geq 2} \frac{(-1)^k \varepsilon^{k-1} s^k}{k(k-1)} + \frac{1}{2} \log(1 + \varepsilon s) \\ + \sum_{r \geq 1} \frac{B_{2r} \varepsilon^{2r-1}}{2r(2r-1)(1 + \varepsilon s)^{2r-1}}, \end{aligned} \quad (29.7)$$

where  $B_{2r}$  denotes a Bernoulli number, unrelated to the model factor  $B_J$ . Set  $P_0 = -a/d$ . For  $j \geq 1$ , substitute  $s = \sum_{i < j} P_i \varepsilon^i$  into the remaining expression in (29.7);  $P_j$  is minus its coefficient of  $\varepsilon^j$  divided by  $d$ . In particular,

$$P_1 = -\frac{1}{d} \left( \frac{a^2}{2d^2} - \frac{a}{2d} + \frac{1}{12} \right).$$

For fixed  $K$ , the resulting approximation has error

$$q_0(X) = N_0 - 3 + \sum_{j=0}^K P_j N_0^{-j} + O\left(\frac{N_0^{-K-1}}{d}\right). \quad (29.8)$$

Indeed,  $P_0 = O(1)$  and every fixed  $P_j$  with  $j \geq 1$  is  $O(1/d)$ . A sufficiently long finite Stirling expansion leaves a residual  $O(N_0^{-K-1})$ , while the derivative is  $d + o(1)$ . This justifies each fixed order of the formal recursion and does not assert its convergence.

At sufficiently high precision the complex correction in  $\log B_J$  matters. Lambert corrections to  $q_0$  alone therefore do not deliver arbitrary coefficient accuracy in (29.2); use  $q_J$  with sufficiently large  $J$  for that purpose.

## 29.4 Integer thresholds and rounding

Define  $T(X) = \min\{n : h_n \geq X\}$ . The leading equivalent implies  $h_n/h_{n-1} \sim (n+3)/\rho$ , so the sequence is eventually strictly increasing. Fix  $J$  and put  $s_J = 11(J+1)/2$ . For some  $K > 0$  and all sufficiently large  $n$ ,

$$|\log h_n - \log M_J(n)| \leq K n^{-s_J}.$$

Define  $q_{J,+}$  and  $q_{J,-}$  on their large- $X$  tails by

$$\begin{aligned} \log M_J(q_{J,+}) + K q_{J,+}^{-s_J} &= \log X, \\ \log M_J(q_{J,-}) - K q_{J,-}^{-s_J} &= \log X. \end{aligned}$$

Both functions being inverted are eventually increasing. An index below  $\lceil q_{J,+} \rceil$  has its upper bound below  $\log X$ ; the index  $\lceil q_{J,-} \rceil$  has its lower bound at least  $\log X$ . Therefore, for all sufficiently large  $X$ ,

$$\lceil q_{J,+}(X) \rceil \leq T(X) \leq \lceil q_{J,-}(X) \rceil, \quad q_{J,-} - q_{J,+} = O\left(\frac{q_J^{-s_J}}{\log q_J}\right). \quad (29.9)$$

Large  $X$  excludes the finite initial segment. The bracket often resolves rounding, but values arbitrarily close to an integer boundary preclude an unconditional exact-rounding promise from an unquantified finite asymptotic formula. Numerical values of  $K$  and a valid starting threshold are not supplied here.

## 30 Non D finiteness and non P recursiveness

This section adds a consequence of the convergent expansion, not of numerical fitting or a single nonreal leading exponent. It applies both to the order-seven problem proved here and to the order-five companion (Part II). We use subscripts 2 and 3 to distinguish their generating functions and radii.

**Theorem 30.1.** *For each  $m \in \{2, 3\}$ , the sectorial germs obtained from  $H_m$  by successive nonnegative windings about  $\rho_m$  are linearly independent in every finite initial collection. Representatives are identified after shrinking their common sector. In particular,  $H_2$  and  $H_3$  are not  $D$ -finite over  $\mathbb{C}(z)$ , and the exact integer sequences A333497 and A336009 are not  $P$ -recursive over  $\mathbb{C}[n]$ .*

[write] **Added 5 October 2026, batch 103.** This theorem is new to the repository. Its inputs are a convergent expansion on every finite angular interval of a logarithmic sheet, the universal recurrence  $D(E)U = \kappa U^2$ , and a nonzero complex amplitude. For  $m = 2$  they are Part I's Theorem 1.1 ((1.7), (4.1),

(3.13)) and, independently, Part II's Theorem 13.1. For  $m = 3$  they are proved in this Part and, independently, by Part I's Theorem 8.1 ((8.3), (8.11),  $C_4 \neq 0$ ). So each half of the theorem rests on two independent proofs of its local input. The intake read the proof below in full and found it sound; an independent check of the write, made afterwards, read it again line by line (end of Section 41).

A function is D-finite if it satisfies a nonzero homogeneous linear differential equation with rational-function coefficients. A sequence is P-recursive if, eventually, it satisfies a nonzero linear recurrence with polynomial coefficients in its index. The proof will reduce a putative finite-order differential equation to a finite-dimensional solution space at an ordinary point. No regular singularity assumption is made.

### 30.1 The two local inputs

For  $m = 3$ , set  $\alpha_3 = \alpha$  from (22.3) and use (25.7) with its convergent series and  $c \neq 0$ , proved above. For  $m = 2$ , the input is the established local theorem in the separate Report 113 (Part II). Explicitly, let

$$h_0^{(2)} = h_1^{(2)} = h_2^{(2)} = 1, \quad h_{n+3}^{(2)} = \sum_{k=0}^n \binom{n}{k} h_k^{(2)} h_{n-k}^{(2)},$$

so  $H_2''' = H_2^2$  with initial derivatives equal to one. Write  $t = \rho_2 - z$  and

$$\alpha_2 = \frac{13 + i\sqrt{71}}{2}, \quad Q_2(b) = (3-b)(4-b)(5-b), \quad D_2(b) = Q_2(b) - 120.$$

The companion proves that  $c \neq 0$  and a convergent bivariate series  $U$  satisfy

$$H_2(\rho_2 - t) = 60t^{-3}[1 + U(ct^{\alpha_2}, \bar{c}t^{\bar{\alpha}_2})], \quad D_2(E)U = 60U^2, \quad (30.1)$$

where  $E = \alpha_2 q_1 \partial_{q_1} + \bar{\alpha}_2 q_2 \partial_{q_2}$ ,  $U(0,0) = 0$ , and both linear coefficients are one. Thus the  $m = 2$  claim uses that separate theorem as an input; it is not established by the order-seven proof alone.

[write] **Added 5 October 2026, batch 103.** “Report 113” is Part II of this report. The input (30.1) is Part I's Theorem 1.1: (1.7) with  $\mathcal{F} = 1 + U$ ,  $c = C$ ,  $a_{10} = a_{01} = 1$ , and (4.1), which is the coefficient form of  $D_2(E)U = 60U^2$ ; Theorem 1.1 states convergence on every fixed finite angular interval of a chosen logarithmic sheet, which is what the second subsection of Section 30 uses.

For the rest of this section write  $\alpha = \alpha_m$ ,  $\kappa = 60$  or  $840$ , and  $p = 3$  or  $4$  as appropriate. Both expansions have the form  $\kappa t^{-p}(1 + U)$ . Their nonzero relative exponents have positive half-integral real parts, with only finitely many monomials below any fixed real level. Every differentiated series converges on a smaller polydisk. The nonvanishing of the complex amplitude is essential in both cases. In order seven it is the stronger exclusion of the entire real-mode family, rather than just the pure pole, that provides it.

### 30.2 Any fixed finite number of windings

Put  $w = \log t$ . For an input with exponent  $\gamma$  of positive real part,

$$|ce^{\gamma w}| = |c| \exp\{\operatorname{Re} \gamma \operatorname{Re} w - \operatorname{Im} \gamma \operatorname{Im} w\}.$$

On any bounded interval of  $\operatorname{Im} w$ , all inputs to  $U$  tend uniformly to zero as  $\operatorname{Re} w \rightarrow -\infty$ . Hence, for every integer  $N \geq 0$  and fixed  $0 < \varepsilon < \pi$ , there is  $r_N > 0$  such that the expansion is holomorphic and locally absolutely convergent on

$$\operatorname{Re} w < \log r_N, \quad -\varepsilon < \operatorname{Im} w < 2\pi N + \varepsilon.$$

It agrees with the original germ on the real branch by the local theorem and the identity theorem. A vertical segment from  $w = \log r$  to  $w = \log r + 2\pi ij$ ,  $r < r_N$ , implements  $j$  windings. The transformation  $z = \rho - t$  preserves their orientation. The integer pole factor has trivial monodromy; a relative power  $t^\beta$  acquires the multiplier  $e^{2\pi i j \beta}$ .

Thus branches  $H_0, \dots, H_N$  are defined on one common sector  $0 < |t| < r_N$ ,  $|\arg t| < \varepsilon$  by replacing  $w$  by  $w + 2\pi ij$ . The radius may decrease with  $N$ . There is no assertion that a fixed-radius circle supports arbitrarily many windings, and none is needed below.

### 30.3 Uniqueness after grouping equal exponents

We need an elementary uniqueness principle that allows different monomials to have the same complex exponent. Suppose

$$F(t) = \sum_{\nu} b_{\nu} t^{\beta_{\nu}}, \quad 0 < t \leq t_0,$$

where the real parts of the exponents lie in  $\frac{1}{2}\mathbb{N}_0$ , there are only finitely many terms below any fixed real level, and  $\sum_{\nu} |b_{\nu}| t_0^{\operatorname{Re} \beta_{\nu}} < \infty$ . If  $F$  vanishes for all sufficiently small positive  $t$ , then every coefficient obtained by grouping equal complex exponents is zero.

To prove this, suppose a nonzero grouped coefficient exists and choose the smallest real level  $\lambda$  at which one occurs. There are finitely many distinct imaginary parts  $\tau_1, \dots, \tau_s$  there. The tail at larger real levels is  $O(t^{\lambda+1/2})$ : its absolute value is at most

$$t^{\lambda+1/2} t_0^{-\lambda-1/2} \sum_{\nu} |b_{\nu}| t_0^{\operatorname{Re} \beta_{\nu}}.$$

Divide by  $t^{\lambda}$  and set  $x = -\log t$ . The finite exponential sum  $T(x) = \sum_{a=1}^s B_a e^{-i\tau_a x}$  then satisfies  $T(x) = O(e^{-x/2})$ . But

$$\lim_{R \rightarrow \infty} \frac{1}{R} \int_X^{X+R} |T(x)|^2 dx = \sum_{a=1}^s |B_a|^2,$$

since distinct-frequency cross terms have zero limiting mean. Decay of  $T$  makes the same mean zero, a contradiction.

This principle applies to a finite linear combination of the branches. For  $P(X) = \sum_{j=0}^N \xi_j X^j$ , the extra coefficient multiplier obeys

$$|P(e^{2\pi i \beta})| \leq \sum_{j=0}^N |\xi_j| e^{-2\pi j \operatorname{Im} \beta}.$$

At sufficiently small  $t_0 > 0$ , each corresponding series is absolutely convergent on its fixed sheet. This verifies the required absolute summability. Grouping is necessary: in order seven, for instance,  $12 \cdot 11 = 12\alpha + 12\bar{\alpha} = 132$ , so distinct multivariate monomials can share an exponent.

### 30.4 Infinitely many distinct nonzero characters

Restrict to the pure positive-complex axis:

$$V(q) = U(q, 0) \quad (m = 2), \quad V(q) = U(0, q, 0) \quad (m = 3).$$

Writing  $V(q) = \sum_{k \geq 1} v_k q^k$ , we have  $v_1 = 1$  and

$$D(\alpha q \partial_q) V = \kappa V^2. \tag{30.2}$$

Restriction is legitimate because every polynomial in the Euler operator preserves the coordinate axes. If  $V$  were a polynomial of degree  $K \geq 1$ , the left side would have degree at most  $K$  and the right side exactly  $2K$ , with nonzero top coefficient. Therefore infinitely many  $v_k$  are nonzero. It is unnecessary to claim that every  $v_k$  is nonzero.

The exponent  $k\alpha$  belongs to its pure-complex monomial alone. For  $m = 2$ , the equality  $a\alpha + b\bar{\alpha} = k\alpha$  implies  $a - b = k$  by imaginary parts, then  $2\operatorname{Re}\alpha b = 0$  by real parts, so  $b = 0$  and  $a = k$ . For  $m = 3$ , the equality  $12a + b\alpha + c\bar{\alpha} = k\alpha$  gives  $b - c = k$  and then  $12a + 11c = 0$ . Nonnegativity again forces the pure monomial. Thus exponent collisions elsewhere cannot cancel any of these sectors.

Now suppose  $\sum_{j=0}^N \xi_j H_j = 0$  on the common sector and put  $P(X) = \sum_{j=0}^N \xi_j X^j$ . Divide by the common factor  $\kappa t^{-p}$  and expand:

$$P(1) + \sum_{\nu \neq 0} a_\nu c^\nu P(e^{2\pi i \beta_\nu}) t^{\beta_\nu} = 0,$$

where  $c^\nu$  denotes the appropriate amplitude product. Grouped uniqueness and the pure-exponent property give

$$v_k c^k P(e^{2\pi i k \alpha}) = 0 \quad (k \geq 1).$$

Since  $c \neq 0$ ,  $P$  has roots  $e^{2\pi i k \alpha}$  for infinitely many  $k$ . Their moduli  $e^{-2\pi k \operatorname{Im} \alpha}$  are strictly decreasing, hence these roots are distinct. Therefore  $P = 0$  and all  $\xi_j = 0$ . A relation holding initially as germs at a point of the sector extends to the whole sector by the identity theorem. This proves the asserted branch independence.

### 30.5 The differential equation contradiction

Suppose  $H$  satisfied a nonzero homogeneous rational linear ODE of order  $d$ . Order zero is impossible since  $H \neq 0$ , so  $d \geq 1$ . Normalize its leading coefficient. A sufficiently small punctured disk about  $\rho$  contains no poles of the normalized coefficients;  $\rho$  itself may be ordinary, regular singular or irregular singular. Choose the preceding construction with  $N = d$  and shrink  $r_d$  to lie in that disk. All winding paths consist of ordinary points of the hypothetical linear equation, so continuation preserves its identity. The  $d + 1$  branches all satisfy it on their common sector.

At any ordinary point  $z_*$  of that sector, its analytic solution germs form a  $d$ -dimensional complex vector space: the map to  $(H(z_*), \dots, H^{(d-1)}(z_*)) \in \mathbb{C}^d$  is a linear bijection by analytic initial-value existence and uniqueness. Those  $d + 1$  germs must be linearly dependent with constant complex coefficients, contradicting what was just proved. This establishes non-D-finiteness without local classification of a singular point.

### 30.6 From a polynomial recurrence to a linear ODE

To transfer the conclusion to the integer sequence directly, suppose  $\sum_{j=0}^s p_j(n) h_{n+j} = 0$  for all  $n \geq N$ , where the polynomials  $p_j$  are not all zero. Set  $f_n = h_n/n!$ . Dividing by  $n!$  gives

$$\sum_{j=0}^s q_j(n) f_{n+j} = 0, \quad q_j(n) = p_j(n) \prod_{a=1}^j (n+a).$$

These polynomials are not all zero. Let  $F(z) = \sum_{n \geq 0} f_n z^n = H(z)$  and  $\vartheta = z d/dz$ . The polynomial-coefficient differential operator

$$L = \sum_{j=0}^s z^{s-j} q_j(\vartheta - j)$$

satisfies  $LF = R(z)$  for a polynomial  $R$ , because its coefficient at  $z^{n+s}$  is the displayed recurrence for every  $n \geq 0$ . It is a nonzero operator: on  $z^M$  its summands have distinct powers  $z^{M+s-j}$ , and a nonzero  $q_j(M-j)$  cannot vanish for all sufficiently large integers  $M$ . A sufficiently high derivative annihilates  $R$ , producing a nonzero homogeneous differential operator annihilating  $H$ . The composition is still nonzero because its leading derivative coefficient remains nonzero. This contradicts non-D-finiteness and proves non-P-recursiveness. No factorial-product closure result is needed.

[write] **Added 5 October 2026, batch 103.** A precision on the step “its coefficient at  $z^{n+s}$  is the displayed recurrence for every  $n \geq 0$ ”: for each  $n \geq 0$  that coefficient equals the left side of the recurrence at  $n$ , which vanishes for  $n \geq N$ ; the coefficients of  $z^k$  with  $k < s$  and those with  $n < N$  are finitely many and form the polynomial  $R$ .

## 31 Reproducibility and numerical status

### 31.1 What the exact checks establish

The companion ZIP contains this TeX source, its PDF, a deterministic build script, an exact checker and its fixtures, and local validation scripts. It is a reader package, without third-party source PDFs. Source publications remain available through the bibliography.

The exact checks independently regenerate the recurrence terms, compare the externally checked prefix, and test the displayed polynomial identities and constants. In particular, they cover the indicial factorization, the normalized equation, the energy coefficient, the displayed low-degree psi coefficients, the transfer normalization and the rational inequalities excluding  $C = 0$ . Finite symbolic checks support the algebra; the analytic arguments for convergence, actual-orbit membership, continuation and transfer are the proofs in the body of this report.

The validation harness uses explicit exceptions rather than removable Python assertions. It replays checks normally and with Python optimization, rejects named corrupted fixtures, tests strict manifest inventory and hash failures, and replays from an unrelated working directory. The package build requires two clean PDF builds to be byte-identical and rejects overfull boxes or unresolved references. Its ZIP builder fixes timestamps, ordering and file modes. Reproduction requires the stated installed dependencies; a fresh directory is not a newly provisioned operating system.

[write] **Added 5 October 2026, batch 103.** In the repository the companion’s files are `code/03-order7-*.py`, `data/03-order7-*` and `03-order7-checks-README.md`; the TeX source, PDF, `README.txt`, `CHECKSUMS.sha256`, `checks/MANIFEST.json` and `checks/requirements.txt` were not shipped. The sealed integrity checks need the delivered layout; the report’s README gives the rerun routes, including the Windows portability patch the harness needs.

### 31.2 Exploratory fits

Exact recurrence values can be fitted to a fixed-order version of (22.6), solving for  $\log \rho$ ,  $\operatorname{Re} C$  and  $\operatorname{Im} C$  while using the exact relation  $A = -\rho^{12}/18052070400$ . An exploratory high-index degree-four fit suggests

$$\begin{aligned}\rho &\approx 5.17917581591984130267752293, \\ C &\approx -0.05798262629139505 + 0.00380586253147502 i.\end{aligned}$$

These decimal digits are not certified enclosures. Agreement across fitting triples and degrees, or high arithmetic precision, does not supply a rigorous truncation bound. Fitted-index residuals are

not independent validation. Only the symbolic enclosure (22.4) is certified here; even the proof that  $C \neq 0$  is independent of the fit.

For scale only, the exact lower and upper radius bounds are approximately 5.06079 and 5.38356. The cited paper’s Table 1 labels its entries  $\rho_m^{-1}$ , yet the displayed  $m = 3$  number 5.1792 lies in this radius range. The exact bound  $\rho > 5$  already proves that this displayed number cannot be its reciprocal. This is a label mismatch, not evidence for additional certified decimal places.

[write] **Added 5 October 2026, batch 103.** The heading of the journal’s table is source 03’s reading, not re-checked at the intake (Section 32.1, item 6). Part I makes the corresponding remark at  $m = 2$ .

## 32 Scope and further research

This proves the specified fourth-order initial-value problem and the  $m = 3$  instance of the B-tree history conjecture. The real-profile step depends on the stated published theorem. The construction does not silently extend the elementary third-order argument from the order-five companion to fourth order, nor does Astashova’s stated third- and fourth-order theorem settle the higher-order family. A full literature-priority claim is outside the source checks made here.

Several concrete questions remain.

1. **Certified constants and effective errors.** Use interval integration of a rescaled system and explicit analytic tail bounds to enclose  $\rho$  and  $C$ , quantify a positive lower bound for  $|C|$ , and produce useful numerical remainder constants and threshold cutoffs. The present symbolic nonvanishing proof is qualitative in  $|C|$ .
2. **Higher derivative orders.** For  $H^{(m+1)} = H^2$ , first establish that the particular positive initial orbit tends to the pole equilibrium; then analyze its stable spectrum and possible nonlinear resonances. The general real-profile problem cannot be resolved merely by reusing the fourth-order theorem. Higher-dimensional Emden–Fowler systems can exhibit different normalized dynamics.
3. **Dependence on initial data.** Describe the movable singularity and stable amplitudes for other positive initial conditions, including the real-mode family and its exceptional cases. The energy integral isolates one amplitude but does not by itself determine the two real components of  $C$ .
4. **Global continuation.** Locate singularities beyond the first circle and describe the multivalued continuation and its sheets. A local slit expansion and a  $\Delta$ -domain do not classify the whole Riemann surface.
5. **Effective asymptotic truncation.** Optimize finite truncations for numerical use while retaining rigorous error bounds. Convergence of the local psi series does not justify an infinite coefficient expansion at fixed index.

### 32.1 Further questions and research

[write] *Added 5 October 2026, batch 103, under the standing rule. Source 03 contains no claim found to be wrong.*

1. **Certified constants and effective errors** (source 03, question 1; Part I’s Question 3). Enclosures of  $\rho_4$ ,  $C$  and a lower bound for  $|C|$ . Open; only (22.4) is certified, and the fitted  $\rho \approx 5.1791758$  and  $C$  of Section 31 are exploratory (Part I has no  $r = 4$  numerics to compare with).

2. **Higher derivative orders** (source 03, question 2; Part I's Question 2). Open for  $4 \leq m \leq 28$ ; for  $m \geq 29$  answered negatively by Theorem 9.1.
3. **Dependence on initial data** (source 03, question 3; Part I's Question 5), including the real-mode family. Open.
4. **Global continuation** (source 03, question 4; Part I's Question 4). Re-scoped by Theorem 30.1: the continuation around  $\rho_4$  (and around  $\rho$  for A333497) has infinitely many independent branches. Further singularities and the sheet structure remain open.
5. **Effective asymptotic truncation** (source 03, question 5). Open; the tridisk radius  $121/6720$  of Section 25 is explicit in  $q$  but a radius in  $t$  needs a bound on  $|c|$ .
6. **External claim kept as source 03's.** The  $m = 3$  entry 5.1792 of the journal's Table 1 is printed under  $\rho_m^{-1}$  (Section 31); the bound  $\rho > 5$  proves it cannot be a reciprocal, but the reading of the table itself was not re-checked at the intake.
7. **Hypotheses of the cited real theorem.** Section 23 checks the hypotheses of Astashova's Theorem 3 [2] as source 03 reads that paper; Part I relies on the same theorem for  $r = 4$  (Section 8). The intake did not re-read the cited paper.

## Part IV

# Order fifty-nine by a cyclic-sign cone (source 04, bundle Report 117)

## 33 Reading Part IV: provenance, credit and notation

*This section and every [write] note of Part IV were added when bundle Report 117 joined the report (batch 103, 5 October 2026). Sections 34–44 print the manuscript in full.*

### 33.1 The order dictionary

Three parameters name the same case, and the source’s “order 59” is Part I’s ODE order thirty:

| conjecture parameter | B-tree order  | ODE order (Part I) | derivative order (Part IV) |
|----------------------|---------------|--------------------|----------------------------|
| $m = 29$             | $2m + 1 = 59$ | $r = m + 1 = 30$   | $d = r = 30$               |

The three normalizations of the conjecture agree: Part I’s (1.2) and (9.1) and Part IV’s (34.5). Part IV’s  $A = (d)_d = 59!/29!$  is Part I’s  $A_r$  of (9.2) at  $r = 30$ ; its characteristic polynomial  $P(\lambda) = \prod_{a=30}^{59} (\lambda + a) - 2A$  is Part I’s  $\chi_{30}(\mu)$  of (9.5); and its Fowler matrix  $B$  is diagonally similar to Part I’s  $J_{30}$ : with  $u_j = A(d)_j Z_j$  (Part I’s  $Z = Y - \mathbf{1}$ ),  $J_{30} = \Delta^{-1} B \Delta$  for  $\Delta = \text{diag}(A(d)_j)$ .

### 33.2 Provenance

Source 04 is Report 117 of the session bundle: the archive `B_Tree_Order_59_Counterexample_to_the_Universal_Asymptotic_Conjecture_Source.zip` (372,599 bytes, wrapper directory `report117/`), arrival commit `60f54ea06`, placed by `9c995cefe` (batch 103) as the files with prefix `04-order59-`. Its main file `report117.tex` (859 lines; 14-page PDF) is titled *A counterexample to the universal B-tree-history conjecture: a cyclic-sign obstruction at B-tree order 59*, dated 2 October 2026, with the status line “Self-contained proof and exact reproducibility companion”. It names no author or tool (empty PDF Author), has no “private review” wording and no pin. Not staged, and retrievable from the arrival commit: the manuscript and PDF, `README.txt`, `CHECKSUMS.sha256`, `checks/MANIFEST.json` and `checks/check_manifest.py` (a byte copy of the staged `code/03-order7-checks-check_manifest.py`).

*What was changed in printing.* Labels carry the prefix `his:o59:`; citation keys were mapped: `bw`→[5], `lanford`→[10], `oeis3`, `oeis4`→[11, 12]. The source’s two appendices are printed as Sections 43 and 44 (titles marked “Report 117, Appendix A/B”), because Part I’s appendix closes the report; the source’s words “Appendix A” in its Section 38 now point to that section. Macros as in Part II. No statement, proof or number was altered.

### 33.3 Credit: what Part IV proves again and what it adds

Source 04 was written without sight of Part I. Its abstract’s “We disprove the universal form of Conjecture 1” is, in this repository, a second route: Part I’s Theorem 9.1 already proves failure for every  $r \geq 30$ , that is every  $m \geq 29$ , and Theorem 34.1 is its first case  $r = 30$  (dated note in Section 33.6). The source says itself that it “does not claim here that 30 is the first order”, and Part I’s exact Routh check (Section 9) shows that order  $r = 29$  is spectrally stable.

| Source 04                                                                                       | Relation to Part I                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 34.1                                                                                    | the case $r = 30$ of Theorem 9.1, by a different route; its scalar form (no $H \sim A(\rho - x)^{-30}$ ) is Part I's remark after that theorem's proof                                                                                                                                                                                                        |
| Proposition 35.1: $\min_j r_j \leq \rho_* \leq \max_j r_j$ for every $d$                        | <b>new:</b> Lemma 8.2 gives only finiteness; (1.6) is the case $d = 3$ and (22.4) the case $d = 4$ ; at $d = 30$ it gives $41.082 \dots \leq \rho_* \leq 43.636 \dots$                                                                                                                                                                                        |
| Lemma 35.2, (35.4)                                                                              | the Abelian argument of Section 9                                                                                                                                                                                                                                                                                                                             |
| Cyclic cone $K$ , Lemmas 36.1, 36.2                                                             | $K$ equals Part I's cone $\mathcal{C}_3$ (Remark 33.1); its invariance by Euler steps is a second route to the cone-invariance use of [13], namely (9.16); the strict sign-regularity of $T_r$ used for (9.17) still rests on [13] [corrected at the independent check; dated note after Remark 33.2]; likely prior art: Mallet-Paret and Smith (Remark 33.3) |
| Proposition 37.1, Section 38, Corollary 38.1: complete, simple, hyperbolic spectrum at $d = 30$ | <b>new as a proved statement:</b> Part I proves an unstable pair for every $r \geq 30$ and has the full count at $r = 30$ only as an exact Routh check (Section 9)                                                                                                                                                                                            |
| Lemma 39.1, Proposition 39.2, $E^s \cap K = \{0\}$                                              | second route to (9.17) at $r = 30$ , where Part I's $E_{\text{slow}}$ is $E^s$ (the top three moduli are the unstable ones), without [1]                                                                                                                                                                                                                      |
| Section 40: ordinary stable manifold, tangent-cone exclusion                                    | the argument of Section 9, with the ordinary (hyperbolic) case [10, Theorem 7.3.1(1),(4)] in place of the generalized one                                                                                                                                                                                                                                     |
| Section 43: thirty-line exact certificate                                                       | reproducible as printed (rerun at intake)                                                                                                                                                                                                                                                                                                                     |

The certificate is tight: the unstable pair is  $0.004879 \dots \pm 9.107358 \dots i$  (intake root approximation, for orientation only), and  $q = 911/100$  brackets  $\omega_* \approx 9.1074$  from below; Part I's finite bridge uses  $b_* = 228/25 = 9.12$  from above. The intake reran the seven inequalities exactly.

### 33.4 Notation of Part IV

| Symbol                                     | In Part IV (source 04)                                     | Part I; false reading to avoid                                                                   |
|--------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| $d, m$                                     | derivative order $d = m + 1$ ; $d = 30$ from Section 37 on | Part I's ODE order $r$ ; <i>not</i> a grade                                                      |
| $A; r_j$                                   | $(d)_d$ ; $[A(d)_j]^{1/(d+j)}$                             | $A$ is Part I's $A_r$ ; $r_j$ is <i>not</i> an ODE order                                         |
| $\rho_*; \rho$                             | actual blowup point; a candidate pole location             | Part I's $\rho_r$ ; the $\rho$ of Section 9                                                      |
| $p, p_\rho, p_T; w_j$                      | exact pole solutions; $H^{(j)} - p^{(j)}$                  | Part I's $G_r$ ; $w$ is not Part I's or Part II's $w$                                            |
| $b_j$                                      | $p^{(j)}(0) = A(d)_j \rho^{-d-j}$                          | the <i>reciprocal</i> of Part I's $b_j = Y_j(0)$ in (9.15)                                       |
| $t, \tau, u_j$                             | $\rho - x$ ; $-\log t$ ; $t^{d+j} w_j = A(d)_j Z_j$        | Part I's $s = \tau + \log \rho$ and $Z = Y - \mathbf{1}$                                         |
| $V, K$                                     | cyclic sign count; $\{V \leq 2\}$                          | Part I's $s^-$ counts linear changes; $K = \mathcal{C}_3$ (Remark 33.1); $K$ is not an amplitude |
| $B$                                        | Fowler matrix                                              | diagonally similar to Part I's $J_{30}$ ; not $B(u)$                                             |
| $P(\lambda), Q(\lambda)$                   | $\prod(\lambda + a) - 2A$ ; $\prod(\lambda + a)$           | Part I's $\chi_{30}$ and $P_{30}$ ; not $Q_d$ , $Q(v)$ or $P(u)$                                 |
| $\lambda_k = \sigma_k + i\omega_k$         | roots; modulus curve and its phase                         | $\lambda$ is an eigenvalue, not Part I's exponent $\lambda$ ;                                    |
| $\omega(\sigma), \theta(\sigma), S_0, S_1$ |                                                            | Part I writes $\mu = a + ib$                                                                     |
| $F(y), \Theta(y), \omega_*, q$             | certificate data; $q = 911/100$                            | $\omega_*$ is Part I's $b(0)$ ; $\Theta$ is not Part II's surface                                |
| $R, I, S$                                  |                                                            |                                                                                                  |
| $E^s, E^u, h$                              | stable and unstable subspaces; graph map                   | at $d = 30$ , Part I's $E_{\text{slow}}$ , $E_{\text{fast}}$                                     |
| $\alpha_j, \beta_j; \delta_j, s_j; c$      | Euler weights; phase increments and sums; an amplitude     | not exponents; $s_j$ is not Part I's $s$                                                         |

### 33.5 Three remarks of the write

*Remark 33.1* ([write] The two cones coincide). For every  $v \in \mathbb{R}^d$ ,  $V(v) \leq 2$  if and only if  $s^-(v) \leq 2$ , where  $V$  is Part IV’s cyclic count (36.1) and  $s^-$  is Part I’s linear count (9.13). Hence  $K = \mathcal{C}_3$ .

*Proof.* Delete the zero coordinates of  $v$ ; let  $\varepsilon_1, \dots, \varepsilon_k \in \{\pm 1\}$  be the remaining signs in order. If  $k \leq 1$ , both counts are zero. Otherwise  $s^-(v)$  counts the indices  $i < k$  with  $\varepsilon_i \neq \varepsilon_{i+1}$ , and  $V(v)$  counts these together with the closing comparison of  $\varepsilon_k$  and  $\varepsilon_1$ . So  $s^-(v) \leq V(v) \leq s^-(v) + 1$ . Moreover  $V(v)$  is even: going once around the cycle returns to the starting sign, so the sign changes an even number of times. If  $V(v) \leq 2$ , then  $s^-(v) \leq V(v) \leq 2$ . If  $s^-(v) \leq 2$ , then  $V(v) \leq 3$ , and evenness gives  $V(v) \leq 2$ .  $\square$

*Remark 33.2* ([write] The invariance step for every order). Part I’s centred system  $Z' = M(s)Z$  (Section 9) reads  $Z'_j = -(r+j)Z_j + (r+j)Z_{j+1}$  for  $j < r-1$  and  $Z'_{r-1} = -(2r-1)Z_{r-1} + (2r-1)(Y_0(s)+1)Z_0$ , with continuous coefficients and nonnegative couplings  $(r+j)$  and  $(2r-1)(Y_0+1)$ . It is therefore a system (36.2) with indices modulo  $r$ . Part I shows  $Z(0) \in \mathcal{C}_3$  by the concavity of  $\log b_j$ , an argument valid in every order. Lemma 36.2 and Remark 33.1 then give  $V(Z(s)) \leq V(Z(0)) \leq 2$  for all  $s \geq 0$ , in every order  $r$ , without the strong  $k$ -positivity theorem [13]; together with  $Z(s) \neq 0$ , which is uniqueness as in Part I, this is (9.16). The spectral half of Part I’s proof, (9.17) for every  $r \geq 30$ , is not replaced by this remark: Part IV’s Proposition 39.2 uses the complete spectral ordering, which the source proves at  $d = 30$  only; that half also still uses [13], through the strict sign-regularity of  $T_r$  required by [1] [added at the independent check].

(Dated note, 5 October 2026: the row “Cyclic cone  $K$ ” of the table in Section 33.3 first read “its invariance by Euler steps is a second route to the strong- $k$ -positivity step [13]”. Part I uses [13] twice: for the cone invariance (9.16), which Lemma 36.2 replaces in every order by this remark, and for the strict sign-regularity of  $T_r = \exp(\delta J_r)$ , the hypothesis of [1] behind (9.17), which it does not replace. The row was corrected, and the last clause of this remark added, at the independent check recorded at the end of Section 41.)

*Remark 33.3* ([write] Prior art for the cyclic count, and the order-59 orbit). Counting sign changes cyclically, and its nonincrease along linear systems whose  $j$ th equation is coupled only to the next coordinate with a nonnegative coupling, is the positive-feedback case of the discrete Lyapunov function of J. Mallet-Paret and H. L. Smith, *The Poincaré–Bendixson theorem for monotone cyclic feedback systems*, J. Dynam. Differential Equations **2** (1990), no. 4, 367–421 (they couple each coordinate to the preceding one, assume strict coupling signs  $\delta_i \partial f_i / \partial x_{i-1} > 0$  with  $\delta_i \in \{\pm 1\}$ , and count  $\#\{i : \delta_i x_i x_{i-1} < 0\}$ , which is the plain cyclic count when every  $\delta_i = +1$ ). The reference was checked bibliographically at the write and again at the independent check; its exact internal statement numbers were not. Lemma 36.2 is a weak form of that monotonicity in the positive-feedback case (their couplings are strictly signed; Lemma 36.2 allows vanishing couplings); source 04 does not cite it, and no novelty is claimed for the lemma here. The uncentred Fowler system of Section 41, question 1,  $U'_j = U_{j+1} - (d+j)U_j$ ,  $U'_{d-1} = U_0^2 - (2d-1)U_{d-1}$ , has exactly the monotone cyclic feedback structure (positive feedback  $2U_0$  on the positive orthant). Mallet-Paret and Smith’s theorem says that the  $\omega$ -limit set of a bounded orbit of such a system is either a nonconstant periodic orbit or is built from equilibria and orbits connecting them. If its hypotheses can be verified for the actual normalized orbit at  $d = 30$  (boundedness away from the boundary of the orthant is the first thing to check), it would bear directly on Part I’s Question 1 and Part IV’s question 1: the obstruction proved here excludes convergence to the pole equilibrium, and a periodic profile would be the natural remaining alternative. This is a pointer, not a claim; Section 41.3 lists it.

(Dated note, 5 October 2026: the first sentence of Remark 33.3 first read “Counting sign changes cyclically, and its nonincrease along linear systems whose  $j$ th equation is coupled only to the next coordinate with a coupling of fixed sign, is the discrete Lyapunov function of J. Mallet-Paret and H. L. Smith, [...] (the reference was checked bibliographically at the write; its exact internal statement numbers were not)”, and the parenthesis of its second sentence read “(Mallet-Paret and Smith work under their own standing sign hypotheses on the coupling, which the write did not compare in detail)”. A fixed sign is not enough for the plain cyclic count: Mallet-Paret and Smith weight each change by the sign of its coupling, and the plain count is their function only when every coupling is positive; with a negative coupling the plain count can increase, from 0 to 2 in the check’s control example. The remark was corrected at the independent check recorded at the end of Section 41; its credit and its pointer are unchanged.)

### 33.6 The source’s abstract

We disprove the universal form of Conjecture 1 of Burghart and Wagner, *On the histories of B-trees* (2026). The counterexample is the parameter  $m = 29$ , or B-tree order 59: its exponential generating function satisfies  $H^{(30)} = H^2$  and  $H^{(j)}(0) = 1$  for  $0 \leq j < 30$ . Writing  $A = 59!/29!$ , we prove that this particular solution cannot satisfy  $H(x) \sim A(\rho - x)^{-30}$  at any finite positive  $\rho$ . An elementary Abelian argument shows that the conjectured coefficient asymptotic would force exactly that behavior. The dynamical obstruction uses a closed, forward-invariant cone of vectors with at most two cyclic sign changes, the complete ordering of the linearized spectrum, and exclusion of the entire stable subspace from this cone. Tangency of the local stable manifold then excludes the actual all-ones orbit. Seven exact integer or rational inequalities certify the needed spectrum; no numerical root-finding or numerical integration enters the proof. The result establishes neither a replacement coefficient asymptotic nor a periodic attractor. No claim of literature priority or external peer review is made.

[write] **Added 5 October 2026, batch 103.** Part I’s Theorem 9.1 already disproves Conjecture 1 in this repository, for every  $m \geq 29$ ; Theorem 34.1 is a second route to its first case  $m = 29$  ( $r = d = 30$ ). What source 04 adds is listed in Section 33.3. Its non-claims (no replacement asymptotic or periodic attractor, no first-transition claim, no classification of orders, no priority, no peer review, not a formal verification) are kept below.

## 34 The statement being contradicted

### 34.1 Model, indexing, and the family

For an integer  $d \geq 2$ , define integers  $h_n$  by

$$h_0 = \cdots = h_{d-1} = 1, \quad h_{n+d} = \sum_{k=0}^n \binom{n}{k} h_k h_{n-k} \quad (n \geq 0). \quad (34.1)$$

Their exponential generating function is the local analytic solution

$$H(z) = \sum_{n \geq 0} h_n \frac{z^n}{n!}, \quad H^{(d)} = H^2, \quad H^{(j)}(0) = 1 \quad (0 \leq j < d). \quad (34.2)$$

Local analytic existence follows by writing the polynomial first-order system for the derivative jet. Comparing Taylor coefficients proves (34.1), so these two definitions identify the same function. In particular, every coefficient is positive.

Burghart and Wagner’s odd-order B-tree model has  $d = m + 1$  and B-tree order  $2m + 1 = 2d - 1$  [5, Section 4, equation (4)]. Here  $h_n$  counts reduced  $(2m + 1)$ -historic trees with  $n$  vertices, equivalently B-tree shape histories of length  $n + m$ . A history counts successive shapes, rather than insertion permutations or just final shapes. Their Conjecture 1 asserts, for every integer  $m \geq 1$ , that some  $\rho_m > 0$  satisfies

$$h_n \sim \frac{(2m + 1)!}{(m!)^2} (n + m)! \rho_m^{-n-m-1}. \quad (34.3)$$

The parameter conversion is essential: derivative order 30 means B-tree order 59, not B-tree order 30.

For comparison, the  $d = 3$  member of (34.1) is OEIS A333497, and the  $d = 4$  member is OEIS A336009 [11, 12]. They belong to B-tree orders 5 and 7, respectively. The present counterexample belongs to the same differential-equation family at  $d = 30$ . No OEIS identifier for this order-59 member is asserted.

Set

$$A = A_d = (d)_d = \prod_{a=d}^{2d-1} a = \frac{(2d-1)!}{(d-1)!}, \quad (34.4)$$

where  $(a)_j = a(a+1) \cdots (a+j-1)$  and  $(a)_0 = 1$ . In the notation  $d = m + 1$ , (34.3) is

$$h_n \sim \frac{A}{(d-1)!} (n + d - 1)! \rho^{-n-d}. \quad (34.5)$$

This is precisely the coefficient asymptotic of the candidate pole  $p_\rho(z) = A(\rho - z)^{-d}$  after multiplication by  $n!$ .

### 34.2 Main theorem and proof architecture

**Theorem 34.1.** *Let  $d = 30$ ,  $A = (30)_{30} = 59!/29!$ , and let  $H$  be the solution of (34.2). There is no finite positive  $\rho$  such that  $H$  exists on  $[0, \rho)$  and*

$$H(x) \sim A(\rho - x)^{-30} \quad (x \uparrow \rho). \quad (34.6)$$

*Consequently there is no  $\rho > 0$  for which (34.5) holds. The  $m = 29$  instance, and hence the universal statement of Burghart–Wagner Conjecture 1, is false.*

[write] **Added 5 October 2026, batch 103.** This is the case  $r = 30$  of Part I’s Theorem 9.1, which covers every  $r \geq 30$  (dictionary in Section 33.1); the scalar form (34.6) is Part I’s remark at the end of Section 9. The proof below is a second route (Section 33.3).

The conclusion concerns a specified initial-value solution. An extra unstable eigenpair at a candidate pole is insufficient by itself: a special orbit might still approach its stable manifold. The proof here rules that out by the following chain.

1. For every candidate  $\rho$ , subtract the exact pole jet from the all-ones solution. Convexity of the initial logarithmic pole jet puts this difference in the closed cone  $K = \{V \leq 2\}$ .
2. A directed cyclic linear system preserves  $K$ . Positive Fowler rescaling preserves it too, so the entire relevant normalized orbit lies in  $K$  and never vanishes.
3. At  $d = 30$ , the complete spectrum has a positive real root and one unstable conjugate pair. All stable modes have at least four cyclic sign changes. A separate flow-and-limit argument proves the stronger statement  $E^s \cap K = \{0\}$  for the whole stable subspace.

4. A nonzero orbit in  $K$  cannot approach the equilibrium along a stable manifold tangent to  $E^s$ . This contradicts the derivative limits forced by the candidate pole, and in particular by the coefficient conjecture.

All finite arithmetic needed for step 3 is exposed in Section 38. The analytic arguments, including the cone and the actual-orbit exclusion, are proved in the text; finite computer checks do not replace them.

## 35 Positive existence and the coefficient-to-pole bridge

### 35.1 The positive real solution is nonvacuous

The coordinates  $H, H', \dots, H^{(d-1)}$  stay positive throughout their positive real interval of existence. Indeed,  $H^{(d)} = H^2 \geq 0$ ; successive integration from the positive initial values proves the claim and shows that each state coordinate is increasing.

For context, the solution does have a finite positive maximal endpoint. This fact is not needed to infer the contradiction from the coefficient conjecture, but it locates the obstruction at a genuine blow-up problem.

**Proposition 35.1.** *Put  $r_j = [A(d)_j]^{1/(d+j)}$  for  $0 \leq j < d$ . The positive real maximal existence time  $\rho_*$  of (34.2) satisfies*

$$0 < \min_{0 \leq j < d} r_j \leq \rho_* \leq \max_{0 \leq j < d} r_j < \infty. \quad (35.1)$$

*Every state coordinate diverges as  $x \uparrow \rho_*$ .*

*Proof.* For every  $T > 0$ ,  $p_T(x) = A(T-x)^{-d}$  is an exact positive solution, and  $p_T^{(j)}(0) = A(d)_j T^{-d-j}$ . The derivative-jet system is cooperative in the positive orthant: its off-diagonal derivatives are nonnegative. Componentwise comparison follows, for example, by the first-contact argument with a strictly positive perturbation and then letting the perturbation tend to zero.

At  $T = \min_j r_j$ , the pole's initial jet dominates the all-ones jet, so the solution stays below that pole on the common interval. It cannot cease to exist at an earlier time: all its coordinates would then be bounded by the finite pole coordinates, allowing continuation. This proves the lower bound. At  $T = \max_j r_j$ , comparison gives the reverse inequality; existence beyond  $T$  would contradict divergence of the lower pole bound at  $T$ . This proves the upper bound.

If  $H$  were bounded at the finite maximal endpoint, then  $H^{(d)} = H^2$  and repeated integration would bound every state coordinate, allowing continuation. Thus  $H \rightarrow \infty$ . Boundedness of any higher state derivative on this finite interval would in turn bound  $H$  by repeated integration. Monotonicity therefore makes every state derivative diverge.  $\square$

[write] **Added 5 October 2026, batch 103.** Proposition 35.1 is new to the repository for general  $d$ : Part I's Lemma 8.2 proves only that  $\rho_*$  is finite. At  $d = 3$  it is Part I's (1.6) (min at  $j = 1$ , max at  $j = 0$ ), at  $d = 4$  Part III's (22.4) ( $j = 2$ ,  $j = 0$ ), and at  $d = 30$  it gives  $41.0820 \dots \leq \rho_* \leq 43.6364 \dots$  (min at  $j = 12$ , max at  $j = 0$ ; intake computation).

### 35.2 An Abelian implication with the exact normalization

**Lemma 35.2.** *If (34.5) holds for a positive  $\rho$ , then the analytic solution exists on  $[0, \rho)$ , and, for every fixed  $j \geq 0$ ,*

$$H^{(j)}(x) \sim A(d)_j (\rho - x)^{-d-j} \quad (x \uparrow \rho). \quad (35.2)$$

*Proof.* Let  $a_n = h_n/n!$  and let  $b_n = [z^n]p_\rho(z)$ . The binomial series gives

$$b_n = \frac{A}{(d-1)!} \frac{(n+d-1)!}{n!} \rho^{-n-d} > 0. \quad (35.3)$$

The conjecture says exactly  $a_n/b_n \rightarrow 1$ . Since  $b_n$  has radius  $\rho$ , so does  $\sum a_n z^n$ . Inside this radius the differentiated and multiplied series satisfy (34.2); uniqueness identifies them with the initial-value solution.

Fix  $\varepsilon > 0$ . Beyond some index  $N$ ,  $(1-\varepsilon)b_n \leq a_n \leq (1+\varepsilon)b_n$ . Sum after multiplying by  $x^n$ , where  $0 < x < \rho$ . The finitely many omitted terms remain bounded as  $x \uparrow \rho$ , whereas  $p_\rho(x) \rightarrow \infty$ . Division by  $p_\rho(x)$ , followed by  $\varepsilon \downarrow 0$ , proves  $H/p_\rho \rightarrow 1$ . For each fixed derivative order  $j$ , the ratio of the corresponding coefficients is still

$$\frac{[z^n]H^{(j)}}{[z^n]p_\rho^{(j)}} = \frac{((n+j)!/n!)a_{n+j}}{((n+j)!/n!)b_{n+j}} \rightarrow 1.$$

The same tail argument proves (35.2). □

This is a necessary Abelian implication. It uses neither differentiation of an asymptotic equivalent nor a complex-domain singularity-analysis converse. It also supplies the entire common interval needed below; no independent claim about the Taylor radius is being assumed.

[write] **Added 5 October 2026, batch 103.** The Abelian argument of Part I, first subsection of Section 9.

The stronger scalar obstruction in Theorem 34.1 needs one further observation. If one assumes only (34.6), with  $d$  in place of 30, then  $H^{(d)} = H^2 \sim A^2(\rho-x)^{-2d}$ . Repeated integration from any fixed interior point yields

$$H^{(j)}(x) \sim \frac{A^2}{(d+j)_{d-j}} (\rho-x)^{-d-j} = A(d)_j (\rho-x)^{-d-j}, \quad 0 \leq j < d. \quad (35.4)$$

Every integrated power still diverges, so the constants of integration are negligible. The elementary integration rule follows directly by sandwiching the positive top derivative between  $(1 \pm \varepsilon)$  times its asymptotic power. Thus this route also avoids unjustified asymptotic differentiation.

## 36 A closed cyclic-sign cone

For a real vector  $v = (v_0, \dots, v_{d-1})$ , delete all zero coordinates and read the remaining signs cyclically in their original order. Let  $V(v)$  be the number of changes between opposite signs, including the last-to-first comparison; set  $V(0) = 0$ . Define

$$K = \{v \in \mathbb{R}^d : V(v) \leq 2\}. \quad (36.1)$$

The count is always even. Positive diagonal maps and nonzero scalar multiplication preserve it. The word cone here means invariance under positive scalar multiplication; convexity is neither asserted nor used.

**Lemma 36.1.**  *$K$  is closed. More generally, every sublevel set  $\{V \leq 2r\}$  is closed.*

*Proof.* If a vector has more than  $2r$  cyclic changes, there are  $2r+2$  cyclically ordered nonzero entries with alternating signs. These signs persist in a neighborhood of the vector. The complement of the stated sublevel set is therefore open. For  $K$ , four alternating nonzero entries suffice. Zero coordinates elsewhere cause no difficulty. □

**Lemma 36.2** (Directed-cycle invariance). *Suppose a continuous real linear system, on a finite time interval, has the form*

$$v'_j = a_j(t)v_j + b_j(t)v_{j+1}, \quad b_j(t) \geq 0, \quad (36.2)$$

*where indices are modulo  $d$ . Its forward evolution cannot increase  $V$  and in particular preserves  $K$ .*

*Proof.* On a compact interval, forward Euler approximation with sufficiently small positive steps has updates

$$\tilde{v}_j = \alpha_j v_j + \beta_j v_{j+1}, \quad \alpha_j > 0, \quad \beta_j \geq 0.$$

Interpolate the old coordinates linearly around a circle. Division by  $\alpha_j + \beta_j > 0$  makes the new coordinate a sample on the arc from old coordinate  $j$  to coordinate  $j + 1$ . The sampling points occur in cyclic order. A linear arc contributes a sign change exactly when its nonzero endpoint signs are opposite; zero endpoints or zero segments add no change after zeros are deleted. Consequently the interpolated circle has no more sign alternations than the original cyclic list, and sampling it in cyclic order cannot increase their number.

Each Euler step therefore preserves every sublevel  $\{V \leq 2r\}$ . Euler products converge to the exact solution for continuous linear coefficients on the compact interval. Closedness, from Lemma 36.1, passes the property to that limit. Applying this to successive compact intervals proves the assertion wherever the system exists.  $\square$

Negative diagonal coefficients and large positive off-diagonal coefficients are allowed. The special cyclic-neighbor structure is important: no claim about arbitrary positive matrices is being used. This weak invariance statement has been proved directly, without requiring a stronger strict zero-number theorem.

[write] **Added 5 October 2026, batch 103.** By Remark 33.1 the cone  $K$  is Part I's  $\mathcal{C}_3$ , and by Remark 33.2 this lemma replaces the use of [13] for (9.16) in every order; Part I's other use of [13], the strict sign-regularity of  $T_r = \exp(\delta J_r)$  feeding [1], is not replaced. The cyclic sign count is the positive-feedback case of the discrete Lyapunov function of Mallet-Paret and Smith (Remark 33.3), not cited by source 04. (Dated note, 5 October 2026: this note first read “this lemma replaces the strong  $k$ -positivity step of Part I's proof for every order” and “The cyclic sign count is the discrete Lyapunov function of Mallet-Paret and Smith”. Part I uses [13] twice, and the lemma replaces only the cone-invariance use; the plain cyclic count is Mallet-Paret and Smith's function only in its positive-feedback case. Corrected at the independent check recorded at the end of Section 41.)

### 36.1 Why the specified initial condition belongs to the cone

Fix any  $\rho > 0$ , and put

$$p(x) = A(\rho - x)^{-d}, \quad w_j(x) = H^{(j)}(x) - p^{(j)}(x), \quad 0 \leq j < d. \quad (36.3)$$

Since  $(d)_d = A$ ,  $p^{(d)} = p^2$ . Its initial jet is

$$b_j = p^{(j)}(0) = A(d)_j \rho^{-d-j}.$$

The logarithms have strictly increasing successive differences:

$$\log b_{j+1} - \log b_j = \log(d + j) - \log \rho. \quad (36.4)$$

Thus the strict sublevel set  $\{j : b_j < 1\}$  is an interval of consecutive indices, possibly empty or all indices. The positive entries of  $w(0) = (1 - b_j)_j$  form one block; its negative entries are in the cyclic complement. Equalities give zero coordinates, which are deleted. Hence

$$w(0) \in K \quad \text{for every } \rho > 0. \quad (36.5)$$

On the common positive interval the difference solves

$$w'_j = w_{j+1} \quad (j < d-1), \quad w'_{d-1} = (H+p)w_0. \quad (36.6)$$

Both  $H$  and  $p$  are positive there, so Lemma 36.2 gives  $w(x) \in K$  throughout that interval. This is an initial-data-specific constraint, valid simultaneously for every possible pole location.

For  $d = 30$ ,  $w$  never vanishes as a vector. If its full jet vanished at an interior point, uniqueness for the polynomial differential system would give  $H = p$  back to zero. But having  $b_0 = b_1 = b_2 = 1$  would require both  $d/\rho = 1$  and  $(d+1)/\rho = 1$ , which is impossible. Equivalently, the linear evolution in (36.6) is invertible and its initial vector is nonzero.

## 37 Fowler coordinates and complete spectral ordering

From now on fix  $d = 30$  and  $A = (30)_{30}$ . Suppose for contradiction that (34.6) holds. Set

$$t = \rho - x, \quad \tau = -\log t, \quad u_j(\tau) = t^{d+j} w_j(x). \quad (37.1)$$

The integration argument (35.4) implies  $u(\tau) \rightarrow 0$  as  $\tau \rightarrow \infty$ . Positive coordinatewise rescaling and the preceding section give

$$u(\tau) \in K \setminus \{0\} \quad \text{for all } \tau \geq -\log \rho. \quad (37.2)$$

Because  $dx/d\tau = t$ , direct differentiation gives the autonomous polynomial system

$$\begin{aligned} u'_j &= -(d+j)u_j + u_{j+1} & (0 \leq j < d-1), \\ u'_{d-1} &= -(2d-1)u_{d-1} + 2Au_0 + u_0^2. \end{aligned} \quad (37.3)$$

Write this as  $u' = Bu + e_{d-1}u_0^2$ . The off-diagonal entries of  $B$  are 1 along the forward chain and  $2A$  on the closing edge, so  $e^{\tau B}$  preserves  $K$  for  $\tau \geq 0$ .

Eliminating coordinates of an eigenvector yields

$$P(\lambda) = \prod_{a=30}^{59} (\lambda + a) - 2A, \quad (37.4)$$

and, for a root  $\lambda$ , an eigenvector normalized by  $v_0 = 1$  is

$$v_j(\lambda) = \prod_{\ell=0}^{j-1} (\lambda + 30 + \ell), \quad 0 \leq j < 30. \quad (37.5)$$

The translation root  $\lambda = 1$  is retained. There is also a real root  $\lambda = -90$ . Both will be included in the full splitting.

### 37.1 The upper modulus curve

Write  $Q(\lambda) = \prod_{a=30}^{59} (\lambda + a)$ . We first show

$$|Q(\sigma)| < 2A \iff -90 < \sigma < 1. \quad (37.6)$$

On  $[-59, -30]$ , every factor has modulus at most 29, and thus the product is at most  $29^{30} < A$ . On  $[-90, -59]$  all factors are nonpositive and their product of magnitudes strictly decreases from  $2A$  to zero. On  $[-30, 1]$  all factors are nonnegative and this product strictly increases from zero to

$2A$ . Outside  $[-90, 1]$  it exceeds  $2A$ . The endpoint identities follow from  $\prod_{a=30}^{59}(a+1) = 2A$  and  $\prod_{a=30}^{59}(90-a) = 2A$ .

For each  $-90 < \sigma < 1$ , the modulus  $|Q(\sigma + i\omega)|$  increases strictly from below  $2A$  to infinity as  $\omega > 0$  increases. There is therefore a unique  $\omega(\sigma) > 0$  such that

$$\prod_{a=30}^{59} \sqrt{(\sigma + a)^2 + \omega(\sigma)^2} = 2A. \quad (37.7)$$

It is smooth by the implicit function theorem, and tends to zero at both endpoints. Outside the closed strip, each modulus factor can only increase when an imaginary part is added, so there are no missing roots there. At an endpoint only  $\omega = 0$  is possible.

Along the curve define its continuously summed phase by

$$\theta(\sigma) = \sum_{a=30}^{59} \arg(\sigma + a + i\omega(\sigma)),$$

with each argument in  $(0, \pi)$ . Put

$$S_0 = \sum_{a=30}^{59} \frac{1}{(\sigma + a)^2 + \omega^2}, \quad S_1 = \sum_{a=30}^{59} \frac{\sigma + a}{(\sigma + a)^2 + \omega^2}.$$

Implicit differentiation of the logarithmic modulus gives

$$\omega' = -\frac{S_1}{\omega S_0}, \quad \theta' = -\omega S_0 - \frac{S_1^2}{\omega S_0} < 0. \quad (37.8)$$

As  $\sigma$  runs from  $-90$  to  $1$ , the phase decreases continuously from  $30\pi$  to zero.

**Proposition 37.1.** *There are exactly fourteen upper-half-plane roots  $\lambda_k = \sigma_k + i\omega_k$  of  $P$ , characterized by  $\theta(\sigma_k) = 2\pi k$  for  $1 \leq k \leq 14$ . They satisfy*

$$1 > \sigma_1 > \sigma_2 > \cdots > \sigma_{14} > -90. \quad (37.9)$$

*Together with their conjugates and the real roots  $1, -90$ , these are all thirty roots. Every root is simple.*

*Proof.* On the modulus curve,  $Q = 2A$  exactly when its total phase is an integer multiple of  $2\pi$ . Strict decrease proves existence, uniqueness and ordering for the fourteen interior multiples. The real-axis discussion leaves only  $1$  and  $-90$  as real roots. This already gives thirty roots. For an upper root,

$$\frac{Q'(\lambda)}{Q(\lambda)} = \sum_{a=30}^{59} \frac{1}{\lambda + a} = S_1 - i\omega S_0 \neq 0.$$

The conjugate case follows. At each real endpoint all reciprocal factors have the same nonzero sign, so  $Q'$  is nonzero there as well.  $\square$

Thus no Jordan blocks, hidden real roots or ties between distinct conjugate-pair real parts remain to be dealt with later.

[write] **Added 5 October 2026, batch 103.** Proposition 37.1 and Corollary 38.1 below give the complete spectrum at  $r = 30$  as a proved statement. Part I proves at least one unstable pair for every  $r \geq 30$  and simplicity for every  $r$ , and has the complete count at  $r = 30$  only as an exact Routh check.

### 38 The compact exact phase certificate

The only finite arithmetic required by the proof determines where the imaginary axis meets the upper modulus curve. Define

$$F(y) = \prod_{a=30}^{59} \left(1 + \frac{iy}{a}\right), \quad \Theta(y) = \sum_{a=30}^{59} \arctan \frac{y}{a} \quad (y \geq 0). \quad (38.1)$$

There is a unique  $\omega_* > 0$  with  $|F(\omega_*)| = 2$ ; it equals  $\omega(0)$  from (37.7). We will certify

$$2\pi < \Theta(\omega_*) < 4\pi. \quad (38.2)$$

Take  $q = 911/100$ , and compute exact integers

$$R + iI = \prod_{a=30}^{59} (100a + 911i), \quad S = 100^{30}A. \quad (38.3)$$

Then  $F(q) = (R + iI)/S$ . The following seven strict inequalities use only integers or rational numbers:

$$R > 0, \quad I > 0, \quad R^2 + I^2 < 4S^2, \quad (38.4)$$

$$6 < \sum_{a=30}^{59} \left( \frac{q}{a} - \frac{q^3}{3a^3} \right), \quad \sum_{a=30}^{59} \frac{q}{a} < 7, \quad (38.5)$$

$$\prod_{a=30}^{59} (a^2 + 100) > 4A^2, \quad \sum_{a=30}^{59} \frac{10}{a} < 8. \quad (38.6)$$

They are independently reproducible by the short program in Section 43 and the fuller companion checker. Neither transcendental-function evaluation nor floating-point arithmetic is involved in these comparisons. As optional exact rational enclosures for orientation, the same computations give

$$1.994 < R/S < 1.995, \quad 0.0049 < I/S < 0.0050.$$

The finite decimals here denote exact rationals, not floating-point error estimates; the seven inequalities above remain the certificate.

For  $x > 0$ , integration of  $1 - x^2 < 1/(1 + x^2) < 1$  gives

$$x - \frac{x^3}{3} < \arctan x < x.$$

Thus (38.5) implies  $6 < \Theta(q) < 7$ . Using  $3 < \pi < 22/7$ , this interval lies within  $(3\pi/2, 5\pi/2)$ . Since  $I > 0$ , its sine is positive. The only possibility in this interval is therefore  $\Theta(q) > 2\pi$ . The modulus inequality (38.4), and strict increase of  $|F(y)|$ , give  $q < \omega_*$ . The first inequality in (38.6) is  $|F(10)|^2 > 4$ , so  $\omega_* < 10$ . Finally,

$$2\pi < \Theta(q) < \Theta(\omega_*) < \Theta(10) < 8 < 4\pi. \quad (38.7)$$

This proves (38.2) with strict margins.

**Corollary 38.1.** *B is hyperbolic. Its unstable eigenvalues are exactly  $1, \lambda_1, \overline{\lambda_1}$ ; its stable eigenvalues are  $-90$  and  $\lambda_k, \overline{\lambda_k}$  for  $2 \leq k \leq 14$ . Its real stable subspace has dimension 27 and its real unstable subspace has dimension 3.*

*Proof.* At  $\sigma = 0$ , the modulus-curve phase is  $\theta(0) = \Theta(\omega_*) \in (2\pi, 4\pi)$ . By strict decrease of  $\theta$ , the root with phase  $2\pi$  has positive real part, whereas every root with phase  $2\pi k$ ,  $k \geq 2$ , has negative real part. No root lies on the imaginary axis. Include the two real roots and use Proposition 37.1.  $\square$

For orientation, an independently computed rational Routh table for  $P(\lambda)/(\lambda-1)$  has first-column signs

$$\underbrace{+ \cdots +}_{28 \text{ entries}} \quad - \quad +.$$

This second finite check agrees with the spectral conclusion. The proof above does not require Routh's theorem or a large determinant table. A numerical root approximation is not an input to the proof.

[write] **Added 5 October 2026, batch 103.** This Routh table is the one of Part I's "Independent finite spectral checks" (Section 9). Here it corroborates a proof; there it was the only complete count.

### 39 The entire stable subspace misses the cone

The ordering just proved has a precise sign interpretation. For an upper root  $\lambda_k$ , set

$$\delta_j = \arg(\lambda_k + 30 + j) \in (0, \pi), \quad s_0 = 0, \quad s_j = \sum_{\ell < j} \delta_\ell.$$

Append  $v_{30} = 2Av_0$  to the eigenvector (37.5). Its unwrapped phase is  $s_{30} = 2\pi k$ , so the appended coordinate closes the cyclic sign comparison with a positive multiple of  $v_0$ .

**Lemma 39.1.** *For a generic phase  $\phi$ , meaning that no displayed coordinate vanishes, every vector  $\text{Re}(cv(\lambda_k))$  with  $\arg c = \phi$  and  $c \neq 0$  has exactly  $2k$  cyclic sign changes. An eigenvector for  $-90$  has 30 cyclic sign changes.*

*Proof.* The real coordinates have signs of  $\cos(\phi + s_j)$ , since their amplitude factors are positive. The consecutive phase arcs have lengths  $\delta_j < \pi$  and together cover an interval of length  $2\pi k$ . Each arc contains at most one zero of cosine, and the full interval contains exactly  $2k$  zeros. At a generic phase none is an endpoint. Every zero crossing changes the sampled sign across that arc, and every sampled sign change requires a crossing. This gives exactly  $2k$  cyclic changes, including the closing arc to  $v_{30} = 2Av_0$ . For  $-90$ , all factors  $-90 + 30 + j$  are negative, so the coordinates alternate signs. Since  $d = 30$  is even, the closing edge also changes sign, giving thirty changes.  $\square$

Only generic phases will be needed. No convention that assigns signs to zero coordinates is introduced.

**Proposition 39.2.** *For the full real stable subspace  $E^s$  of  $B$ ,*

$$E^s \cap K = \{0\}. \tag{39.1}$$

*Proof.* Suppose  $0 \neq z \in E^s \cap K$ . The linear flow  $e^{\tau B}z$  remains in  $K$  by Lemma 36.2. Decompose  $z$  into its components in the simple real eigenspace and simple conjugate-pair real eigenspaces. Among its nonzero components select the one with largest real part. Proposition 37.1 makes this component unique up to its conjugate pair.

If it is the pair indexed by  $k \geq 2$ , there is a  $c \neq 0$  such that

$$e^{-\sigma_k \tau} e^{\tau B} z = \text{Re}\left(ce^{i\omega_k \tau} v(\lambda_k)\right) + o(1) \quad (\tau \rightarrow \infty). \tag{39.2}$$

All other components decay relatively, by the strict real-part ordering. Choose a generic phase and a sequence  $\tau_n \rightarrow \infty$  on which  $\arg c + \omega_k \tau_n$  is that phase modulo  $2\pi$ . Such a sequence is an arithmetic progression of spacing  $2\pi/\omega_k$ ; no density assumption on discrete rotations is needed. The left side of (39.2) belongs to  $K$ , because its rescaling factor is positive. Its limit is a nonzero generic vector of that pair, with  $2k \geq 4$  changes by Lemma 39.1. This contradicts closedness of  $K$ .

If the leading component is the real mode  $-90$ , the same positive rescaling yields a nonzero real eigenvector with thirty changes, again contradicting closedness. These cases exhaust the stable subspace and prove (39.1).  $\square$

This proposition supplies an indispensable step. Counting signs in individual eigenvectors alone would not control arbitrary linear combinations. Here invariance of the cone, strict spectral ordering, and a leading-mode limit establish the statement for the entire 27-dimensional stable subspace.

[write] **Added 5 October 2026, batch 103.** A second route to Part I's (9.17) at  $r = 30$ , where Part I's  $E_{\text{slow}}$  is  $E^s$  because the three largest moduli belong to the unstable eigenvalues  $1, \lambda_1, \bar{\lambda}_1$ . Part I uses the spectral theorem of [1] instead.

## 40 Excluding the actual nonlinear orbit

The vector field (37.3) is polynomial, with quadratic remainder at its equilibrium 0. Corollary 38.1 makes that equilibrium hyperbolic. The ordinary local stable-manifold theorem gives a  $C^1$  graph

$$W_{\text{loc}}^s = \{s + h(s) : s \in E^s \text{ near } 0\}, \quad h(s) \in E^u, \quad h(0) = 0, \quad Dh(0) = 0, \quad (40.1)$$

and every trajectory tending to 0 eventually lies in this graph. For an explicit reference, apply Lanford's Theorem 7.3.1(1),(4) to the local time-one map [10, printed p. 74, PDF p. 76]. Its derivative is  $e^B$ , its stable and unstable spaces are  $E^s, E^u$ , and no multiplier has modulus one. Choose the spectral cutoff between the largest stable multiplier modulus and 1. In the notation of that theorem,  $\rho_- < 1 < \rho_+$ , which is the ordinary hyperbolic stable case. Local ODE existence supplies this map on a neighborhood of the origin; global completeness is unnecessary. A convergent trajectory's sufficiently late future stays in the required neighborhood.

**Lemma 40.1.** *The graph (40.1) has no sufficiently small nonzero point in  $K$ .*

*Proof.* Otherwise there are  $z_n = s_n + h(s_n) \in K \setminus \{0\}$  tending to zero. Pass to a subsequence such that  $z_n/\|z_n\|$  converges to a unit vector  $v$ . It lies in  $K$  by positive-scaling invariance and closedness. Since  $h(s) = o(\|s\|)$  and the fixed projection onto  $E^s$  along  $E^u$  is bounded,  $\|s_n\| \leq C\|z_n\|$ . Consequently  $h(s_n)/\|z_n\| \rightarrow 0$ , so  $v \in E^s$ . This contradicts Proposition 39.2.  $\square$

*Completion of Theorem 34.1.* The assumed scalar pole asymptotic implies (35.4), and hence  $u(\tau) \rightarrow 0$ . The actual all-ones orbit has  $u(\tau) \in K \setminus \{0\}$  by (37.2). For all sufficiently large  $\tau$ , the stable-manifold theorem puts it in (40.1), contradicting Lemma 40.1. Thus the scalar asymptotic is impossible. If the published coefficient asymptotic held, Lemma 35.2 would imply that same scalar asymptotic (and all normalized derivative limits directly). It therefore fails for  $d = 30$ , which is  $m = 29$  and B-tree order 59.  $\square$

The argument needs no asymptotic leading-eigenmode theorem for a nonlinear trajectory. In particular, it makes no nonresonance assumption and no assumption that the slowest stable coefficient is nonzero. Approaching a stronger stable submanifold, higher-order cancellation, or unusually fast decay cannot evade the tangent-cone contradiction. Neither a genericity argument nor a dimension count has been substituted for the all-ones initial condition.

[write] **Added 5 October 2026, batch 103.** The tangent-cone exclusion is Part I’s argument in Section 9. At  $r = 30$  the equilibrium is hyperbolic, so the ordinary stable manifold suffices; Part I uses Lanford’s generalized stable manifold because it does not assume hyperbolicity for every  $r \geq 30$ .

## 41 What is established, and what remains open

### 41.1 Logical scope

The counterexample settles the universal formulation by one member of the natural family. It does not invalidate every smaller-order case. The exact seven-inequality certificate concerns  $d = 30$  only; this report does not claim a complete stability classification of integer orders. In particular, it does not use or require a separate finite computation through  $d = 150$ , and it does not claim here that 30 is the first order with an extra unstable pair. The eigenvalue 1 is always present; any discussion of a later “loss of stability” must specify that it concerns additional modes beyond this translation direction.

No replacement for (34.5) has been proved for order 59. The theorem excludes convergence to the pure-pole equilibrium for this orbit; it does not identify an alternative limiting object. Oscillatory transients, numerical periodic-looking plots and a local Hopf calculation would not, on their own, prove a periodic attractor or its basin for the specified solution. The existence of non-power solutions for other exponents or other initial conditions also does not supply that missing conclusion.

The sources used here establish the combinatorial model, the exact published conjecture, and the local stable-manifold theorem. The finite certificate, cone argument and orbit exclusion are presented explicitly in this report. The companion’s independent implementations and negative tests are mathematical and reproducibility checks, not formal proof-assistant verification or external peer review. No global literature-novelty determination is asserted.

### 41.2 Further work directed by the obstruction

The most useful next questions concern the actual high-order behavior, rather than refining a disproved pure-pole formula.

1. **Identify the real normalized orbit.** For  $U_j = (\rho_* - x)^{d+j} H^{(j)}$ , the uncentered Fowler equations are  $U'_j = U_{j+1} - (d+j)U_j$  and  $U'_{d-1} = U_0^2 - (2d-1)U_{d-1}$ . One needs rigorous bounds and a description of the actual orbit’s limit set. If a periodic profile is proposed, prove its existence, stability and attraction of this initial condition. The present exclusion proves none of these three properties.
2. **Recover coefficient asymptotics.** A real blow-up description alone would not justify coefficient transfer. Establish suitable complex continuation, control all singularities of smallest modulus, and prove uniform error bounds. A genuinely log-periodic singular profile would require transferring its harmonics with their individual gamma factors; an assumed profile cannot be used as a theorem.
3. **Build valid inverse and threshold estimates.** After a replacement coefficient law is proved, derive the inverse counting scale and any phase-dependent corrections, together with error control strong enough for integer threshold brackets. Inversion of the false pure-pole model at order 59 is not justified.
4. **Resolve the transition across orders.** Exact spectral certificates for successive integer  $d$  can locate additional crossings; continuous parameter families can distinguish spectral

crossings from genuine nonlinear bifurcations. Then determine which initial data approach which limiting profiles. Stability at a smaller order, by itself, proves neither the all-ones pole behavior nor the coefficient conjecture at that order.

These are research targets, not consequences silently appended to the counterexample.

### 41.3 Further questions and research

*[write] Added 5 October 2026, batch 103, under the standing rule. Source 04 contains no claim found to be wrong. Its four questions above are Part I's Questions 1 and 2 at  $m = 29$ .*

1. **The actual limit set at order 59** (source 04, question 1; Part I's Question 1). Open. Pointer: the uncentred Fowler system is a monotone cyclic feedback system, so the Poincaré–Bendixson theorem of Mallet-Paret and Smith (Remark 33.3) may restrict the  $\omega$ -limit set of the normalized orbit to a periodic orbit or a set of equilibria and connecting orbits, once boundedness and the other hypotheses are checked. Not attempted at the write.
2. **Coefficient asymptotics and inverses at order 59** (source 04, questions 2–3). Open; inversion of the false pure-pole model is not justified.
3. **The transition across orders** (source 04, question 4; Part I's Question 2). Part I's exact Routh check shows  $r = 29$  spectrally stable and  $r = 30$  with exactly two right-half-plane roots besides 1; whether the all-one orbit converges to the pole equilibrium for  $4 \leq m \leq 28$  is open.
4. **The spectral exclusion for every order by Part IV's route.** Proposition 39.2 needs the complete spectral ordering. The modulus-curve argument of Section 37 is written for  $d = 30$ ; whether it, with Lemma 39.1, gives  $E_{\text{slow}} \cap \mathcal{C}_3 = \{0\}$  for every  $r \geq 30$  without [1] was not examined. Part I's Theorem 9.1 does not depend on the answer.

**Independent check of the write (dated note, 5 October 2026).** An independent adversarial check of Parts II–IV, made by the intake after the write (6603ab5a8), reread eight items: Theorem 30.1 with its proof; the write's precision note on the step from a polynomial recurrence to a linear ODE; Remarks 33.1, 33.2 and 33.3, with Lemma 36.2; the radii 1183/960 and 121/6720, with the write's note that a radius in  $t$  needs a bound on  $|c|$ ; the order dictionary of Section 33.1; and the bounds  $16800^{1/6} \leq \rho_4 \leq 840^{1/4}$ , with the extreme  $r_j$  of Proposition 35.1. It found all eight items valid, with no counterexample and no gap in any proof chain. Three passages of the write said more than the mathematics gives and are corrected where they stand. Part I uses [13] twice: for the cone invariance (9.16), and for the strict sign-regularity of  $T_r = \exp(\delta J_r)$ , the hypothesis of [1] behind (9.17). Lemma 36.2 replaces the first use only, but the row “Cyclic cone  $K$ ” of the table in Section 33.3 and the note after Lemma 36.2 said that it replaced the whole strong  $k$ -positivity step (dated notes keep both first wordings). Remark 33.2 itself was careful; its last sentence now also names the second use. Remark 33.3 said that the cyclic count is Mallet-Paret and Smith's discrete Lyapunov function for couplings “of fixed sign”. They assume strictly signed couplings and weight each sign change by the sign of its coupling, so the plain cyclic count is their function only when every coupling is positive (dated note after the remark). The note after Lemma 36.2, which repeats the identification, carries the same qualification. Part I's note after (9.16) now says that Part I still uses [13] for  $T_r$ , and refers to Remark 33.3 for the cyclic count instead of identifying it. No other statement or number changed.

The check used neither the delivered programs nor the write's. It read the proof of Theorem 30.1 line by line. The proof is a branch-counting argument: the continuations  $H_0, \dots, H_N$  of  $H$  around  $\rho$  are linearly independent for every  $N$ , while a linear ODE of order  $d$  has a  $d$ -dimensional solution

space at an ordinary point. The check confirmed that the inputs (a convergent expansion on every finite angular interval,  $D(E)U = \kappa U^2$ ,  $c \neq 0$ ) are proved for  $m = 2$  and  $m = 3$ . It recomputed  $D_2(b) = -(b+1)(b^2 - 13b + 60)$  and  $D_3(b) = (b+1)(b-12)(b^2 - 11b + 70)$ , hence  $\alpha_2 = (13 \pm i\sqrt{71})/2$  and  $\alpha_3 = (11 \pm i\sqrt{159})/2$ . It checked the isolation of the pure complex axis, the degree argument giving infinitely many  $v_k \neq 0$ , the tail bound and the Bohr mean in the grouped uniqueness lemma, the strictly decreasing moduli  $\exp(-2\pi k \operatorname{Im} \alpha)$  of the characters, and the passage from a recurrence to an operator,  $[z^{n+s}]LF = \sum_j q_j(n)f_{n+j}$  with  $q_j(n) = p_j(n) \prod_{a=1}^j (n+a)$ , including the write's precision note. With its own recurrence code it computed  $h_0, \dots, h_{600}$  for A333497, which agree term for term with `data/independent-exact_h_0_600.txt`, and the corresponding terms of A336009, which agree with the 31-term OEIS prefix in `data/03-order7-checks-fixtures-oeis_A336009_prefix.json`. As corroborating evidence only, not as part of the proof, it searched both sequences for polynomial recurrences by linear algebra over  $\mathbb{F}_p$ ,  $p = 2147483629$ . It set up 204 systems: for each sequence, every order  $s \leq 60$  starting at  $n = 0$  and every order  $s \leq 40$  starting at  $n = 150$ , each at the largest degree that the 601 terms allow with about 25 spare equations (from  $n = 0$ : degree 575 at order 0, 286 at order 1, 142 at order 3, 42 at order 12, 12 at order 40, 7 at order 60). All 204 systems have full column rank, so no recurrence of those orders and degrees exists modulo  $p$ , hence none over  $\mathbb{Q}$  or  $\mathbb{C}$ . The same code finds the known order-two recurrences of the Apéry numbers and of  $n![z^n]e^z/(1-z)^3$  (positive controls). This search covers finitely many orders and degrees; Theorem 30.1 covers all of them and does not depend on it.

For Remark 33.1 the check tested every  $v \in \{-1, 0, 1\}^d$  with  $d \leq 9$  (29,523 vectors): no counterexample, and  $V$  always even. For Lemma 36.2 it applied 200,000 random Euler steps ( $\alpha_j > 0$ ,  $\beta_j \geq 0$ ,  $d \leq 9$ ) and  $V$  never increased; with a negative coupling,  $V$  can rise from 0 to 2, so the sign hypothesis is needed. Integrating Part I's nonlinear system (9.4) at  $r = 30$  (fourth-order Runge–Kutta, step  $2 \cdot 10^{-4}$ ) for  $\rho \in \{30, 35, 41, 42.3, 43, 45\}$  gave  $V(Z(s)) \leq 2$  along every run, as Remark 33.2 predicts. It re-derived the radii:  $1183/960 = 1/(8M)$  with  $M = 120/1183$  from  $|D(\beta)| \geq 1183/2$ , and  $121/6720 = 1/(12M)$  with  $M = 560/121$  from  $|D(\beta)| \geq 363/2$  (the brute-force minimum over low indices is 840 in both cases), and  $c_0 = -1/18052070400$ . It confirmed the angular factor behind the write's note that a radius in  $t$  needs a bound on  $|c|$ . It checked the order dictionary:  $2m+1 = 59$ ,  $r = m+1 = 30 = d$ ,  $(30)_{30} = 59!/29!$ , and Burghart and Wagner's constant  $59!/(29!)^2 = A/(d-1)! = 1773968723472921360$ . It checked the pole coordinates at  $16800^{1/6}$  and  $840^{1/4}$  that give the bounds on  $\rho_4$ , with the comparison argument closing both directions, and recomputed the extreme  $r_j$  of Proposition 35.1 at  $d = 3, 4$  and  $30$  (at  $d = 30$ ,  $41.082019 \dots$  at  $j = 12$  and  $43.636458 \dots$  at  $j = 0$ , as printed). Its account of Mallet-Paret and Smith, their bibliographic data and their sign hypotheses, rests on an independent citing paper and on recollection; the paper itself was not reread. This was a careful reading with exact computations, not a formal verification.

## 42 Reproducibility and reading guide

The accompanying archive has a single top-level directory, `report117/`. It contains this PDF, the standalone L<sup>A</sup>T<sub>E</sub>X source, reproducible build and packaging scripts, and the finite exact checker. The source papers are linked in the bibliography and are not copied into the archive.

The compact certificate and the independent rational Routh check use Python standard-library integers and `fractions.Fraction` only. The checker uses explicit failure guards, so its checks remain active under Python's optimization flag. Its companion harness tests both ordinary and optimized execution, deliberately corrupted fixtures and inventories, and replay in a fresh directory. A strict closed manifest rejects missing, changed and unexpected files. Integrity hashes detect changes relative to their inventory; they are not an authenticated signature.

See `README.txt` for the exact commands and dependency versions, and `checks/README.md` for each checked identity and the negative test scope. The machine-readable `verification_results.json` records the delivered validation run. Run the commands with `-B` to avoid adding bytecode caches to the closed package. A reader should verify the supplied checksum inventory, not regenerate it merely to make a modified package pass.

The PDF build fixes timestamps and metadata, compiles in two independent clean temporary directories, rejects overfull boxes and unresolved references, and requires byte-identical PDFs. The ZIP builder uses sorted paths and fixed timestamps and file modes. Byte identity is claimed within the tested TeX/font and Python/compression toolchains; different toolchain versions can change bytes without changing the mathematics. A fresh-directory replay is not a newly provisioned operating system. The final PDF is also rendered for a page-by-page visual check, separate from the proof and arithmetic checks.

[write] **Added 5 October 2026, batch 103.** In the repository the companion's files are `code/04-order59-*.py`, `data/04-order59-*` and `04-order59-checks-README.md`; its `checks/check_manifest.py` is the byte-identical `code/03-order7-checks-check_manifest.py`. The PDF, TeX source, `README.txt`, `CHECKSUMS.sha256` and `checks/MANIFEST.json` were not shipped; the report's `README` gives the rerun routes and the two Windows patches the harness needs.

## 43 A short standalone exact certificate (Report 117, Appendix A)

The following standard-library program regenerates all seven inequalities. Its integer pair update is multiplication by  $100a + 911i$ ; the simultaneous assignment uses the old  $R, I$  in both right-hand expressions. Every comparison is exact. Failure guards are ordinary conditionals, so `python -O` cannot remove them.

```
from fractions import Fraction as F
from math import prod

d = 30
A = prod(range(d, 2*d))
q = F(911, 100)
R, I = 1, 0
for a in range(d, 2*d):
    R, I = 100*a*R - 911*I, 911*R + 100*a*I
S = 100**d * A
lower = sum(q/a - (q/a)**3/3 for a in range(d, 2*d))
upper = sum(q/a for a in range(d, 2*d))
checks = {
    "real positive": R > 0,
    "imaginary positive": I > 0,
    "modulus at q below 2": R*R + I*I < 4*S*S,
    "lower phase bound exceeds 6": lower > 6,
    "upper phase bound below 7": upper < 7,
    "modulus at 10 exceeds 2":
        prod(a*a + 100 for a in range(d, 2*d)) > 4*A*A,
    "phase bound at 10 below 8":
        sum(F(10, a) for a in range(d, 2*d)) < 8,
}
```

```

for name, ok in checks.items():
    if not ok:
        raise RuntimeError(name)
print("PASS: all seven exact inequalities")

```

This short replay illustrates the complete critical arithmetic; `checks/check.py` adds independent reconstructions, frozen exact fixtures, identity checks and strict integrity validation. The analytical inference from these seven comparisons to the spectrum is Section 38, not an unstated property of the program.

## 44 A proof dependency checklist (Report 117, Appendix B)

For a line-by-line audit, the essential dependencies are:

1. **Published normalization:**  $d = m + 1$ , B-tree order  $2d - 1$ , the all-ones jet, and the factorial and radius powers in (34.5).
2. **The Abelian bridge:** positive coefficient tails imply the real limits of all needed derivatives; no singularity-analysis converse is used.
3. **Closedness at zeros:**  $K$  uses deleted-zero cyclic variation, and every prohibited sign pattern is witnessed by nonzero alternating coordinates.
4. **Forward invariance:** Euler steps sample consecutive cyclic arcs and pass to the continuous flow through the closed cone.
5. **The actual initial jet:** logarithmic convexity holds for every candidate  $\rho$ , positivity makes the closing coupling nonnegative, and uniqueness excludes the exact-pole exception.
6. **Complete spectral control:** the modulus curve gives all roots, simplicity and strict real-part ordering; seven strict rational/integer inequalities put exactly one conjugate pair beyond the translation direction in the unstable space.
7. **Whole-space exclusion:** linear-flow invariance and a phase-chosen limit control all stable linear combinations, not just individual eigenvectors.
8. **Nonlinear exclusion:** hyperbolicity gives the local stable graph; its tangency and  $E^s \cap K = \{0\}$  prohibit a nonzero normalized orbit in  $K$  from converging to the equilibrium.

The supplementary blow-up enclosure and the independent Routh table are useful checks and context, but neither is required for this logical chain from the conjecture to a contradiction.

## A A compact verification guide

[write] **Added 5 October 2026, batch 103.** This appendix belongs to Part I. Parts II–IV, which precede it, have no appendices of their own: source 04’s two appendices are printed as Sections 43 and 44.

The detailed third-order proof dependencies can be checked in the following order.

1. Verify the change of variables (2.1) and the exact cancellation in (2.5). Compactness must allow the harmless boundary  $p = 0$ , while keeping  $q$  away from zero.
2. Reconstruct the real solution by (2.6). Finite  $\rho$  and the constant 60 follow without a numerical approximation.
3. Check the nonresonant Poincaré hypotheses for (3.1); a smooth or formal linearization would not suffice for the claimed convergence.
4. Check the signs in (3.5) and (3.6), then the exact time identity (3.9). The coefficient denominators in  $S$  and  $B$  have real parts growing with total degree.
5. Apply analytic implicit inversion at  $(\mathcal{R}, \eta) = (1, 0)$ , where the derivative is 1. Nonzero orbit constants and (3.12) prove  $C \neq 0$ .
6. Separate the proof of the radius from the proof of uniqueness on the circle. Strict positivity for each of  $H, H', H''$  gives the common margin in (5.2) and the dominating disk (5.3).
7. Transfer finitely many terms only. Remove analytic diagonal monomials, retain exact Gamma ratios, and expand one extra grade to justify the noninteger-tail remainder.
8. Invert the logarithmic expansion using monotonicity and a ceiling window. Neither fitted constants nor a root extremely near an integer justify dropping that window.

For the fourth-order result, additionally check the monotone-density derivative upgrade, the three nonresonant exponents, the conserved energy, and the backward-Metzler exclusion of a pure real mode. For the uniform obstruction, verify the finite rational phase bounds and the infinite-range inequalities, the whole-subspace sign-variation theorem, and the generalized stable-manifold hypotheses. The last step must use the slow/fast spectral gap; ordinary hyperbolic stability alone would not cover every high order.

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